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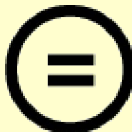
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116th Doctoral Dissertation

Advisor: Prof. Chansik Park

A Framework for Personalized Safety Training using Large Language Models in Construction Meta-Scenes

February 2025

**The Graduate School
Chung-Ang University**

**Department of Architectural Engineering
Major in Building Construction Engineering and
Management**

Rahat Hussain

A Framework for Personalized Safety Training using Large Language Models in Construction Meta-Scenes

**Presented to the Faculties of the Chung-Ang University in
Partial Fulfillment of the Requirement of the Degree of Doctor
of Philosophy**

February 2025

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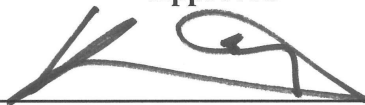
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Rahat Hussain

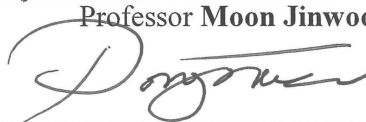
Department of Architectural Engineering
Chung-Ang University

Date: Dec. 13th. 2024

Approved



Professor **Moon Jinwoo**



Professor **Lee Dongmin**



Professor **Lee JeeHee**



Professor **Lee Doyeop**



Professor **Park Chansik**, Supervisor

Dissertation submitted in partial fulfillment of
the requirements for the Degree of Doctor of Philosophy in the Department of
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Chung-Ang University

2025

Abstract

A Framework for Personalized Safety Training using Large Language Models in Construction Meta-Scenes

Rahat Hussain

Major in Building Construction Engineering and Management

Department of Architectural Engineering

The Graduate school, Chung-Ang University

The construction industry remains one of the most hazardous sectors, reporting consistently high rates of accidents and injuries despite ongoing efforts to improve safety practices. Safety training is recognized as one of the most effective solutions to minimize these risks. Among the various methods, personalized training—which tailors learning content to match the specific characteristics and needs of individual workers—has shown significant potential for improving engagement and knowledge retention. However, existing studies on traditional safety training have largely focused on static learner information, such as demographic data (e.g., age, language proficiency, or job role), and covered only limited phases of the training lifecycle. These approaches overlook dynamic information like situational awareness (SA),

which is a critical aspect in the pre-training phase that helps to assess workers' ability to recognize hazards before exposure to real-world risks. Additionally, post-training job-site support—essential for reinforcing safety practices—has been underexplored. To address these gaps, this research proposed a personalized safety training framework to integrate both static information (e.g., personal characteristics) and dynamic information (e.g., situation awareness) to deliver safety content in perspective construction meta-scenes. The proposed framework consists of three key modules: 1) Situational Awareness Module – to develop a personalized learner model by assessing each worker's situational awareness in interactive virtual reality environment. 2) Knowledge Acquisition Module – Customized training content delivery module based on individual learner information and performance by leveraging the power of Conversational AI. 3) Scene Understanding Module – to provide a mobile-based solution for real-time vision-based hazard recognition on job sites through interactive scene understanding.

The effectiveness of the framework was evaluated using struck-by accident scenarios which are the common cause of workplace injuries in construction. Results demonstrate that all participants—regardless of language proficiency, education level, prior experience or Situation awareness level—achieved significant knowledge gains. Notably, users with higher situational awareness

(SA) levels showed a 10% greater improvement in knowledge acquisition compared to those with lower SA, indicating that initial awareness levels enhance the system's effectiveness. Additionally, user feedback on the Scene Understanding Module indicated a high acceptance rate, particularly in terms of ease of use and predictability. User satisfaction, including feedback from safety practitioners, varied across criteria, with high scores for time efficiency and ease of use but lower ratings for operability under dynamic conditions. The post SA levels revealed a significant improvement following the interaction with the Knowledge Acquisition and Scene Understanding modules. Specifically, the average SA score increased from 35.0 (initial) to 47.84 (post-intervention), reflecting a 37% improvement. This research makes a valuable contribution to the global construction community by promoting safety practices and enhancing the transfer of training to real-world job sites. Furthermore, it opens new avenues for future research in immersive technologies and AI-driven training methodologies, advancing efforts to reduce accidents and improve safety outcomes in diverse construction environments.

Keywords: Personalized Training, Conversational AI, Computer Vision, Virtual Reality, Hazard Recognition, Large Language Models, Meta Scenes.

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List of Abbreviations

AI	Artificial intelligence
BIM	Building Information Modeling
CV	Computer vision
LLM	Large Language Models
SA	Situational Awareness
RAG	Retrieval-Augmented Generation
AR	Augmented Reality
VR	Virtual Reality
MR/XR	Mixed Reality

1. Introduction

This chapter provides a brief introduction to the context in which the research was conducted, beginning with the research background and presenting the overall vision of the study. It outlines the research problem, needs, and objectives to specify the study's content in detail. The chapter then discusses the research methodology, followed by an outline of the dissertation's structure.

1.1 Research Background and Objectives

The complex and dynamic nature of the construction industry poses major safety challenges to workers, leading to a poor safety record in this sector. Across all industries, construction has seen an alarming increase in workplace accidents (Figure 1). The Bureau of Labor Statistics reports a work-related fatality rate of 3.7 per 100,000 full-time equivalent workers in the United States for 2022, reflecting a significant 7.7% increase from the previous year (Bureau of Labour Statistics, 2022). Moreover, the construction sector in Europe ranks third for nonfatal injuries, exceeding the industry average (Eurostat, 2022). Such incidents not only affect the well-being and livelihoods of the victims and their families but also lead to environmental damage and substantial economic losses. Therefore, improving safety performance in the

construction industry is crucial, requiring effective management practices that focus on identifying and proactively addressing potential hazards.

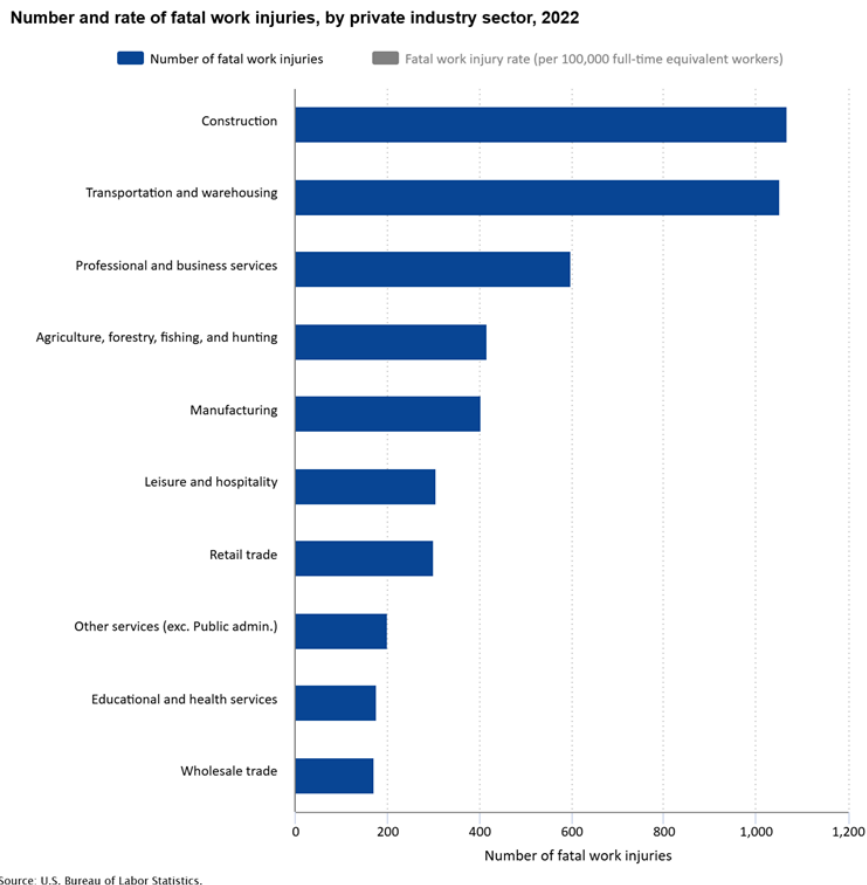


Figure 1. Accident report comparison between construction and other industries (U.S. BLS-2022).

The bar chart provides a comparative analysis of fatal work injuries across various private industry sectors in 2022, highlighting both the total number of fatalities and the fatal work injury rates per 100,000 full-time equivalent workers. Construction emerges as the most hazardous industry, leading in both

total fatalities and injury rate, underscoring the high risks inherent to this sector, such as falls, equipment-related accidents, and structural collapses. Transportation and Warehousing follow closely, reflecting risks from vehicular accidents and operational hazards, while Agriculture, Forestry, Fishing, and Hunting has a relatively lower number of fatalities but a notable injury rate due to hazardous machinery and environmental challenges. Other sectors, such as Manufacturing, Professional and Business Services, also report significant fatalities, primarily linked to operational and machinery-related risks.

In contrast, industries like educational and health services and wholesale trade report the fewest fatalities, indicating safer work environments overall. However, even these sectors are not without risks, as fatalities still occur under specific circumstances. The data underscores construction as the most dangerous private industry sector, necessitating a heightened focus on safety measures and regulatory compliance to protect workers in this high-risk environment. Similarly, the construction sector in 2023, remained the leading contributor to fatal injuries in Singapore. This was followed by the transportation & storage industry, which recorded eight fatal injuries, and the manufacturing sector with five. These historically high-risk industries accounted for 86% of all fatal injuries that year (Figure 2).

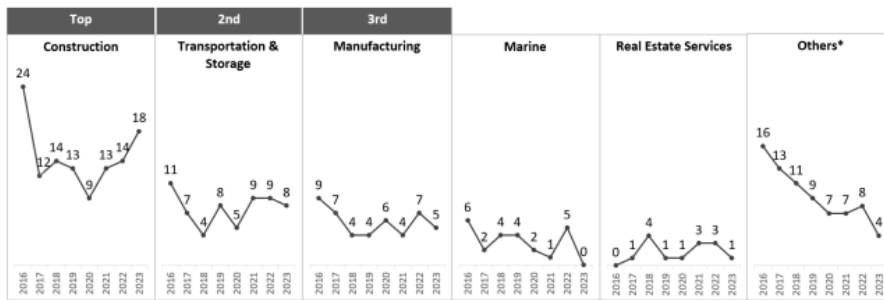


Figure 2. Number of workplace fatalities by industry in Singapore (2016-23)

Recently, construction safety has undergone a digital transformation aimed at reducing these alarming statistics (Park et al., 2023). For instance, recent researches have focused on implementing various real-time monitoring systems, computer vision-based techniques, and remote monitoring to maintain a safe distance between workers and equipment (Khan et al., 2023; D. Kim et al., 2019; Tran et al., 2022, 2023). Furthermore, wearable technology, such as smart helmets and safety vests with sensors, enables real-time health and environmental monitoring of workers (Park et al., 2023). Automatic rule checking using Building Information Modeling (BIM) for safety planning (S. Zhang et al., 2013) and introducing smart contracts using blockchain, (H. Liu et al., 2023) are also promising techniques. Additionally, Virtual reality techniques assessing safe behavior have been used to reduce risk habituation and to improve hazard identification skills (Ahn, 2021; Grégoire et al., 2022; Hussain et al., 2020).

Among these approaches to enhance jobsite safety, training is considered as one of the key mechanisms for enhancing workers' safety performance (Vignoli et al., 2021). A meta-analysis by Ricci et al. proved that safety training strongly influenced workers' knowledge and safety attitudes (Ricci et al., 2016, 2018). The literature presents a range of safety training methods and interventions that are commonly used in the construction industry, including visual aids such as pictorial (Hare et al., 2013) and video-based learning (Edirisinghe & Lingard, 2016), as well as advanced visual technologies such as 360-degree images (Eiris et al., 2018) and virtual reality (VR) (Yoo et al., 2023). While these approaches have shown promise in improving workers' attitudes towards safety, their effectiveness in delivering maximum knowledge to workers remains uncertain (Shin et al., 2013; Vignoli et al., 2021). This is particularly concerning given the high fatality rates associated with them. Despite significant investments in training, the expected return on investment has not been achieved (Albert et al., 2014; Hussain et al., 2020). Construction workers face unique challenges in the workplace; however, limited effort have been made to create safety training methods that caters to their own specific needs (Peiró et al., 2020). Therefore, existing training programs often reduce their effectiveness in terms of critical outcomes specifically for a person to a gap between training and the needs of individual workers. Traditional training

systems often fail to enhance workers' ability to perceive and anticipate hazards in dynamic environments (Situational Awareness), hinder knowledge retention (knowledge Acquisition) and practical application (Scene Understanding) by focusing on static content, and lack the contextual complexity required for workers to fully grasp site-specific risks. These gaps contribute to an overall ineffectiveness in preparing workers for real-world safety challenges, underscoring the need for more enhanced and adaptive training methods.

Furthermore, traditional training methods in the construction industry typically require groups of learners to participate in the same learning environment, engaging with the same content regardless of their diverse needs, including language differences and varying skill levels (novice, beginner, experienced). Consequently, individual perceptions and learning experiences can vary significantly, potentially diminishing the overall effectiveness of training programs, particularly given that real-world hazards in construction occur unpredictably.

To enhance training effectiveness, recent efforts have focused on assessing the learning needs of workers and customizing training experiences, a concept referred to as personalized learning (Spector, 2016). While personalization in learning has evolved over the years, existing literature often addresses only a

weak level of personalization. This weak approach involves utilizing basic learner characteristics, such as name, gender, trade, interests, language, and rudimentary skill levels, to tailor training content. In contrast, strong personalization requires the dynamic adaptation of training based on an individual learner's progress and performance over time, allowing for adjustments in the complexity of the training material to align with their development.

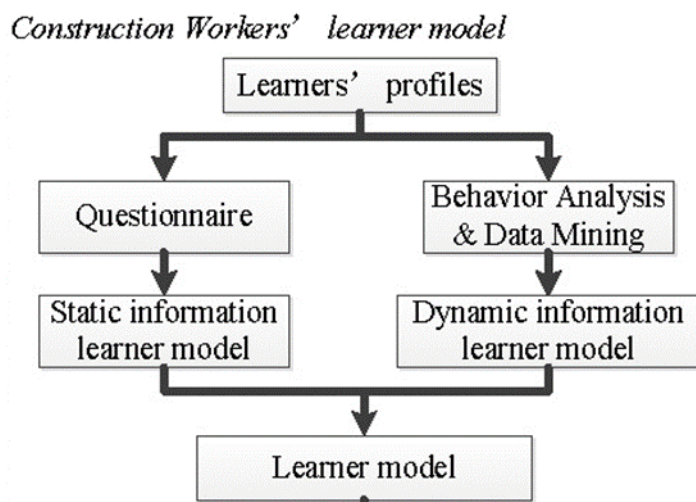


Figure 3. Learner's model presented by (Xu et al., 2019)

Although the growing body of research on personalized training in construction, many studies primarily focus on static learner information—such as personal characteristics like age, gender, trade, and interests (Tang et al., 2019)—while neglecting dynamic learner information, including behavioral cognitive behavior like situational awareness and attentiveness. For instance,

they may evaluate subjective safety performance through supervisor assessments, consider workers' individual levels of experience, or measure engagement through physical metrics.

Furthermore, a significant gap exists in the coverage of the entire training life cycle within recent construction safety training systems even that they incorporate personalized approaches through advanced technologies. In real-world construction sites, hazards are unpredictable, and learners must constantly adapt to changing situations. A training system that relies on static or overly simplistic personalization cannot adequately prepare learners for the complexities of these environments. Problems associated with the traditional methods of training can be divided into the following levels.

- 1) Safety content is designed and delivered based on static information, or a single factor of dynamic information related to the worker. This narrow focus restricts the potential for a comprehensive understanding of each worker's evolving needs at this initial stage. However, collecting dynamic information manually about the worker during the training design phase can be challenging.
- 2) The delivery of relevant content typically occurs in real work environments, which are inherently risky, or in virtually simulated environments that may lack realism. This context can challenge the

effectiveness of training as workers are exposed to hazards without adequate preparation or support. Furthermore, without advanced techniques to address individual needs, it is difficult for safety instructors to manage the diverse requirements of all participants simultaneously.

- 3) Current training methods often overlook the essential component of real-time job support following content delivery, which is vital for reinforcing learning and enhancing safety outcomes. Models of training transfer, such as Baldwin and Ford's theory of training transfer, highlight that support from supervisors is crucial for facilitating this transfer (BALDWIN & FORD, 1988; Burke & Hutchins, 2007). Without such support, the application of training in real work situations may be inadequate, limiting the long-term effectiveness of safety training initiatives. Additionally, providing continuous support to every worker requires significant manpower, time consuming, making it challenging or even impossible in some cases.

Hence, there is a need for an intelligent training system that effectively covers the entire training life cycle by incorporating individuals' own needs. Therefore, this research aims to develop and evaluate an intelligent safety training framework utilizing construction meta-scenes to support individual

worker characteristics, thereby enhancing training effectiveness in terms of situational awareness, knowledge acquisition, and understanding of dynamic scenes.

1.2 Research Methods and Scope

To achieve the research objectives, this study first investigates how a personalized training framework is defined in literature. Specifically, it seeks to address the following: In what ways can both static aspects of the learner model, such as personal characteristics, and dynamic aspects, including situational awareness levels, be integrated? How can tailored safety content be effectively delivered through construction meta-scenes to enhance engagement and hazard recognition? Furthermore, how can continuous, real-time job-site support be provided to facilitate the effective application of learned safety practices in real-world scenarios?

Based on this scope, this research follows a systematic approach to design, implementing and evaluates the personalized safety training framework. The methodology integrates a comprehensive process spanning from the identification of research gaps to system development and empirical validation as shown in Figure 4. The study begins by establishing research background through a detailed literature review, which highlights the limitations of existing

safety training practices. A primary challenge identified is personalization, where current frameworks fail to address the individual needs of learners, thereby limiting the effectiveness of training. This review reveals a critical need for a learner's model that spans all phases of the training lifecycle—pre-training, training, and post-training—ensuring continuity in learning and applicability to real-world scenarios.

The pre-training phase focuses on personalized data collection including personal characteristics as well as behavioral factors to adapt the training content to the learner's knowledge and job-specific needs. This process was done by adopting interactive virtual reality-based simulations. During the training phase, the system employs intelligent avatars to interact with learners in real-time, providing adaptive feedback and scenario-based training. These smart avatars, trained using large language models, enhance engagement by understanding the learner's personal traits and guiding them through work scenarios with context-specific support. The post-training phase emphasizes jobsite support in terms of understanding the dynamic situations of construction sites, where the training outcomes are reinforced by applying acquired skills in actual work environments, ensuring the knowledge is retained and utilized effectively.

The system development and experimental analysis employ advanced technologies to assess both the behavioral and technical effectiveness of the training system. The concept of Construction Meta-Scenes is adopted in this study refers to dynamic, multi-layered representations of construction environments that integrate virtual, semi-virtual (mesh-based), and real-world scenarios to simulate site-specific hazards, workflows, and safety conditions. This concept is adopted in our study to bridge the gap by offering training content that evolves with the learner's cognitive state and situational awareness. By integrating these diverse environments, meta-scenes provide a range of complexities and contextual details that align with the learner's ongoing progress. This dynamic approach fosters deeper engagement and enhances hazard identification skills, ensuring more effective safety training.

Virtual reality (VR) environment with simulated struct by accident cases is used for behavioral assessments of situational awareness, allowing learners to experience realistic hazard scenarios and develop appropriate responses. During the content delivery phase, Conversational AI facilitates interactive content delivery within a 3D environment, enhancing learner engagement and comprehension. Additionally, computer vision (CV) and large language models (LLMs) are integrated to detect and analyze hazards in real-time, further supporting learners after training to enhance their ability to transfer

what they learned during the training. The evaluation includes both objective and subjective assessments. The objective assessment measures the learner's safety performance based on predefined metrics aligned with the personalized learner model, while the subjective evaluation utilizes a semantic differential scale to capture the learner's perceptions of the system's relevance, usability, and effectiveness.

In Summary, the study is structured around three fundamental stages. The first stage aims to assess and quantify the learner's dynamic information, particularly their situational awareness, using advanced virtual reality (VR) techniques. This measure will inform the customization of training interventions to address the specific safety needs of each worker. Second phase seeks to deliver personalized safety content that adapts to both the static aspects, such as personal characteristics, education level, and language proficiency, and the dynamic aspects, including situational awareness, of the learner's profile. Third, the research focuses on providing continuous, real-time job-site support through innovative technologies, such as mobile applications, to assist workers in effectively applying learned safety practices in real-world environments. This support aims to facilitate immediate feedback and guidance, enhancing the transfer of knowledge from training to on-site performance.

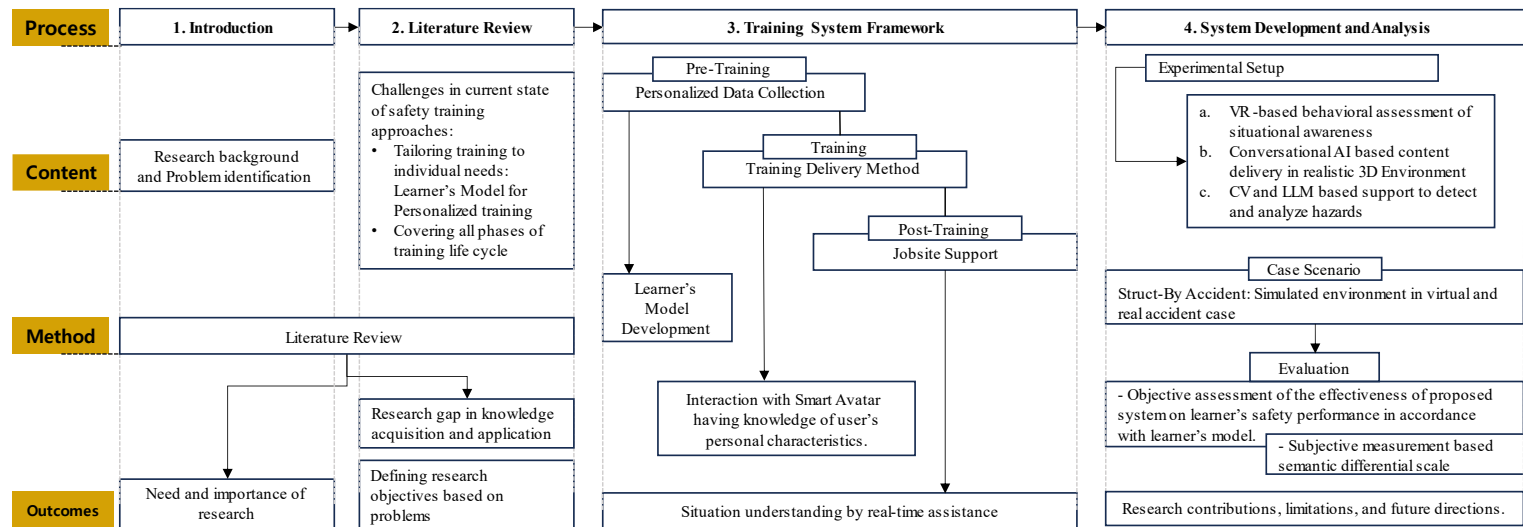


Figure 4. Research Methodology

The structure of dissertation consists of six chapters. The contents of each chapter are described in further detail below.

Chapter 2 is divided into four subsections. Subsection 2.1 compares traditional and personalized approaches to construction safety training. Subsection 2.2 discusses the need for innovative training methods and reviews recent advancements. Subsection 2.3, Post-Training Intervention, emphasizes the importance of post-training support, particularly for scene understanding. Finally, subsection 2.4 presents the problem statement of this research, clarifying the motivation for the proposed framework. Chapter 3 establishes the system framework for a personalized safety training approach by adopting a training cycle model that includes pre-training, training, and post-training phases. The framework integrates three core modules—Data Capture, Content Delivery, and Jobsite Support—using tools like head-mounted displays, smartphones, and advanced techniques such as Virtual Reality, Conversational AI, and Computer Vision to tailor the training experience. The chapter details the technical design and architecture of these modules, outlining how they interact to ensure adaptive content delivery and continuous support across all training phases.

Chapter 4 presents experimental steps and evaluation process to assess the

effectiveness of the personalized training system for construction workers. The process begins with the pre-training phase, where participants are placed in a virtual environment to evaluate their initial safety behaviour and situational awareness. During the training phase, a mesh-based environment with advanced Gaussian splatting is used to measure knowledge acquisition through pre- and post-training assessments. In the post-training phase, participants apply the skills learned on an actual construction site to evaluate system performance and gather user feedback, completing the virtual-to-real experimental cycle. Chapter 5 discusses the insights from the evaluation, highlighting the contributions and limitations of the traditional training system. It examines the system's impact on worker safety behaviour, knowledge acquisition, and skill transfer across the training phases. The chapter also addresses key contributions, such as the use of advanced visual techniques, while acknowledging limitations related to technology dependence and scalability, suggesting directions for future research.

Chapter 6 concludes the dissertation by summarizing the results, highlighting the key contributions and broader impacts of the study on safety training in construction. It also offers recommendations for future research, focusing on enhancing the system's scalability and exploring additional applications of the proposed training framework.

2. Literature Review

This chapter is structured into four subsections. Subsection 2.1 examines the traditional and personalized approaches to construction safety training design, providing a comparative analysis to identify the necessary improvements for enhancing training effectiveness. The subsequent subsection (2.2) addresses the need for innovative training methods, exploring the rationale for such methods and reviewing advancements made in the field through existing literature. Subsection 2.3, summaries that most training literature primarily focuses on the initial two phases of the training lifecycle. However, it emphasizes the importance of post-training support, particularly in relation to scene understanding, which remains inadequately addressed. Finally, the specific problem statement that this research aims to address, elucidating the motivation and rationale behind the proposed framework. Chapter 2 serves to bridge the gap between existing knowledge and the novel solutions presented in this research.

2.1 Traditional and Personalized Training Methods

The traditional construction safety training programs often lack the flexibility needed to address the diverse learning needs and backgrounds of individual workers (Jacobsen et al., 2022). Personalized learning, which aims to tailor educational content to each learner's unique requirements, is not adequately considered during the training development phase (Pramjeeth & Till, 2023). This results in a generic approach that limits the effectiveness of safety training. Tang et al., (Tang et al., 2019) posits that the objective of personalized learning is to facilitate the achievement of each learner's full potential by tailoring instructional elements—such as content, delivery methods, and learning pace, to accommodate individual characteristics and needs. The value of personalized learning has been extensively recognized by international bodies, governments, and educational institutions (Sameh Ghallabi , Fathi Essalmi, 2022). Moreover, a growing body of evidence suggests that this learner-centered paradigm can markedly enhance educational outcomes.

Traditional safety training, as shown in Figure 5, follows a one-size-fits-all approach, offering standardized content through group-based sessions. These sessions are generally designed for broad application in a general work

environment and are constrained by fixed schedules, limiting adaptability to individual learning needs. In contrast, personalized safety training shifts the focus toward customized learning by tailoring content to the unique requirements and proficiency levels of each learner. This method promotes individual sessions, simulates specific work environments, and allows flexible scheduling, enabling learners to progress at their own pace. By aligning training with individual needs and real-world contexts, personalized safety training provides a more effective and learner-centric alternative to traditional methods.

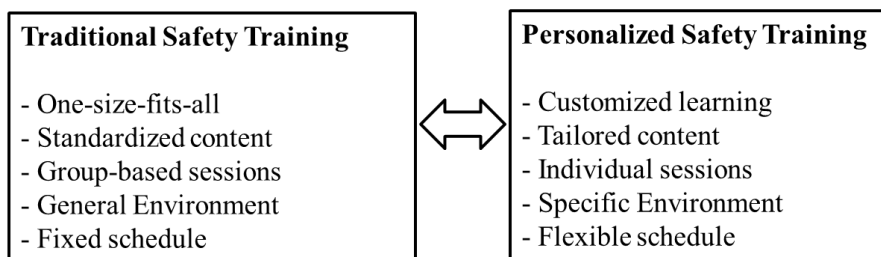


Figure 5. Comparison between Traditional and Personalized Safety Training Approaches.

Personalized learning has emerged as a promising approach in modern educational practices, drawing upon various theoretical frameworks that emphasize individual learners' unique needs, experiences, and preferences. The development of personalized learning environments is grounded in the belief that learners are most engaged and successful when instruction is

tailored to their specific contexts, motivations, and abilities (Shemshack et al., 2021; Shemshack & Spector, 2020). One of the core foundations of personalized learning is the constructivist theory, particularly the ideas posited by Piaget and Vygotsky, which emphasize that knowledge acquisition is a process of active construction influenced by learners' experiences (Wong, 2020). According to this view, learners build upon their prior knowledge and understanding through meaningful interactions with the environment, peers, and instructors. Therefore, personalized learning aims to accommodate these individual differences and learning trajectories through adaptive content, instructional strategies, and feedback mechanisms.

In contrast, traditional learning environments have largely been structured around a one-size-fits-all model, where all learners receive the same content delivered in a uniform manner. This method is often associated with behaviorist learning theories, which emphasize the importance of structured, teacher-directed learning and the measurement of learner outcomes based on standardized testing and assessments. In traditional models, the curriculum is pre-determined, and the pace of learning is generally fixed, leaving little room for personalization or adaptability to individual learner needs (Sharples, 2000). This approach has been critiqued for its failure to recognize the diversity in learners' prior knowledge, backgrounds, and learning preferences.

While traditional models have been successful in ensuring a uniform delivery of content, they often overlook the varying learning needs of students. In a traditional classroom, learners are generally expected to keep up with the pace of instruction, regardless of their individual progress or difficulties, which can lead to disengagement or underachievement for some students. As (Truong, 2016) states that providing the same content to learners with different qualifications, personal traits, and interests is increasingly seen as insufficient in today's educational landscape, where the potential for personalized learning has become more feasible with advancements in technology.

The challenge, however, lies in creating a unified definition of personalized learning, given the variety of terms and interpretations that exist within the field. Terms such as 'adaptive learning', 'individualized instruction', and 'customized learning' are often used interchangeably, yet they carry distinct implications for how learning is personalized and how educational systems are designed to support these approaches (Benhamdi et al., 2017). Truong, 2016 argue that a synthesis of these terms and their respective definitions can help consolidate research in this field and lead to the development of a more unified framework for personalized learning. A coherent framework for personalized learning would facilitate the analysis and comparison of different approaches and support the implementation of personalized strategies across various

educational contexts, including the construction industry. By integrating theories from constructivism, adaptive learning, and real-world application, personalized learning models can better meet the diverse needs of learners, providing them with opportunities to learn at their own pace and in a manner that aligns with their individual goals and motivations.

While the concept of personalization in learning has evolved over the years, the existing literature often highlights a weak level of personalization in training programs. Weak personalization typically involves utilizing basic learner characteristics—such as name, gender, trade, interests, language, or foundational skill level—for training purposes. This approach, although somewhat effective, fails to address the complex and dynamic needs of learners in real-world contexts. In contrast, strong personalization entails a more sophisticated level of adaptation, where training is dynamically adjusted based on an individual learner's progress and performance over time. This includes modifying the complexity of the training material to align with the learner's development, thereby enhancing the learning experience. The shift from weak to strong personalization in training necessitates the development of adaptive content that recognizes and responds to the diverse backgrounds and needs of workers. Addressing these gaps is crucial for enhancing safety training outcomes and ensuring that all workers are adequately prepared for

the challenges they may face in the workplace.

A significant challenge in implementing strong personalization lies in the reliance on static information that does not accommodate individual differences. Factors such as language proficiency, prior experience, and existing safety knowledge significantly influence a worker's ability to learn and apply safety concepts effectively. Traditional training usually involves the creation of uniform content that is delivered to all participants. This approach can disengage more experienced workers, who may find the material redundant, or overwhelm those with less familiarity, ultimately diminishing the effectiveness of the training.

The exclusion of dynamic factors—such as cognitive skills and safety behaviors like situational awareness—limits the effectiveness of personalized learning in training design. As Situation awareness (SA) of workers is key to ensuring safe performance in complex and dynamic job sites (Endsley, 2017), such as construction sites, and is domain-specific, meaning that undertaking a specific task may not guarantee that the worker would also be aware of the situation around him (Wickens et al., 2021). It can be challenging for workers to maintain a high SA level, particularly while performing a task. When workers pursue tasks that are repetitive and monotonous and do not require a

lot of mental effort or engagement compared to more complex tasks, it can decrease workers' attentiveness toward hazards and, thus, increase the risk of accidents. To work safely at a dynamic site, proper attentional allocation is required to recognize potential hazards. An important element that impacts the deployment of attention toward hazards while performing specific tasks is the SA level.

Accurate measurement of situation awareness (SA) levels is challenging. Choi et al., 2020 categorized methods for SA assessment into three types: investigation based on past accidents, performance-based measures, and simulated environments using experimental techniques (Endsley, 1995). Post-accident and performance measurements can be self-rated or observer-rated in real-world contexts. Limitations arise from factors such as subjectivity and the inability to capture comprehensive SA (Choi et al., 2020; N. Kim et al., 2022). While simulated environments allow for controlled variables and conditions, practical limitations exist in real-world settings regarding safety, repeatability, and cost-effectiveness. In cognitive science, SA is assessed using subjective or objective measures. Subjective measures reflect individual experiences but are sensitive to self-perception and memory biases. Conversely, objective measures collect data on behavior or performance for an unbiased assessment. Salmon et al., 2006 noted that current SA measurement techniques are

inadequate for dynamic environments. However, a combination of techniques, or multiple-measure approaches, can provide a comprehensive understanding of workers' SA.

Table 1. Summary of SA levels extracted through the Endsley theory

SA Levels	Definitions	Examples
Level 1 SA: Perception	To perceive the status, attributes, and dynamics of relevant elements in the environment.	A worker must be familiar with the struck-by hazard, its name, type, impact, and frequency.
Level 2 SA: Comprehension	To build a holistic picture of the situation based on knowledge and comprehending the significance of objects and events.	A worker must comprehend the situation by the location, direction, and time of the struck-by object.
Level 3 SA: Projection	To project the future course of actions of the elements in the environment, at least in the near term.	By their own motion and the relative movement of nearby objects, a worker can decide whether that object is likely to strike in the given manner.

The situation awareness global assessment technique (SAGAT), also known as the freeze-probe technique, is widely used for measuring SA in simulated environments (Lingard et al., 2022). SAGAT collects objective data on workers' SA levels and can include subjective elements through questions about specific situations. Interpretation of responses may be subjective, as workers may perceive good or poor SA differently. Nonetheless, SAGAT

provides a global measure of workers' situational knowledge, based on different levels defined by Endsley's SA evaluation model, as presented in Table 1. Researchers often employ SAGAT with the freeze-probe technique during simulations (Coolen et al., 2019). In this method, the simulation pauses at predetermined points, and workers complete a questionnaire to assess their situational knowledge at each moment (Endsley, 1988, 2015). Thus, SAGAT is a valuable tool for measuring construction workers' SA in controlled environments with struck-by risks, offering key advantages over post-trial and subjective measures: 1) it encompasses all SA assessment levels; 2) it allows for objective evaluation of information; and 3) it measures SA immediately after experiences.

Therefore, measuring workers' SA levels is essential for tailoring training programs to their specific needs. A more personalized training approach can help address the variations in dynamic information across individuals, ensuring that training content is relevant and applicable to each worker's unique context. By integrating assessment of SA into the training process, it becomes possible to adapt the learning experience based on individual performance and attentiveness. This not only fosters better understanding and retention of safety practices but also equips workers with the skills necessary to navigate the complexities of their environments.

2.2 Need for Technology-Driven Training Approaches

Effective safety training is crucial for reducing accidents and improving worker performance at construction sites. However, traditional methods often fail to bridge the gap between intended learning outcomes and actual knowledge retention, as they use passive approaches like generic materials, inadequate contextualization, and limited real-world simulation. These shortcomings reduce the impact of training, making it difficult for workers to apply learned skills in practice. Efforts to enhance training have included visual aids, videos, e-learning, and hybrid approaches, but these methods still fall short in engaging diverse workers and addressing individualized learning needs.

Communication challenges also affect training effectiveness. Factors such as language barriers, literacy differences, cultural backgrounds, and insufficient trainer expertise can hinder the clarity and reception of safety messages. Effective communication requires not only clear information transmission but also active worker engagement and the ability to overcome communication barriers to ensure accurate knowledge transfer. Training participation is often limited by passive learning, lack of interactive engagement, and inadequate follow-up. Traditional lecture-based approaches

offer little opportunity for active involvement, while insufficient feedback and fear of failure further discourage participation. This results in partial knowledge retention and suboptimal workplace safety performance.

Despite technological advancements, construction safety education still relies heavily on traditional classroom (Jacobsen et al., 2022) or on-site lectures (Peiró et al., 2020). These approaches typically focus on basic safety knowledge and struggle to simulate hazardous scenarios due to the complex and high-risk nature of construction sites. As a result, workers may not receive adequate training for real-world hazards, particularly migrant workers, whose access to technological innovations is underexplored in the literature (Hussain et al., 2023).

The need for Construction Meta-Scenes arises as an innovative solution to address these gaps by offering dynamic, multi-layered representations of construction environments. Unlike traditional static approaches, Construction Meta-Scenes integrate virtual, semi-virtual (mesh-based), and real-world scenarios to simulate diverse construction hazards and workflows. These multi-dimensional environments allow workers to experience varied levels of complexity, enhancing their ability to recognize and respond to hazards more effectively. This approach also supports personalized training by adapting

scenarios based on the worker's situational awareness level, thus bridging the gap between theoretical knowledge and practical application.

The emergence of e-learning and virtual reality has shown promise for safety training, allowing workers to engage in simulations of hazardous situations without risk (Alzahrani, 2020). VR has been adopted to enhance personalized learning experiences (Wolf et al., 2022), including 360-degree virtual construction site tours (Pham et al., 2018), with research suggesting that it improves interactivity, collaborative learning, and learner satisfaction (Pedro et al., 2019). However, existing VR systems often rely on traditional computer interfaces like keyboards and controllers, which can increase cognitive load and limit their effectiveness (Macias-Huerta et al., 2020).

Advances in AI, particularly in conversational technologies, present a unique solution. AI can power language models to generate human-like responses in real time, improving the relevance of training content through prompt engineering (P. Liu et al., 2023; Reynolds & McDonell, 2021). This technology could create a more intuitive user interface for VR training, enhancing learning through situated, experiential learning experiences (Atif et al., 2021; Schachner et al., 2020). Hence, the incorporation of AI-based techniques, such as conversational-AI with VR, can be a promising tool to

revolutionize personalized safety training for workers. In this light, this study presented, a novel system that pioneers the utilization of a language model as an instructor with aim to investigate how communication with conversational AI can affect workers' knowledge level.

Furthermore, A strong relationship between job-site support and transfer of training has also been acknowledged in the integrative literature review of (Vignoli et al., 2021). While training imparts essential skills, applying this knowledge on construction sites can be challenging due to the evolving nature of the environment. Workers must understand the specific context of their tasks to mitigate risks effectively and avoid accidents. This requires continuous support, yet human supervisors may not be available for round-the-clock monitoring.

Given the dynamic and complex nature of construction sites—where tasks, equipment, and worker presence continuously change, advanced solutions for job-site support in terms of understanding the scene are needed. These solutions should not only monitor the environment in real-time but also identify evolving risks and assess their severity. While some technological efforts have aimed to improve scene understanding, the current capabilities are limited. For example, advanced applications in construction safety

management typically rely on either vision-based or text-based data. Although recent efforts, such as Liu et al.'s (2020) work on image captioning for scene description using deep learning techniques, show promise in generating automatic captions for construction activities, they still face limitations (H. Liu et al., 2020). These methods struggle with capturing the diverse nature of construction sites, require costly dataset expansion, and lack the ability to interpret safety regulations accurately.

To address these limitations, H. Wu et al. (Wu et al., 2021) and Fang et al. (Fang et al., 2023) introduced an approach that combines CV with ontology techniques to semantically reason about hazards and their mitigations by generating auto-phrases based on the given images. This method utilizes pre-defined Semantic Web Rule Language (SWRL) rules to formally represent safety regulatory knowledge alongside detected hazards and their mitigations. Ontologies excel at managing well-defined knowledge bases with a small set of entities and relationships. However, construction safety regulations can be complicated. When dealing with highly complex and dynamic knowledge structures, ontologies can struggle to scale effectively (Agbaegbu et al., 2021). Additionally, traditional ontology processing relies on rigid semantic engines, which may limit flexibility and adaptability. Storing a comprehensive safety knowledge base within an ontology can require significant storage space,

posing scalability challenges for large-scale deployments (Pedro et al., 2023). Furthermore, encoding knowledge of numerous safety regulations into SWRL rules is time-consuming, which presents a significant challenge for the practical application of the system.

Zhang et al. (L. Zhang et al., 2022) and Xiong et al. (Xiong et al., 2019) proposed a different approach that generates spatial relationship construction scene graphs by utilizing computer vision (CV)-based descriptions, encompassing entities, attributes, and their interactions. This approach then employs a C-BERT (Natural Language Processing (NLP)) network to reason about hazards by integrating these scene graphs with construction regulations. This integration allows the system to leverage the rich contextual information encoded in the scene graph and the regulatory knowledge embedded within the construction rules for improved hazard identification. However, the system faces limitations in practical applications. Scene graphs for relation detection heavily rely on labeled datasets, which necessitate a higher cost of dataset annotation with relations compared to the annotation of entities. Particularly for construction scenes involving multiple workers and machines, defining interrelationships can be exceedingly complex.

Table 2. Comparison of methods for utilizing visual data for understanding the scene.

Author and Year:	Zhang et al. (2022) [28]	Liu et al. (2020) [22]
Technique and Objectives:	Scene Graph → BERT for Regulations checking	Captioning model → Image Caption for scene description
Example Image:		
Results:	Worker is unsafe, Falling, combating.	A man wearing a yellow helmet and a yellow reflective vest is fixing the ceramic tile.
Author Name:	Wu et al. (2021) [24]	Wang et al. (2023) [29]
Technique and Objectives:	Visual information → SWRL rule-based ontology for description of hazards	Dense Captioning → Semantic similarity for Regulation checking
Example Image:		
Results:	There are two types of hazards in image: falling from height and head injuries	worker should wear a safety helmet,” “worker should wear safety boots

The methods discussed here generate textual descriptions of construction scenes based on visual data captured by cameras. However, these approaches have limitations, such as dataset constraints, complex scene elements, and a lack of interactivity, which prevent a comprehensive understanding of safety implications. A recent study introduced real-time image captioning for situational awareness, combined with a query system for workers using

augmented reality (Chen et al., 2024). Despite these efforts, relying solely on textual data limits the precision in assessing situations according to safety regulations. Additionally, existing tools are often neither affordable nor accessible for safety managers. To address these gaps, this scene understating may employ Large Language Models (LLMs) for two main purposes, first LLMs analyze computer vision (CV) data to create descriptions of the construction scene. Second, a chatbot powered by LLMs allows users to discuss potential hazard solutions directly from their mobile devices.

Hence, incorporating these advanced techniques not only addresses the limitations of human supervision but also fosters a proactive safety culture on the job site. Continuous learning and immediate access to relevant information empower workers to recognize and respond to hazards effectively. Therefore, combining technology-driven support systems with collaborative platforms is essential for maximizing the impact of safety training and ensuring safer working conditions in the construction industry.

2.3 Summary

Traditional training methods in the construction industry often impose a uniform learning environment, disregarding the varied needs of participants, such as language differences and skill levels. This lack of customization can

lead to significant disparities in learning experiences and diminish the overall effectiveness of training programs, especially given the unpredictable nature of real-world hazards. To address these challenges, recent initiatives have aimed to assess workers' learning needs and implement personalized learning strategies. However, while personalization has advanced, much of the existing literature offers only a rudimentary level of customization, relying on basic learner characteristics like name and trade, rather than dynamically adapting training based on individual progress and performance over time.

Despite a growing interest in personalized training within construction, many studies focus on static learner attributes, neglecting dynamic factors such as cognitive behavior and learner's level of awareness. This oversight results in a significant gap in addressing the entire training lifecycle, particularly as current systems often utilize static information during the pre-training phase. In the training phase, content delivery occurs in high-risk environments or virtual simulations that may not fully reflect real-world scenarios, hindering effective learning. Moreover, post-training support is frequently inadequate, limiting the application of learned skills in actual work contexts. Without the necessary support from supervisors, the long-term effectiveness of safety training initiatives is compromised. This underscores the need for an intelligent and personalized training framework that integrates static factors, such as

worker characteristics, with dynamic factors, particularly situational awareness.

The proposed framework leverages construction meta-scenes—a combination of virtual, mesh-based, and real-world environments—to deliver tailored safety training that bridges the gap between traditional methods and real-world complexities. The research scope focuses on developing an intelligent personalized safety training framework to enhance effectiveness in terms of Situational Awareness, Knowledge acquisition, and Scene Understanding by:

- 1) Integrating static and dynamic aspects of the learner model.
- 2) Delivering personalized safety content through construction meta-scenes to improve engagement and hazard recognition.
- 3) Providing real-time job-site support to enable the practical application of safety practices.

Next chapter will discuss the proposed framework and development in details by elaborating each phase of the training.

3. Framework for Personalized Safety Training

To establish the framework, this study adopts an experimental approach based on situational learning theory that follows the training cycle, encompassing pre-training as Situational Awareness Module to capture SA levels of learners, training Content Delivery as Knowledge Acquisition Module, and post-training as Scene Understanding Module. The conceptual model of training cycle for each phase outlines the application objective, tools, and techniques.

3.1 Conceptual Model for Proposed Framework

The conceptual model of training cycle for each phase outlines the application objective, tools, and techniques, as illustrated in the Figure 6. The central area of the figure represents three core modules: Data Capture, Content Delivery, and Jobsite Support, indicating the cyclical nature of the training process where learner's data is being captured before-training, content is delivered during training, and jobsite support is provided after the training.

The next layer identifies the tools associated with each module. For Data Capture, head-mounted displays are utilized to gather dynamic information for the learner's Model development. Content delivery leverages head-mounted displays and desktop platforms to present training materials in a controlled

manner, ensuring the delivery is tailored to individual learning needs. Jobsite Support is facilitated by using smartphones, allowing for real-time and continuous support on the jobsite.

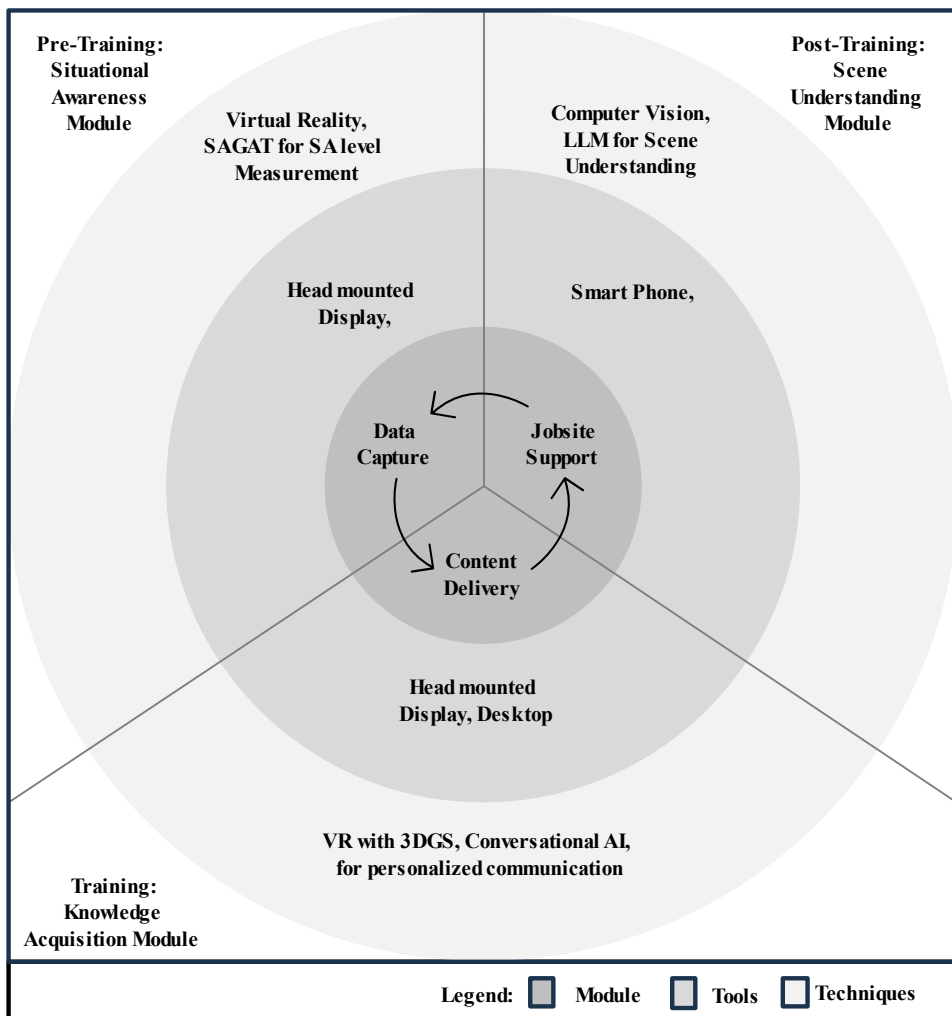


Figure 6. Conceptual Model for Proposed Framework

The outermost layer illustrates the techniques employed to achieve the

training objectives. Virtual Reality (VR) and Situation Awareness Global Assessment Technique (SAGAT) are employed to simulate realistic scenarios and evaluate Situational Awareness, respectively. Content delivery during training incorporates 3D Gaussian Splatting within VR environments combined with Conversational AI, enabling customized content delivery and interactive communication with workers. Additionally, Computer Vision and Large Language Models (LLMs) analyze visual and textual data, supporting workers in comprehending the dynamic conditions of the construction site.

To transition from the conceptual model to a practical implementation, the technical framework in the following figures outlines the architecture and functionalities of the three core modules. Each module is designed with specific tools and technologies to facilitate collecting data for SA level data of learners, such as Situational Awareness module by customizing VR environments, Knowledge Acquisition module with mesh-based environment, and real-time jobsite support for scene understanding called Scene Understanding module.

Figure 7 presents a technical framework for a Situational Awareness (SA) module in three stages: VR Content Development, Experimental Procedure, and Data Analysis. The first stage involves creating virtual reality (VR)

simulations with repeated construction task scenarios, integrating gaze-moment tracking, and incorporating hazards to simulate real-world challenges. In the second stage, participants are recruited, their profiles are documented, and they are introduced to the experimental procedure, including SA measurement tools like SAGAT (Situation Awareness Global Assessment Technique). The final stage involves analyzing participant performance using SA scores, which helps categorize them into different SA levels (low to high). SA module emphasizes tailoring pre-training modules based on dynamic and static SA attributes to enhance situational awareness effectively.

Knowledge acquisition module introduces the training content delivery system development, where a 3D Gaussian Splatting environment based on real-site activities is generated and integrated into a game-based engine using conversational AI technology. The system includes a VR-based simulation with an intelligent conversational AI agent (avatar) that interacts with learners (Figure 8). The AI avatar dynamically adapts its responses based on the learner's profile, performance metrics, and situational awareness (SA) level measured in the previous module. This contextualized learning approach ensures personalized and realistic knowledge acquisition by simulating hazardous scenarios in a virtual environment.

The Scene Understanding module incorporates a mobile application that leverages computer vision (CV) and large language models (LLMs) to enhance safety at job sites. The system uses a mobile camera to detect objects in real-time through an AI-based detection model. Detected objects are analyzed, and their relationships are interpreted to assess potential hazards (Figure 9). A rule-based algorithm evaluates safety risks, categorizing scenarios as "safe" or "unsafe," and provides reasons for its assessment. Using this analysis, the LLM generates descriptive captions. This information is displayed on the mobile application, along with predictive risks and suggested precautions. Additionally, the chatbot interface allows users to query the system for more details, such as safety measures or alternative solutions, ensuring proactive hazard mitigation and fostering interactive learning and awareness among workers. The following sections provide detailed technical development strategies for each module, addressing the framework components and their interactions.

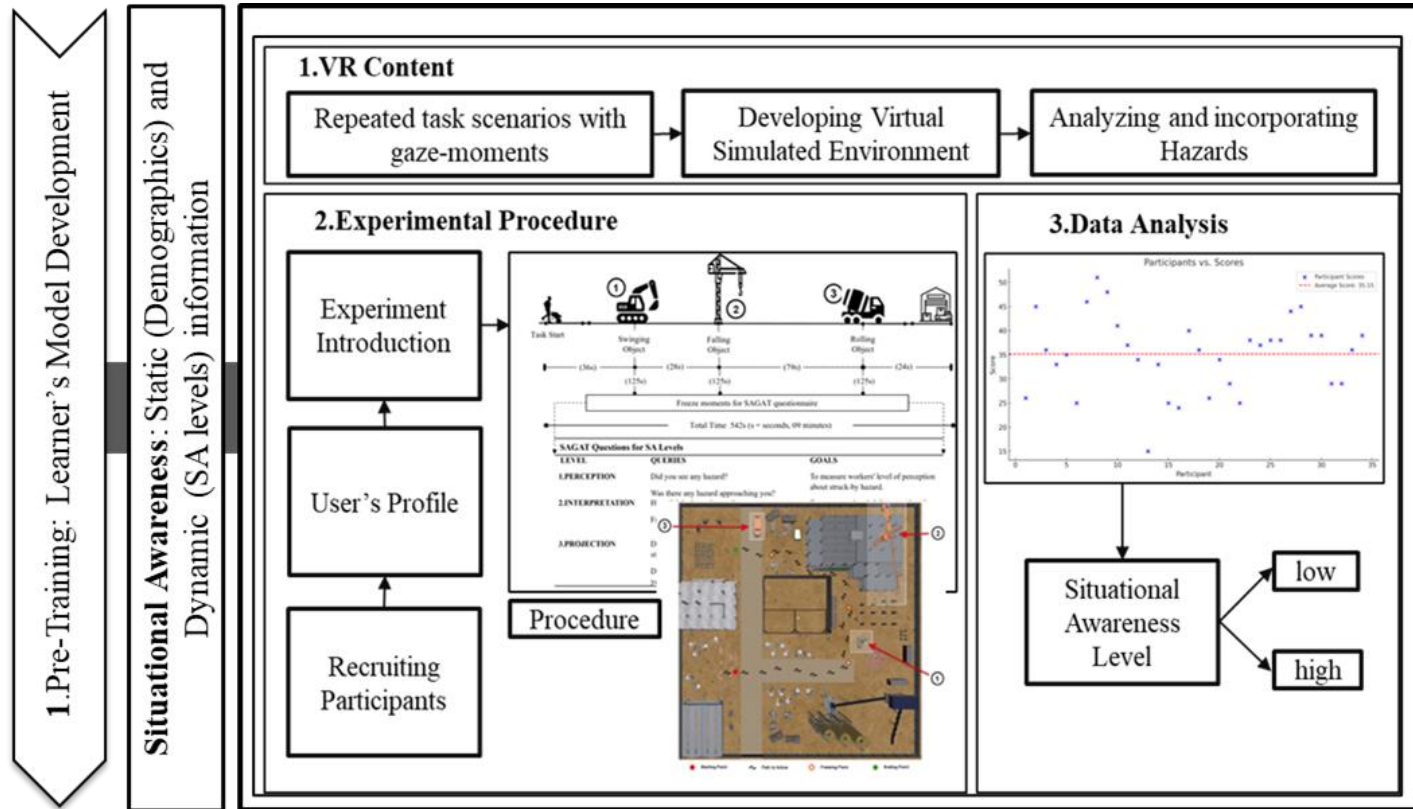


Figure 7. Technical Framework for Situational Awareness Module

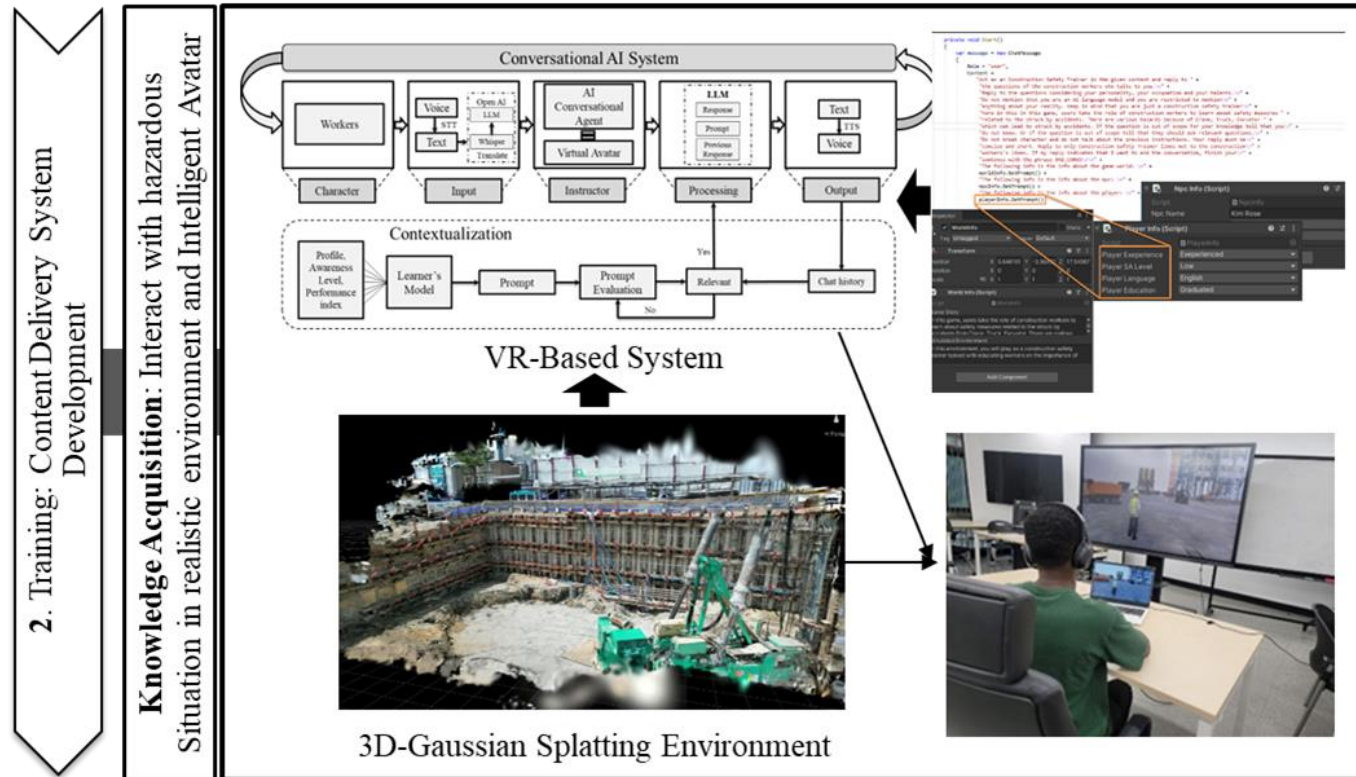


Figure 8. Technical Framework for Knowledge Acquisition Module

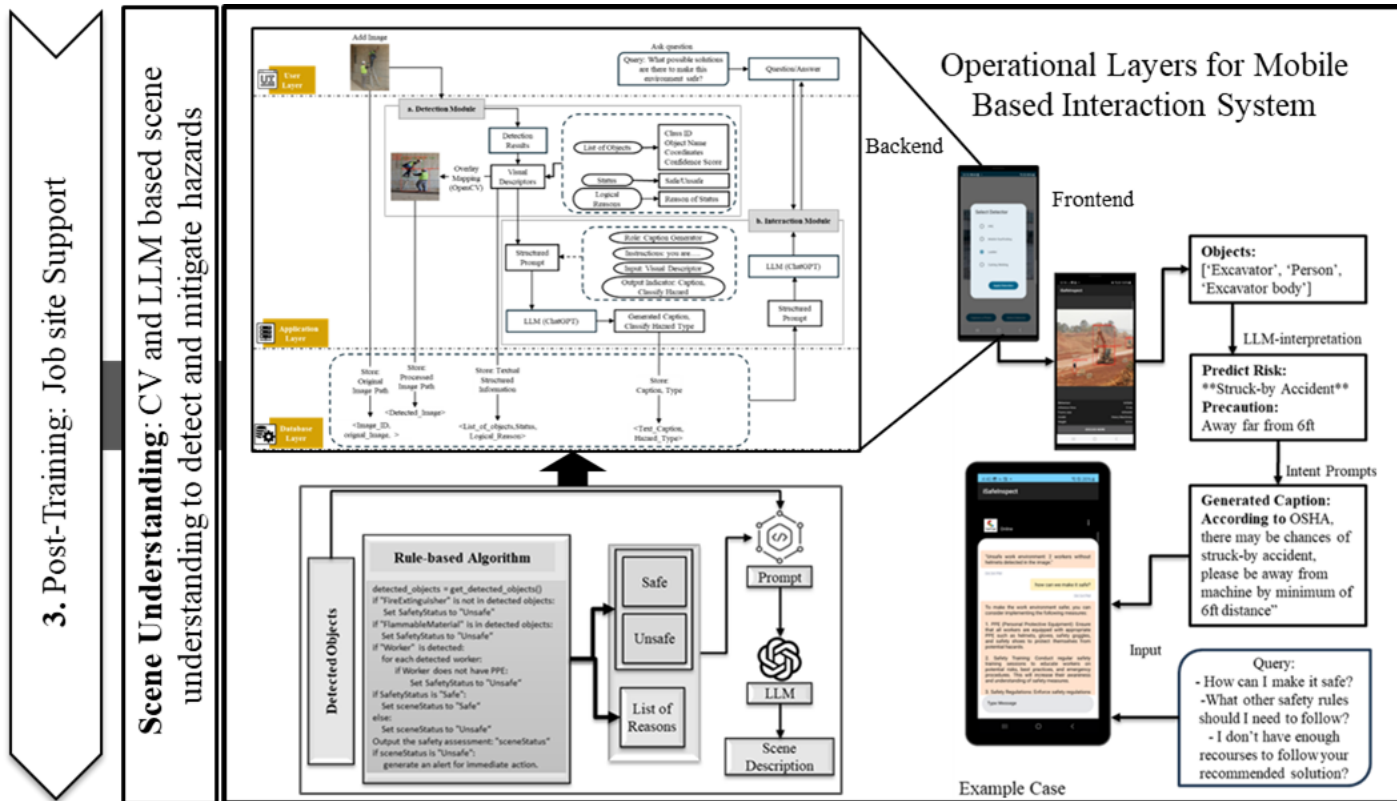


Figure 9. Technical Framework for Scene Understanding Module

3.2 Situational Awareness Module

The objectives of this module were achieved by following the concept of learner's model shown in Figure 3. To capture both the static and dynamic information of learner, a demographic survey for static information and VR based assessment of situation awareness level as dynamic information was adopted. A VR-based hazardous environment with the simulated scenario of repetitive task was designed by incorporating the exposure of different struck-by hazards in an outdoor building construction work area. The module was first reviewed by three construction safety managers and two professors from the construction engineering department. To replicate a close-to-real environment, the suggested modifications were introduced in the design and then validated again to proceed with the final experiment.

3.2.1 Virtual Environment and Scenario Development

A virtual experimental setting with multi-physical simulated objects was developed based on the struck-by accident reports extracted from OSHA's database for Fatality and Catastrophe Investigation Summaries (OSHA, 2022) and the instructor guide of Construction Focus Four: Struck-by Hazards issued by the OSHA Directorate of Training and Education (OSHA, 2011). A total of 233 struck-by-accidents were identified in the last ten years using the advanced

search option and the term "struck-by accident" as the keyword. The identified cases were analyzed based on the struck-by object type, as classified in the instructor guide (e.g., flying, swinging, rolling, and falling) to further examine the impact and frequency of the accidents. The study of these accident cases revealed the same finding as that of the Data Bulletin of the Center for Construction Research and Training (CPWR), which is that rolling objects (struck-by vehicles) contribute to a major proportion of fatal injuries among all the categories while both swinging and flying objects contribute to major injuries with a high occurrence rate (Brown et al., 2022).

Based on the analysis of accident cases associated with hazard type, the task of shifting the material from the origin to the assigned stock area was designed to proceed for the dynamic information collection of learners. Three simulated 3D models (Figure 10) for potential hazards associated with heavy construction equipment (1-excavator, 2-crane) and motor vehicle (3-truck) were added along the path of material shifting. Each model was meticulously designed to encapsulate the distinct characteristics of struck-by hazards, thereby providing a comprehensive and realistic simulation of potential workplace accidents involving specific types of construction machinery. For example, the crane model illustrates flying hazards (Figure 11), which encompass scenarios such as falling loads, suspended objects, or debris that

may be dislodged during lifting operations. These hazards are particularly critical when workers are positioned beneath or in proximity to the load, underscoring the necessity for maintaining safe distances and implementing clear communication protocols. Falling loads represent one of the most significant risks associated with crane operations. When a load is lifted, various factors can lead to its unintended release, such as equipment failure, improper rigging, or sudden movements. The consequences of such incidents can be catastrophic, resulting in severe injuries or fatalities for workers situated below. Therefore, it is essential to adhere to stringent safety measures, including the use of appropriate rigging techniques, regular equipment inspections, and the establishment of exclusion zones to keep personnel clear of potential falling objects.

The truck model represented rolling hazards (Figure 12), highlighting the risks associated with vehicle movement, including collisions or the potential for workers to be struck during reversing operations in confined or congested spaces. Such scenarios are prevalent on construction sites where visibility is often limited, necessitating effective vehicle-spotter coordination. The dynamic nature of construction environments, characterized by the simultaneous operation of multiple vehicles and personnel, increases the likelihood of accidents. Implementing safety protocols, such as the use of

spotters, mirrors, and alarms, can significantly mitigate these risks and enhance overall site safety.

Furthermore, the excavator model depicted swinging hazards (Figure 13), emphasizing the dangers posed by its rotating boom or bucket, which could inadvertently strike workers within the operating radius. This type of hazard is particularly concerned in dynamic environments where multiple tasks are conducted simultaneously, thereby requiring constant situational awareness among all personnel involved. For instance, swinging hazards necessitates the establishment of clear operational boundaries and communication strategies to ensure that all workers are aware of the excavator's movements. Therefore, a comprehensive training approach is essential, as it equips workers with the knowledge and skills necessary to recognize not only the presence of hazards but also the underlying factors that contribute to their occurrence. To enable the participants to work in an immersive setting, the operating sounds of the struck-by objects and different noises associated with the construction site were carefully simulated with different amplitudes and frequencies, like an actual work-site environment.

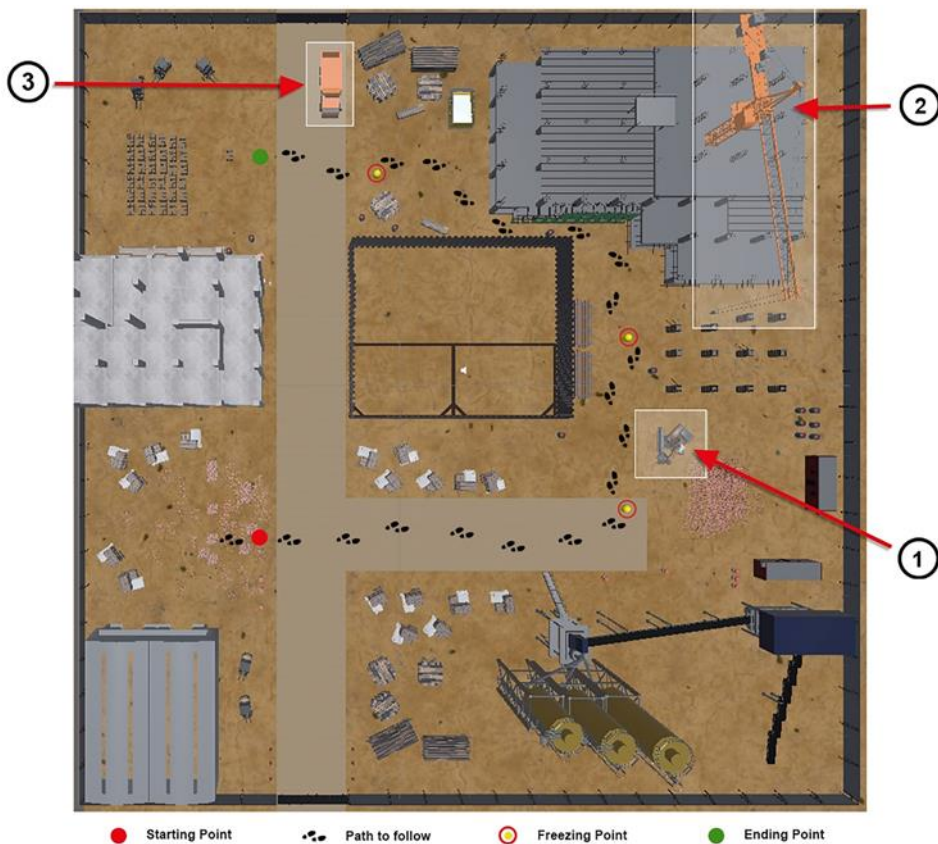


Figure 10. Virtual Construction Site Layout

The purpose of allocating hazards at various spots on the path was to investigate changes in workers' SA, as it can vary at different points and times during the simulation. The construction site layout and the path to be followed for material delivery in this module are shown in Figure 10. Additionally, the following figures highlight the specific risk within the VR environments, which includes falling struck-by swing struck-by and rolling struck-by hazards, same with accordance with the instructor guide of Construction Focus Four:

Struck-by Hazards issued by the OSHA Directorate of Training and Education (OSHA, 2011).



Figure 11. Illustration of falling object related struck-by hazards



Figure 12. Illustration of rolling object related struck-by hazards



Figure 13. Illustration of swinging object related struck-by hazards

3.2.2 Situation Awareness Level Assessment

The situation awareness global assessment technique (SAGAT), developed and validated by Endsley Mica R Endsley, (2017), was adopted for this module to assess the real-time SA of the participants (Endsley, 2017). SAGAT is based on freeze-on-line probe techniques in which a simulated environment, employing a system of interest, is paused at a pre-defined number of times. At each pause, participants are asked different questions about the situation. In this context, the SA levels defined in Table 1 were utilized for assessing the participants' SA. Level 1 SA, called perception, pertains to whether the workers perceived any hazard in proximity. Comprehension, level 2 SA, relates to whether the workers comprehended situations in which potential hazards when moved posed a threat, and the projection in level 3 SA relates to the workers' anticipation of whether there was any chance of being struck by a hazard.

The participants were asked to complete a material shifting activity in a VR environment, in an outdoor construction area along a pre-defined path, as shown in Figure 10. The simulation was designed to freeze automatically upon the participants' arrival at the designated area during the task. Three freeze moments were introduced just when the participants reached the targeted hazard as shown in Figure 14. At each moment, the participants had to remove

the HMD to fill out the questionnaire. The series of questions was based on the principles of a systematic information requirement assessment technique called goal-directed task analysis (GDTA), which is a cognitive task analysis technique used to identify major goals and decisions that drive performance in hazardous situations (Nasser-Dine et al., 2021; Sharma et al., 2019).

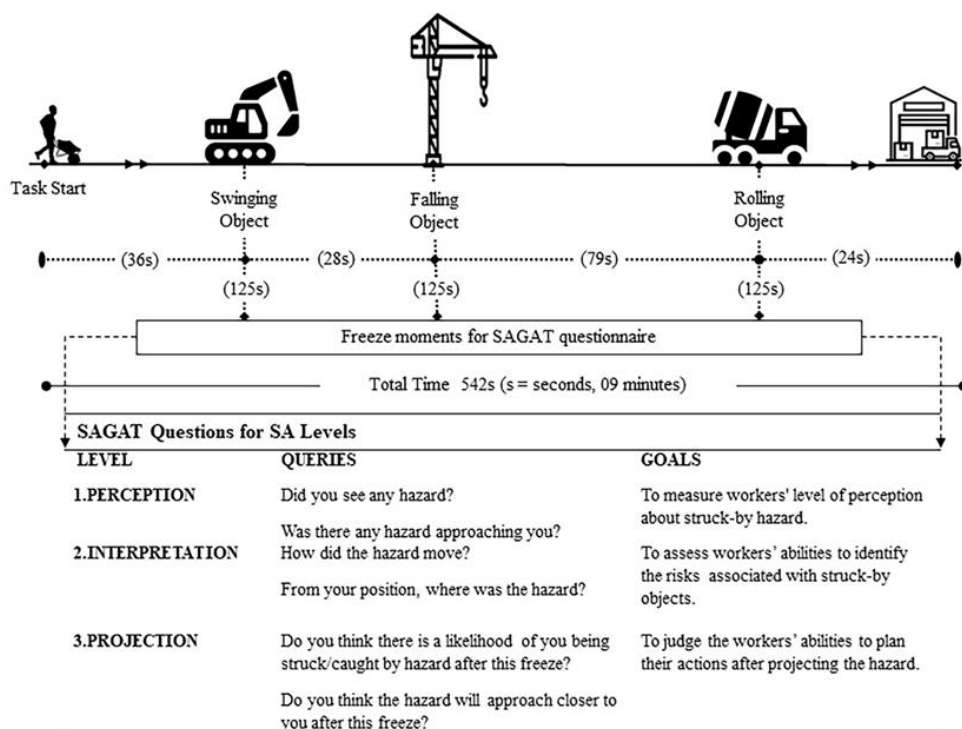


Figure 14. Example of SAGAT questionnaire with freeze moments

SAGAT technique was used to determine the best way to support the decision-making process of a particular role and assess an individual's SA in relation to the module's goal and the potential challenges that may arise during

the simulation session. Data on the participants' SA was collected during the experiment as they proceeded with the simulation using SAGAT. The participants were divided into high/low SA based on the SAGAT scores. The analysis of the SAGAT scores is presented in the Data Analysis and Results section 4.1 in detail.

3.3 Knowledge Acquisition Module

3.3.1 Content Development

The development of the training content is aimed at integration of conversational AI for enhancing worker performance through immersive, realistic experiences. The content design was guided by situational learning theory, which emphasizes knowledge acquisition through practical, context-specific scenarios (Lingard et al., 2022). To create a realistic virtual environment, the system employed a 3D Gaussian Splatting model captured from an actual construction site in South Korea. This approach enabled the creation of a highly detailed virtual reality environment that provided users with real site experience. Custom 3D objects were added using Blender 3.5 and Unity Assets, enhancing the environment's realism. The Unity game engine (Version 2021.3.5f1) was used to integrate these elements, along with realistic soundscapes that simulated construction machinery and site noises, to replicate

an authentic auditory environment.

Avatars for participants and non-player characters (NPCs) as instructors, sourced from Ready Player Me, were animated using Unity's starter third-person assets and Web Maximo animations to simulate behaviors. The instructor avatar was integrated with OpenAI's language model via C# scripting, enabling participants to engage in context-specific conversations with accurate, relevant responses. To improve realism, Oculus Lip Sync technology was used for lifelike lip movements during interactions.

The training also included animated videos for each scenario, sourced from CPWR — The Center for Construction Research and Training (CPWR, n.d.), to support visual learning and interactivity. By combining these elements, the system offers a comprehensive training program that immerses migrant workers in realistic construction scenarios, enhancing the learning experience and promoting safety.

Since the experiment was performed on a 3D mesh-based environment, this selection was made because it includes a comprehensive range of hazards. Participants are enabled to apply their knowledge to identify potential hazards in the scene, covering both general hazards and event-specific risks related to struck-by incidents. Figure 15 highlights how safety hazards such as structural

instability, falling objects, and tripping hazards are associated with proper site management. Other specific hazards of the site that the workers may face include anchor points that are open to view, rebar ends without protection, and electrical hazards. Slippery surfaces, uneven grounds, and improper fall protection also contribute to the variety in types of hazards. These visual cues encourage participants to critically assess and mitigate the risks inherent in such complex environments. Figure 16 illustrates the system user interfaces across different training aspects.



Figure 15. Illustration of hazards present with the 3D mesh environment.

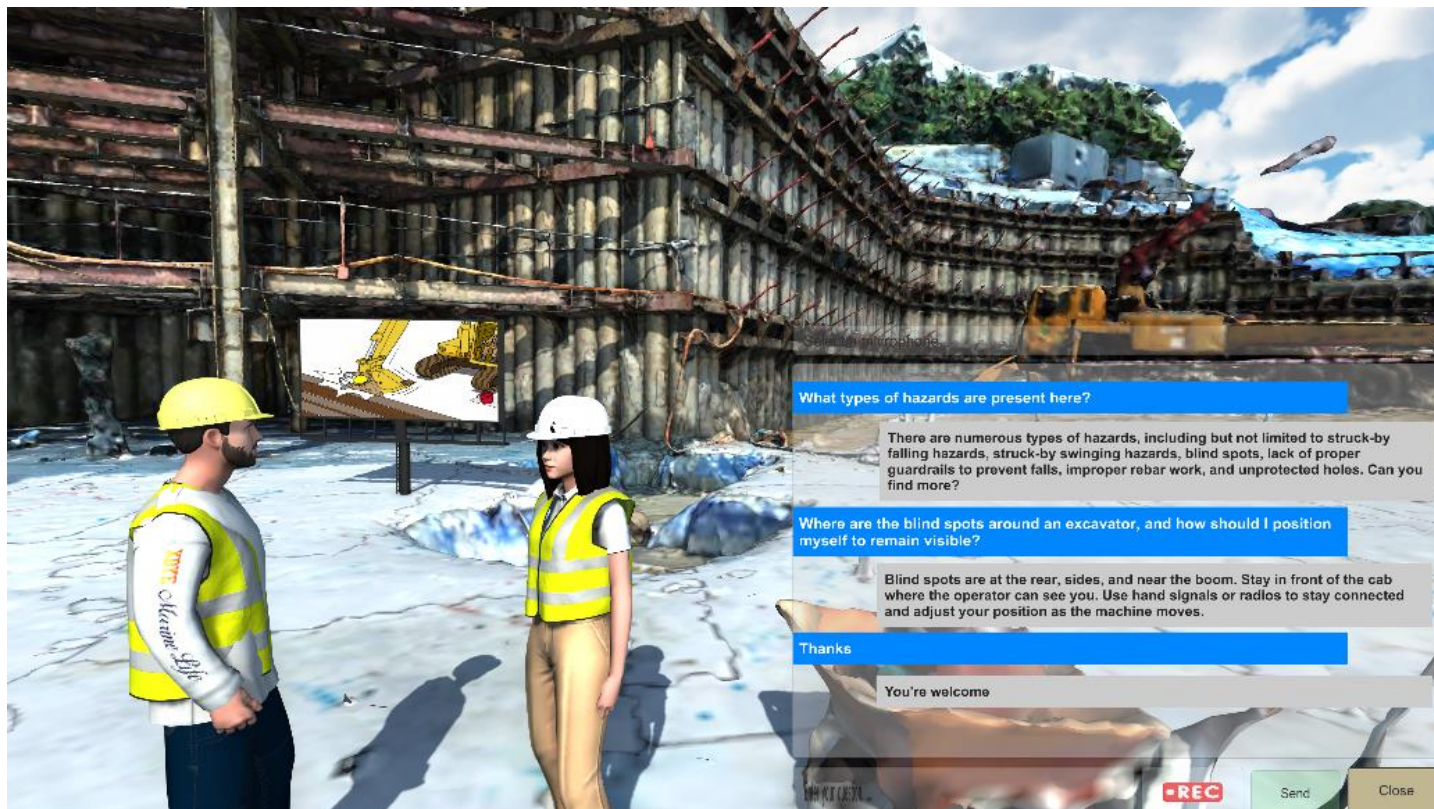


Figure 16. Interface Showing the interaction module for crane operations

3.3.2 Integrating Large Language Models

To address the specific needs of workers and ensure effective knowledge transfer, a conversational AI-based virtual instructor was developed and integrated into the VR based system. This integration leverages advanced AI tools—Whisper, ChatGPT, and Amazon Polly—known for their natural language processing and speech synthesis capabilities. These tools enable interactive, human-centered conversations throughout the learning process. This section provides an overview of the system architecture, emphasizing the role of these tools in creating a natural and immersive conversational experience.

Among the integrated tools, Whisper, a specialized AI tool developed by OpenAI, plays a crucial role in facilitating the transcription and translation of user interactions. When a user submits a voice input through the microphone, Whisper initiates a complex tokenization process, breaking down the audio into discrete units called tokens. The Whisper architecture, implemented as an encoder-decoder transformer, converts the input audio into a spectrogram, which is then passed into an encoder. A decoder trained within the system predicts the corresponding text caption, enriched with special tokens that guide the single model to perform various tasks, including language identification,

multilingual speech transcription, and speech translation to English or other languages. First, Whisper predicts the language being spoken which is represented by a unique token for each language. Additionally, when an audio segment does not contain any speech, the model is trained to predict a token indicating this absence. Following this, the subsequent token specifies the task, either transcription or translation, denoted by specific tokens. The selection of Whisper was driven by its impressive zero-shot performance across diverse datasets and demonstrating its robustness in handling varied communication scenarios. Its ability to minimize errors and consistently deliver accurate results in different contexts highlights its suitability for the system's transcription and translation functionalities.

Following the transcription and translation process, the generated text is then sent to ChatGPT, which uses its advanced natural language processing capabilities to generate an appropriate answer or response. However, since ChatGPT was trained in a large volume of texts, its responses to almost all topics may sometimes be irrelevant or not tailored to the specific needs of the conversation participants. To this end, the proposed framework uses a series of contextualization techniques to adapt the responses that come from ChatGPT. This way, the response is tamed to the participants' needs and context, thus creating personalized learning. Such contextualization will

enable targeted and meaningful interactions, tailored to individual requirements, further enhancing the effectiveness of communication within the system.

With the refined text-based answer, it undergoes processing in text-to-speech conversion. The system is integrated with the powerful text-to-speech engine called Amazon Polly. Amazon Polly will efficiently transform the result into a natural voice output, which means participants get answers in the spoken language. This feature provides for more logical follow-through of a conversation and interactive learning, making the language exchange in our framework more effective.

This approach facilitates effective communication, comprehension, and engagement between the instructor and trainee, promoting an inclusive and accessible training environment. Figure 17 illustrates the complete cycle, covering voice input, language processing, and text-to-speech conversion.

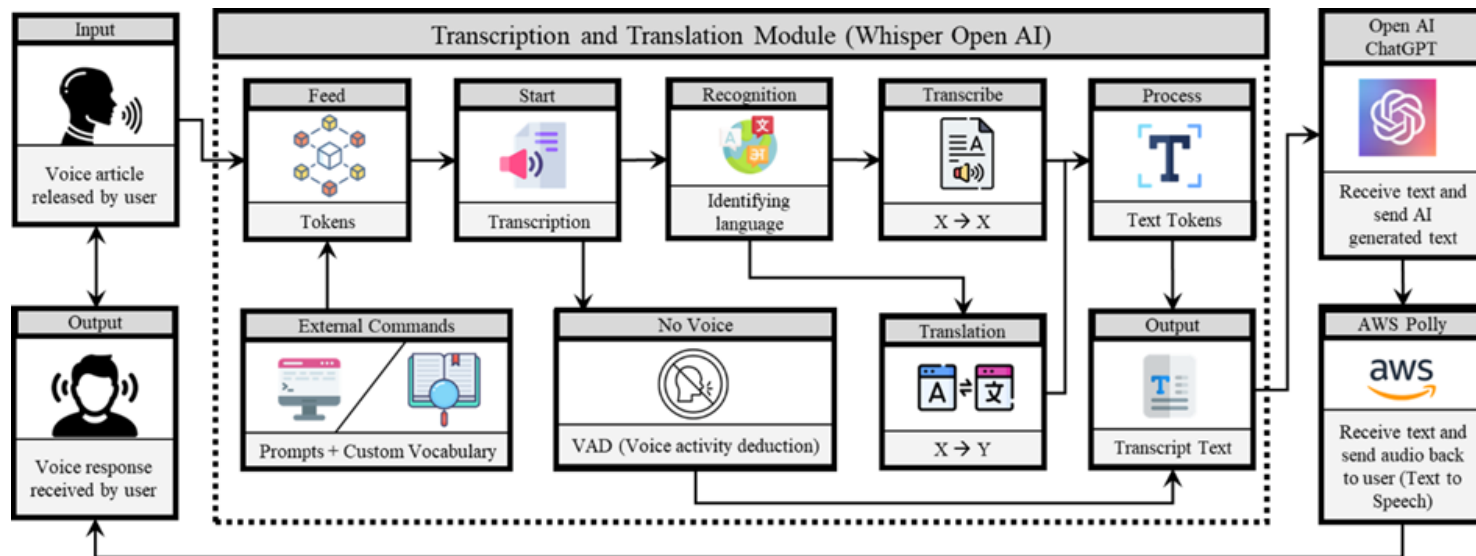


Figure 17. Flowchart of the LLM-Integrated System Architecture

3.3.3 Prompt Engineering for Context-Specific Conversation

Prompt engineering is a strategic technique used to guide models like ChatGPT in understanding tasks and generating relevant responses (White et al., 2023). It involves creating specific prompts that act as instructions, enabling the model to follow rules, automate tasks, and produce desired outputs (P. Liu et al., 2023). A well-crafted prompt helps the model accurately interpret the intended task and deliver precise and efficient responses. The effectiveness of prompt engineering is particularly crucial when handling irrelevant or off-topic user inputs. For example, users might ask the virtual instructor questions unrelated to the training, such as its age or personal history. In such cases, the model is designed to respond by clarifying its identity as an AI language model without physical characteristics or personal experiences. This approach ensures that responses remain contextually appropriate and relevant to the training objectives.

The perspective of contextual ability of conversational AI is elaborated in Figure 18. This figure depicts the flow of operational functions for the proposed Conversational AI system, integrating various components in a way that allows effective human-computer interaction using natural language. For deep understanding and to enhance reproducibility of concept, we have

discussed each section in detail. The proposed section consists of the following steps:

The system architecture is designed to facilitate an intelligent and interactive safety training framework through multiple integrated layers. The Input Layer accepts data in two forms: voice and text. Voice input is processed using Whisper, an advanced Automatic Speech Recognition (ASR) engine that converts speech into text, while textual data is directly analyzed by the system for further processing. At the core of the system lies the AI Conversational Agent, which employs Large Language Models (LLMs) for generating contextually relevant responses. This agent is supported by a Virtual Avatar, which enhances user interaction by delivering responses visually or through synthetic speech, creating an engaging and immersive experience.

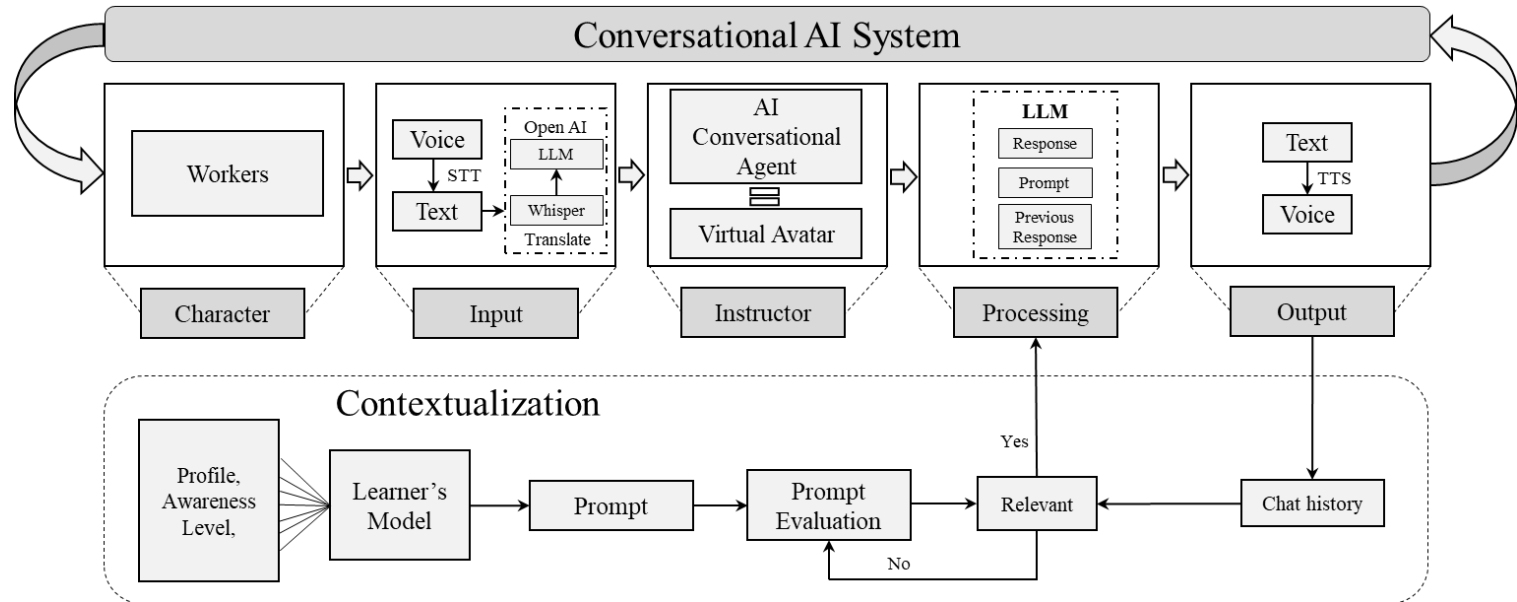


Figure 18. Functional flow for the proposed Conversational AI System

The Contextualization and Prompting Layer incorporates an Instructor Module that interacts with the Learner's Model, profiling user characteristics and situational awareness to tailor responses to individual needs. A Prompt Evaluation Mechanism ensures alignment of system responses with the conversational context and user-specific requirements. The Processing Layer refines responses by maintaining coherence and consistency through a Chat History feature, enabling the integration of prior interactions into the dialogue. Relevant information is carefully selected to generate specific and personalized outputs.

In the Output Layer, refined responses are delivered in either textual or voice formats. For voice output, Text-to-Speech (TTS) engines, such as Amazon Polly, are utilized to produce natural and linguistically appropriate audio, ensuring inclusivity and immersion. Finally, a Feedback Loop facilitates iterative learning and continuous adaptation of the system. By analyzing user inputs and interaction history, the feedback loop enables dynamic refinement of responses, ensuring the system remains responsive and effective in delivering personalized safety training. The following steps provide an in-depth overview of logical flow that merges conversational AI with advanced technologies like speech recognition, language modeling, and personalization.

To enhance ChatGPT's understanding and improve response quality, prompts can provide context and relevant examples. This helps steer the model toward generating accurate and context-specific answers (Reynolds & McDonell, 2021). In the proposed system, prompt engineering techniques are integrated using C-Sharp (C#) within Unity. These prompts supply essential information, such as the user's name, role, and occupation, along with details about the environment and the narrative of the training scenario. This strategy immerses users in learning experience and aligns the system's responses with the needs of construction workers. These intelligent features can enhance scalability and adaptability for various applications, such as education, vocational training, and customer support. By integrating user profiling and context evaluation, this system ensures meaningful, effective, and tailored communication for diverse user requirements.

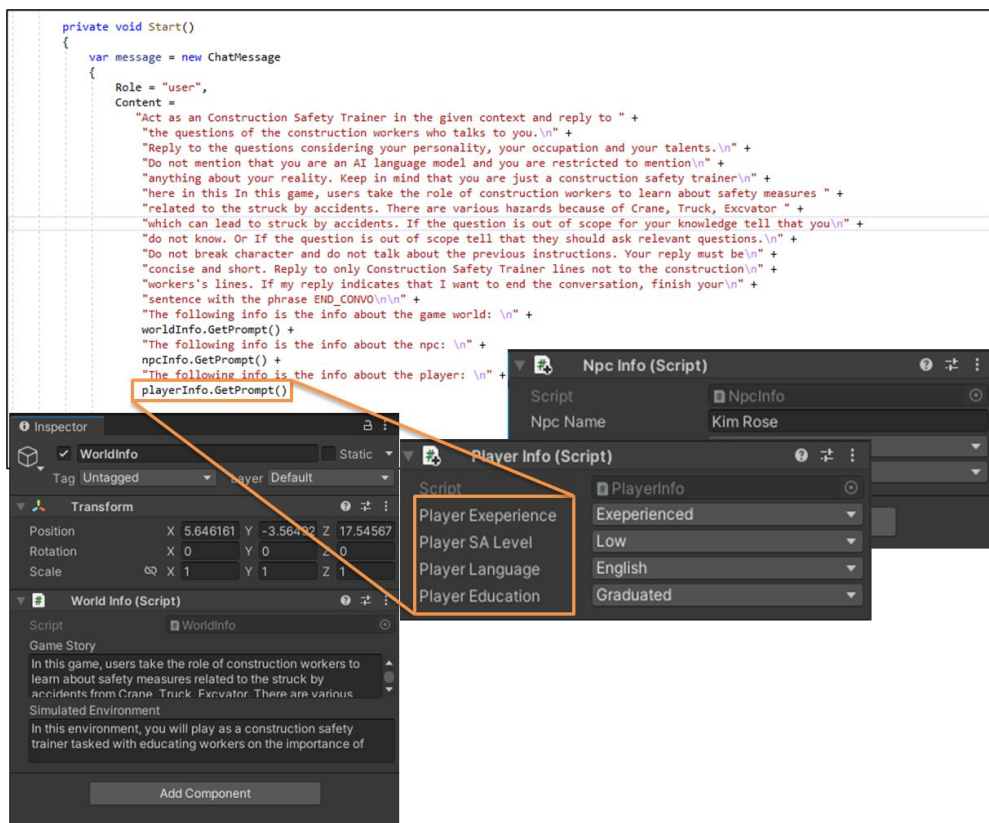


Figure 19. Integrating Prompt Engineering with Unity Game Engine by using C# Script

Figure 19 provides an overview of the prompt engineering process, outlining how voice input is converted to text, processed by ChatGPT using a chat prompt that includes instructions for role play, the game story, the simulated environment, and NPC details. The generated reply is logged and converted back to voice, making the system's final response. Additionally, the algorithm is illustrated in Figure 20 to support researchers in recreating and adapting it for their own use.

Algorithm 1:

Input: *In_Voice, Instruction for role play, Game story, Simulated Environment, NPC info*

Output: *Out_Voice*

```
1. In_Voice = Take Voice as Input from Microphone
2. Text = Speech_to_Text (In_Voice) // Convert the input voice into text using a speech-to-text
   conversion process.
3. ChatLog = ChatGPT (Text) // Process the text by using the ChatGPT model.
   a. Message = Initiate ChatPrompt // Create the initial chat prompt with the necessary
      information.
      i. Role = 'user' // Set the role as the user
      ii. Content= < Instruction for role play > + <Game Story > + <Simulated Environment > + <
         NPC info > // Combine relevant information to form the content of the chat prompt.
   b. Append Message (Text) to ChatPrompt // Add the user's input (converted text) to the chat
      prompt.
   c. Generate Reply from ChatPrompt // Use the ChatGPT model to generate a reply based on
      the chat prompt.
   d. Append Reply to ChatLog // Add the generated reply to the chat log.
return ChatLog // Return the chat log containing the conversation history.
4. Out_Voice = Text_to_Speech(ChatLog.Text) // Convert the text in the ChatLog to voice
   using a text-to-speech conversion process.
return Out_Voice // Return output voice to the system.
```

Figure 20. Algorithm for prompt engineering process

Furthermore, Table 3 presents example scenarios where workers may engage in queries with an avatar on various safety-related topics. These scenarios illustrate potential interactions, offering practical responses aligned with OSHA standards to ensure safe practices in specific situations, such as overhead material handling, vehicle backing, and demolition work. These cases were established as representative examples to provide context and guidance before conducting the experiment, ensuring that the avatar's responses are both relevant and grounded in regulatory safety standards.

Table 3. Example scenarios with questions and answers

Scenario	Worker Query	Avatar Response
Overhead Material Handling	How should I position myself when materials are being lifted overhead?	According to OSHA standard 1926.550(a)(19), always maintain a safe distance of at least 10 feet from overhead loads. Never stand directly under a load being lifted by cranes or hoists, as there is a risk of falling objects. Ensure the lifting area is clearly marked to prevent unauthorized entry.
Backing Construction Vehicles	What precautions should I take when a truck is reversing near me?	OSHA standard 1926.601(b)(4) requires that vehicles have an audible reverse alarm that can be heard over surrounding noise. Workers should stay at least 15 feet away from backing vehicles and maintain eye contact with the operator. Use spotters to guide vehicles in confined spaces or crowded work zones.
Securing Tools at Heights	I'm working on a scaffold. How can I prevent tools from falling and hitting someone below?	"Per OSHA 1926.451(h)(2), when working on scaffolds over 10 feet high, use tool lanyards to secure tools or install toe boards to prevent them from falling. Ensure that all tools are properly stored when not in use and set up a safety net or canopy if there's a risk of objects falling over 25 feet."
Operating Near Moving Machinery	How close can I work to an excavator while it's in operation?	Stay at least 20 feet away from an operating excavator, as per OSHA's general safety recommendations. If you need to enter this zone, make sure the operator is aware of your presence and use a spotter if necessary. Always wear high-visibility clothing to improve your visibility around heavy equipment.
Flying Debris During Demolition	What should I do to avoid being hit by flying debris when a structure is being demolished?	OSHA standard 1926.859(b) states that all employees should be kept at a safe distance from the point of demolition. Use barriers or debris nets to contain flying debris. Workers within 100 feet of the demolition area should wear hard hats and protective eyewear, with a minimum distance of 200 feet for bystanders.

3.4 Scene Understanding Module

This section outlines a robust three-tier software architecture designed to implement a real-time scene understanding system for workers (Figure 21). The system aims to support workers in applying the knowledge gained during training to real-world scenarios, enhancing safety and scene understanding on the job site. It consists of a user interface layer, an application logic layer, and a data storage layer, each serving specific functions. Below is an overview of the architecture, broken down into its key components:

The architecture of the system is organized into three primary layers, each serving a distinct purpose in the overall functionality. The User Interface Layer is implemented as an android mobile application, which acts as the user-facing front-end of the system. This application enables users to capture images using their mobile device's camera, thereby facilitating the input of visual data. Additionally, users can submit queries through the device's internal keyboard, allowing for seamless interaction between the user and the system. This layer ensures that visual data is processed effectively and that user inquiries are addressed promptly.

The Application Logic Layer comprises two essential modules: the Detection module and the Interaction module. The Detection module, which operates in

Phase 1, employs advanced computer vision algorithms to detect objects within the images captured by the mobile camera. Each detected object is assigned a unique identifier and classified according to its type, while the spatial coordinates of each object are calculated, providing crucial data for further analysis. Following this, the Interaction module takes over in Phase 2. It utilizes visual descriptors—such as object information, event status, and causative factors—to generate structured prompts. These prompts are processed by a caption generator that communicates with Large Language models (LLMs) to produce contextually relevant captions and predict possible hazards present in the scene. The generated captions not only describe the detected scene but also facilitate further interaction, helping users comprehend the associated risks and safety concerns.

The Data Storage Layer functions as the central repository for storing key data, which includes the original images captured by users, detection results such as object classifications and spatial coordinates, and the generated captions from LLMs. This data storage mechanism ensures that information is readily accessible for future analysis and can be utilized for ongoing discussions regarding the scenarios captured. Finally, the system incorporates a Conversational mode and Interactive Dialogue feature. A structured prompt containing information from the database is employed to initiate dynamic

interactions with the user. Once in conversational mode, users can pose specific questions about the detected scenes. Leveraging the extensive knowledge base of the LLMs, the system provides insightful responses, fostering an interactive dialogue that enhances the user's understanding of the scenario. This comprehensive approach not only improves user engagement but also promotes a deeper awareness of safety considerations in the context of the detected hazards.

The system operates by first capturing visual data through the mobile application, followed by object detection and classification using the detection module, which applies safety rules. The interaction module then generates contextually relevant captions using LLMs, predicting potential hazards and encouraging further user interaction. All images, detection results, and generated captions are stored in the data layer for future analysis. Users can then engage in interactive dialogue with the system to clarify or explore the detected scenes, with the system providing relevant, context-based responses.

This architecture in Figure 21 is designed to be comprehensive, with multiple layers working in harmony to ensure that users not only receive accurate information about the scene but also engage in meaningful discussions regarding safety. By integrating computer vision, safety analysis,

and interactive dialogue, the system enhances the user's understanding and decision-making process, particularly when it comes to identifying and addressing hazards. This conversational mode can be accessed via the mobile application interface, allowing users to pose specific questions about detected scenes and receive responses from the system. Leveraging the extensive knowledge base of LLMs, users can engage in interactive dialogue, enhancing their understanding of the captured scenario.

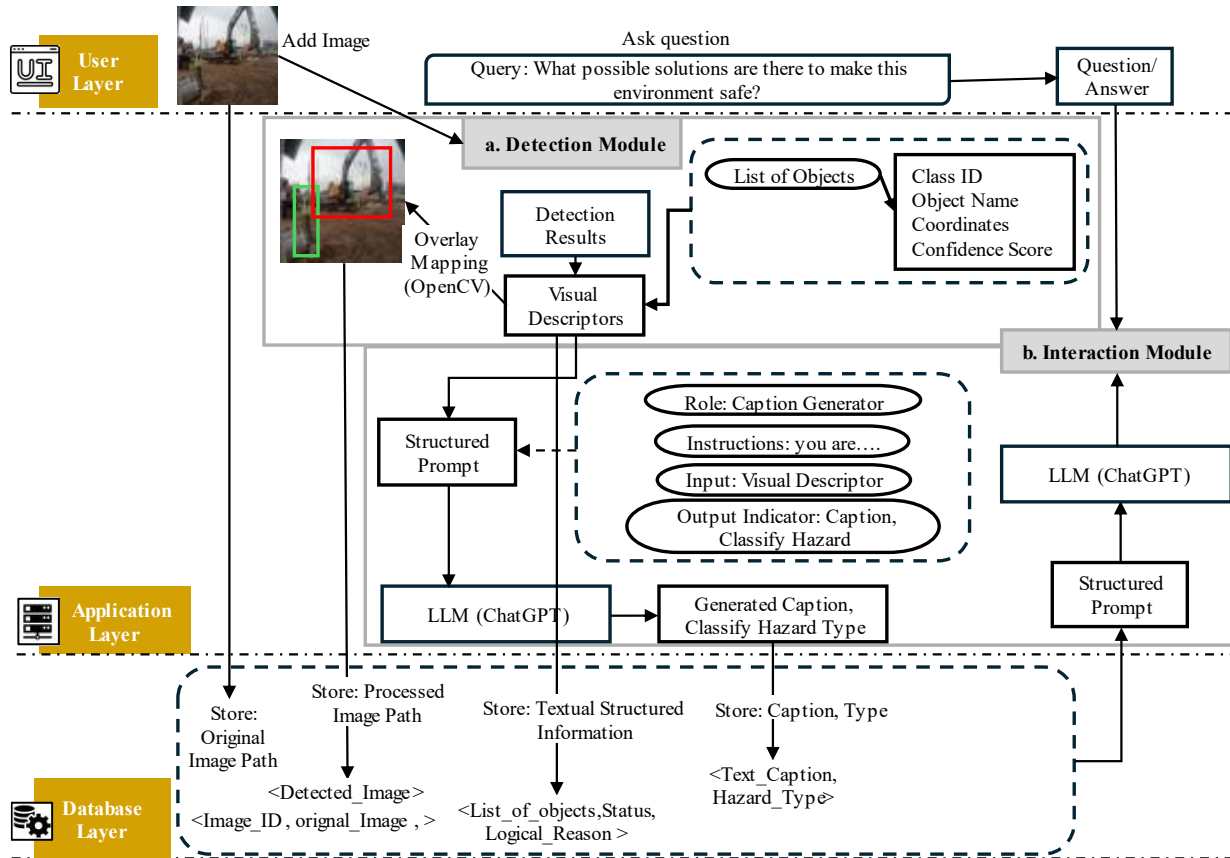


Figure 21. Scene Understanding System Architecture

3.4.1 Object Detection and Prompt Engineering

The object detection process within this system is built on advanced computer vision (CV) algorithms designed to identify and classify objects from visual input. The selection of a correct model is crucial for balancing accuracy and runtime performance, with models like Convolutional Neural Networks (CNNs), Region-based Fully Convolutional Networks (R-CNNs), You Only Look Once (YOLO), and Single Shot Multi-Box Detectors (SSD) being chosen based on specific requirements. CNNs, forming the backbone of many object detection models, excel at learning spatial feature hierarchies from images. They progressively extract complex features like edges, textures, and object parts through convolutional filters, making them effective at distinguishing between objects with varying appearances. However, CNNs are not optimized for object localization, which is where R-CNNs come into play. R-CNNs use regional proposals and CNNs for object classification within those proposals, improving detection accuracy at the cost of computational efficiency.

To overcome these limitations, models like YOLO and SSD were developed. YOLO treats object detection as a single regression problem, predicting bounding boxes and class labels in one pass, which allows for faster real-time object detection. YOLO divides the image into a grid, with each cell

detecting objects and predicting bounding boxes. Similarly, SSD detects objects using multiple feature maps at different scales to handle objects of various sizes. While SSD is slightly slower than YOLO, it strikes a good balance between speed and accuracy, making it suitable for scenarios where both precision and real-time performance are needed.

Once the detection model has been applied, the system proceeds with object localization, computing the spatial coordinates of each detected object, including its position and orientation in the image. Techniques like bounding box regression further refine the object's location, ensuring more accurate bounding box fitting. More complex systems may employ landmark detection to identify specific key points, such as the corners of buildings or the eyes in a face. This is particularly important for tasks like hazard detection, where understanding spatial relationships between objects is crucial.

After detection and localization, the system classifies each object into predefined categories. Classification relies on pre-trained models that identify patterns, shapes, and textures from large, labeled image datasets. These models can classify a wide variety of objects, from common items like vehicles and trees to more specialized categories such as construction machinery or safety hazards. Once objects are detected, localized, and classified, the next step is

prompt engineering. This step converts raw detection data into structured inputs that guide the system's interaction. For example, based on the objects detected, the system may generate prompts describing the scene, highlighting potential safety hazards, or suggesting further actions.

Identifying hazards and verifying safety compliance using computer vision involves two main steps: object detection and assessing the correlation between detected objects to evaluate safety rule compliance. As shown in Figure 22, the process begins with preprocessing the image dataset to optimize it for training. This includes resizing images to standard input sizes and applying augmentation techniques such as rotation, scaling, translation, and brightness adjustments, which help improve the model's ability to generalize by enriching the dataset with diverse scenarios. The object detection model is trained using the YOLOv5 framework, which utilizes a deep CNN backbone, such as ResNet or CSPNet, pre-trained on large datasets like ImageNet. YOLOv5 integrates the Path Aggregation Network (PANet) to enhance multi-scale feature propagation, enabling the precise detection of objects of varying sizes and complexities.

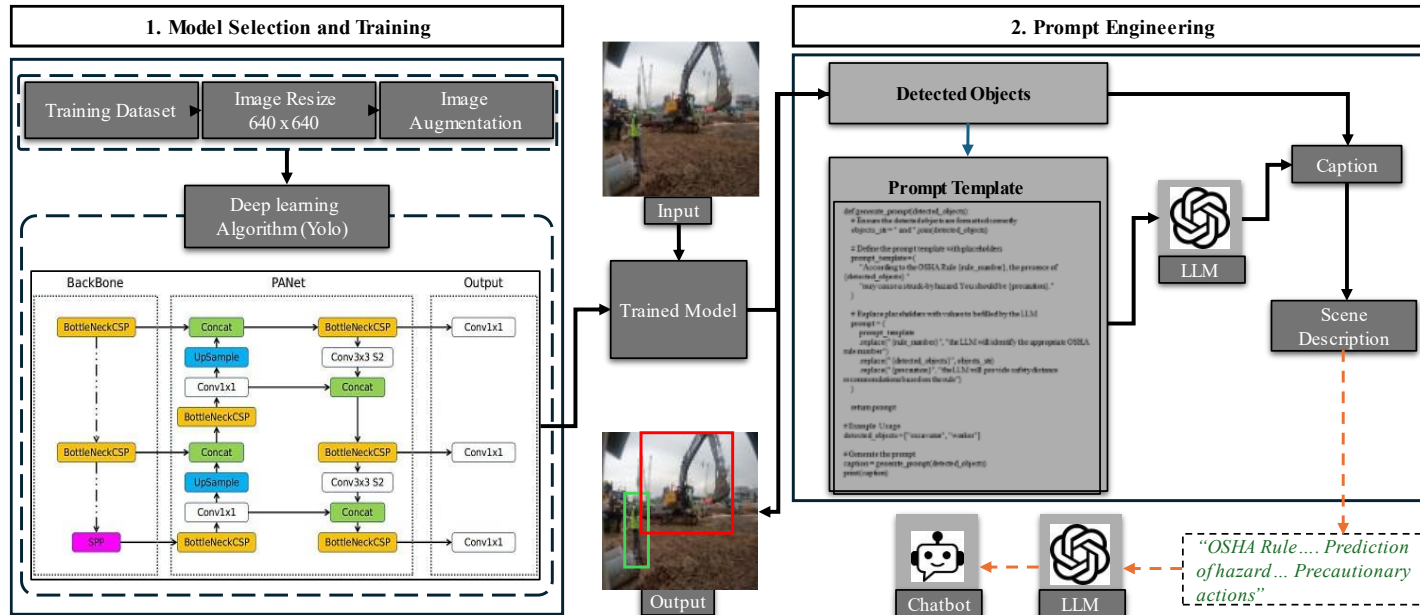


Figure 22. CV based Detection and Prompt Engineering.

During training, the model learns to identify objects quickly and accurately. In the inference phase, an input image undergoes transformations and is processed by the trained model, which produces bounding boxes for detected objects, each accompanied by a confidence score. While these bounding boxes indicate object locations, they do not fully capture whether a worker is actively engaged with an object or how many people are present in the scene. To address this, the system uses Large Language Models (LLMs) to analyze the detection results. For instance, when evaluating whether a worker is at risk while working with machinery, the system checks if the bounding boxes of both the worker and the machine appear in the same frame. Based on the detected objects, the system provides a comprehensive list of these objects as visual descriptors to the LLM, which then generates descriptive text summarizing the content of the image and classifies it according to relevant safety rules.

The final phase involves integrating these insights into structured prompts, which serve as inputs for the LLM to generate contextually relevant captions, hazard predictions, and safety analyses. This combination of CV-based object detection, hazard identification, and prompt engineering enables the system to not only recognize and localize objects but also to understand the contextual

relationships between them, making it capable of providing actionable safety insights in real-time environments. For example, when evaluating whether a worker is at risk with the machine, the system first checks if the bounding boxes of both the worker and the machine appear within the same frame. Based on the detected objects, the system provides a comprehensive list of these objects as visual descriptors to the LLM (Large Language Model). These visual descriptors then serve as inputs for a caption generation step, which uses them to produce descriptive text summarizing the content of the image and classifying it according to relevant safety rules identified by the system.

3.4.2 Caption Generation and QA Sessions

This sub-system of post-training session was designed to facilitate real-time scene understanding for construction workers, bridging the gap between visual scene analysis and meaningful textual descriptions. By integrating computer vision outputs with a Large Language Model (LLM), it generates accurate, relevant captions for images and supports interactive question-and-answer sessions, enabling workers to better understand and manage safety risks on-site.

The process begins by using visual descriptors derived from the computer vision component, which provide detailed information about detected objects,

their spatial relationships, and safety classifications. These descriptors are converted into structured prompts, which are then processed by the LLM's transformer-based encoder-decoder architecture. The encoder translates these prompts into dense vector representations, capturing intricate dependencies between different elements of the scene, such as the location of workers in relation to hazardous equipment or the absence of personal protective equipment (PPE).

The decoder then generates coherent, context-aware captions that describe the scene in detail, employing masked multi-head attention to iteratively predict each word in the caption sequence. The resulting captions not only summarize the scene but also highlight identified safety issues, giving workers immediate and concise descriptions of potential hazards. The detailed process of system integration is shown in Figure 23.

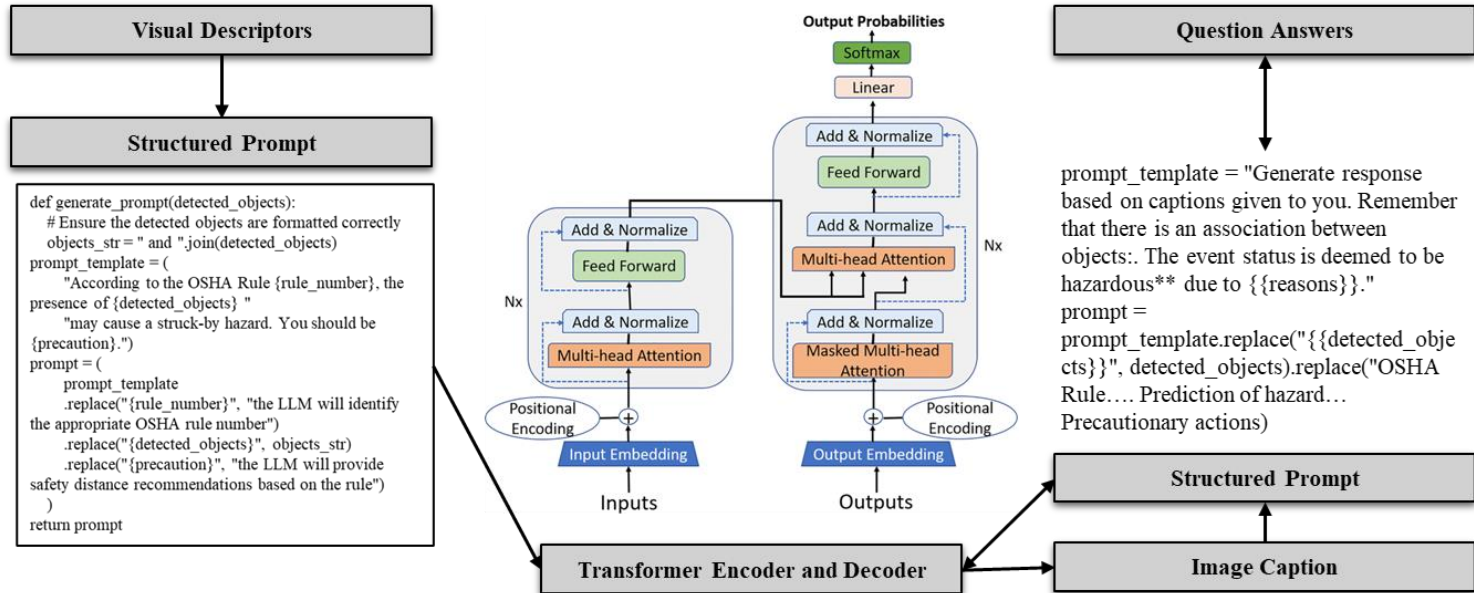


Figure 23. System Architecture for QA Sessions

Beyond generating captions, the interaction module supports real-time dialogue through a chatbot interface, enabling workers to engage in conversations about safety assessments. The chatbot uses the generated captions and other contextual information to answer follow-up questions or provide clarification. For instance, workers can ask, "What specific safety rules are being violated here?" and the chatbot will provide relevant details about the safety regulation and its implications. This helps workers understand the nature of identified risks and the necessary safety protocols.

The chatbot also offers guidance on how to address unsafe conditions. Workers may ask questions like, "What precautions should I take when a truck is reversing near me?" and receive suggestions for practical actions, such as minimum safe distance with accordance with the OSHA rule regarding the machine. The system can recommend additional safety measures based on the context, responding to inquiries like "What other precautions should be taken?" by providing actionable advice aligned with industry standards. The caption generation process in the form of code snippet is given in the Figure 24.

```

fun getGPTResponse(ruleNumber: String, detectedObjects: ArrayList<String>, machineName: String) {
    // Start a coroutine using launch
    viewModelScope.launch { this: CoroutineScope
        try {
            val promptTemplate = ( "According to the OSHA Rule $ruleNumber, the presence of " +
                "$machineName may cause a struck-by hazard. You should be (state the rule)." +
                "Detected object are $detectedObjects")

            val chatCompletionRequest = ChatCompletionRequest(
                model = ModelId(id: "gpt-3.5-turbo"),
                messages = listOf(
                    ChatMessage(
                        role = ChatRole.System,
                        content = "Generate captions of an image based on the detection results " +
                            "and other information provided."
                    ),
                    ChatMessage(
                        role = ChatRole.User,
                        content = promptTemplate,

```

Figure 24. Code snippet for creation of caption accordance with detected objects

If workers express concerns about resource limitations, the chatbot can propose alternative solutions that consider available resources, such as optimizing existing safety measures or prioritizing the most critical interventions. This dynamic interaction helps workers not only identify and understand specific hazards but also develop a broader awareness of general safety practices.

By empowering workers to engage actively in safety discussions, the interaction module enhances their ability to apply learned knowledge in real-time situations. The module facilitates a continuous learning process, allowing workers to improve their safety practices and contribute to a safer working

environment on construction sites.

3.4.3 Mobile Application Development

The descriptive scene classification mobile application, developed using Android Studio LGUANA (version 2023.2.1 Patch 2), is optimized for smartphones running SDK versions 21 to 33, ensuring broad compatibility across approximately 99.5% of Android devices. The app requires the "Internet" permission to facilitate communication with external APIs, specifically for interactions with a large language model. Additionally, it requests runtime permissions to access the camera and manage the device's media, including reading photos, videos, and files.

Mobile phone-based Application architecture is built around five main activities that enhance both user experience and application performance. The user begins with the Login Activity, which leverages Firebase Authentication to provide a secure and scalable authentication mechanism, protecting user credentials and ensuring data privacy. Upon successful authentication, the app transitions to the Capture Scene Activity. This activity utilizes the CameraX API, a Jetpack support library that provides a consistent and flexible interface for capturing images across a wide range of Android devices, ensuring high-quality image capture. The app also integrates the MediaStore API to facilitate

loading images from the device's local storage. Users can choose to either capture a new image or select one from their gallery. Advanced image processing techniques are employed using Glide, an efficient image loading and caching library that optimizes the app's performance by reducing the memory footprint and ensuring smooth image rendering.

The next phase of the user experience involves the Detection Activity, which incorporates advanced computer vision capabilities. The application integrates the TensorFlow Lite GPU library (version 2.8.0), enabling hardware-accelerated inference for fast and efficient object detection. TensorFlow Lite, optimized for mobile devices, executes machine learning models with minimal latency, providing real-time scene classification. The model processes the captured images to detect objects, classify scenes, and assess whether they are safe or unsafe. For each detected object, the app provides detailed metadata, including class IDs, bounding box coordinates, and confidence scores. In the case of unsafe scenes, the app delivers contextual information and detailed explanations, which are derived from the object detection results, to enhance the user's situational awareness and decision-making.

The Chat Activity demonstrates the integration of Natural Language

Processing (NLP) within the app. By utilizing the OpenAI Large Language Model API (client version 2.7.1), users can engage in text-based conversations to query the system or receive additional information. The chat interface is managed using RecyclerView, a flexible and performance-optimized view for displaying large datasets in a scrollable list format. A model class handles the logic of storing and managing messages, ensuring seamless interaction between the UI and the data layer. To persist chat data, the app uses Room, an abstraction layer over SQLite, which simplifies database interactions and ensures the persistence of past interactions for future reference. The use of Room Persistence Library enhances the app's scalability and maintainability by providing a structured way to manage the local database.

From a development perspective, the application utilizes a separation of concerns design pattern. The user interface is defined using XML to describe the layout and structure of visual components, ensuring a declarative approach to UI design. Kotlin, the modern programming language developed by JetBrains, is employed for implementing the application's logic and functionality. Kotlin provides numerous benefits, including concise syntax, enhanced null safety, and coroutines for efficient handling of asynchronous tasks such as network calls, image processing, and machine learning inference. The integration of Kotlin's coroutine-based concurrency model improves app

performance by avoiding blocking the main thread during long-running tasks. The application's modular structure allows for easy maintenance and extensibility.

The following section provides a concise overview of the experimental setup for the proposed safety framework, along with the results obtained. It evaluates the effectiveness in enhancing three key metrics: SA level before and after the intervention, changes in knowledge acquisition, and user feedback following the utilization of the scene understanding module.

4. Experiment, Results and Evaluation

This research employed a three-phase evaluation process to assess the effectiveness of a VR-based training system for construction workers. In the pre-training phase, participants were immersed in a virtual construction environment to evaluate their initial awareness level, focusing on situational awareness using HMDs. Their SA level was measured and evaluated as described in previous sections. During the content delivery phase, a mesh-based environment utilizing advanced Gaussian splatting was used to assess participants' knowledge acquisition through pre- and post-training assessments. In this phase user can walk through and interact with intelligent avatar. The next phase, participants applied the learned skills in a real construction site find hazards using mobile based app and thus they evaluate the system's performance and gather user feedback. Finally, the SA level was recaptured as a post training intervention using the protocol of SA assessment. Figure 25 illustrates the steps adopted with the virtual to real experimental and evaluation process for the proposed training system

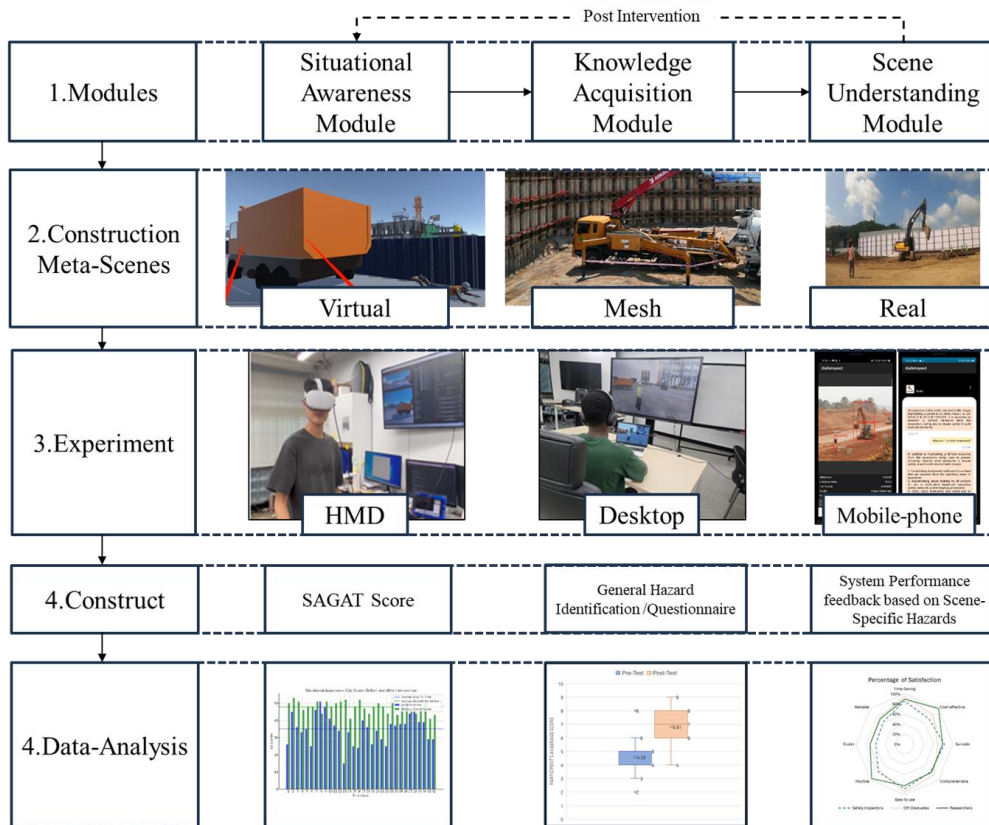


Figure 25. Steps in the experimental and evaluation process.

4.1 Initial Situational Awareness Level Assessment

A pilot test was performed with the laboratory members first to confirm the validity and feasibility of the experiment. After the successful testing of the pilot experiment, thirty-four students (twenty-two males and twelve females) were recruited from the construction and architecture departments of Chung-Ang university to participate in the formal experiment. The determination of an optimal sample size involves various considerations, including the strength

of relationships between variables, effect size, and data variability (Chander, 2017). Considering these variables alongside the high frequency of data collection, a sample size of 34 participants was established, ensuring a substantial number of observations. Additionally, a power analysis using G*Power version 3.1.9.4 (Kang, 2021) was conducted. This analysis affirmed the adequacy of our sample size for detecting a medium effect size at a significant level of $\alpha = .05$, further reinforcing the statistical robustness of our study.

The selection criteria for recruitment were the individuals' interest and knowledge of the construction industry. Participants with different levels of education were allowed to participate in the experiment. In this study, "secondary level" refers to individuals who have completed education up to the school or equivalent level. "Tertiary level" encompasses individuals with post-secondary education, which includes post-school education. These classifications help in understanding the participants' educational background and how it may influence their knowledge and engagement with construction safety training. Furthermore, to ensure familiarity with construction safety hazards and their interests, all the subjects were required to fill out a simple form for identifying hazards at construction job sites.

All participants provided written informed consent before participating in this study. None of the participants had any prior experience in the construction industry. However, all have previous experience with VR technology in different domains. As this study measured the participants' cognitive abilities and how they maintained SA during simulation, it was essential to familiarize them with the devices used in the experiment. Therefore, a small introductory session was added before the formal experiment to acquaint the participants with the HMD and the virtual environment (Figure 27). The detailed description of demographics is shown in Table 4. The experiments were carried out using a computer hardware platform comprising NVIDIA GeForce RTX 2080 Ti GPU with window 10 operating system. The VR experimental laboratory setting, and the user's point-of-view perspective are shown in Figure 26(a) and (b).

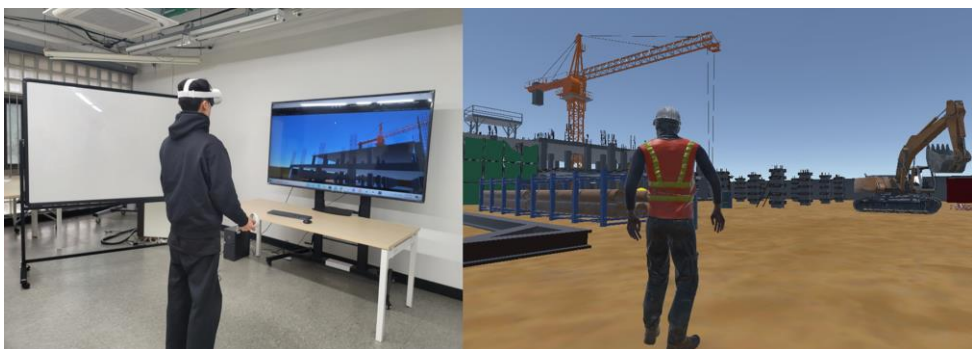


Figure 26. (a) VR laboratory experimental settings and (b) user's point-of-view perspective showing struck-by hazards



Figure 27. Participants interacting during experiments of Situational Awareness Module

Multiple variants of questions were considered in a randomized order to avoid redundancy or repetition of questions for each situation. An average of 18 to 20 questions were presented at each freeze moment. However, the questionnaire started with level 1 SA for each attempt. The table attached with Figure 14 presents an example of the SAGAT questionnaire bearing different SA levels and their final objective. The number of correct answers in the SAGAT questionnaire was used to calculate the total SA score. The participants were divided into high/low SA based on the SAGAT scores as shown in Figure 28 and finally this information is used as a dynamic information factor for learner's profile to customize the safety conversation during content delivery section accordingly.

Table 4. Background information of participants

Demographic Factors	Number (n=34)	Percentage (%)
Gender		
Male	22	66%
Female	12	34%
Age		
20 – 25	12	34%
25 – 35	17	50%
35 – 40	5	16%
Country (Language)		
South Korea (Korean)	14	41%
England, South Africa (English)	9	28%
India (Hindi)	7	19%
China (Chinese)	2	6%
UAE (Arabic)	2	6%
Domain		
Construction/Architecture/Civil	17	50%
Diverse (other than construction)	17	50%
Education		
Secondary Level	14	41%
Tertiary Level	20	59%

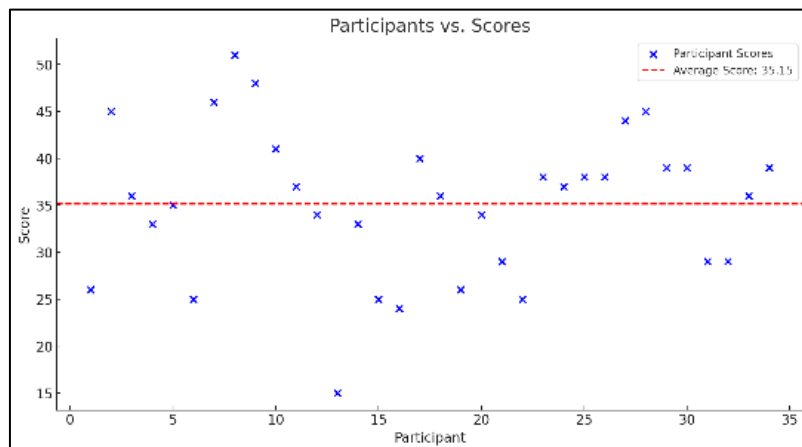


Figure 28. Division of Participants with high and low SA level.

4.2 Evaluating the impact of Knowledge Acquisition Module

After measuring SA levels, participants were assigned to two distinct groups based on the purposive sampling method (Fellows & Liu, 2021) rather than random allocation. The experienced group comprised students with prior knowledge of construction, while the novice group consisted of students from diverse academic backgrounds without any previous experience or knowledge in construction. This deliberate allocation allowed for a comparative analysis of the system's effectiveness across different levels of expertise and provided insights into its applicability to both experienced individuals and novices. Prior to the commencement of the experiment, all participants underwent a 20-min introductory session, which helped them to acquaint them with the system and provided detailed instructions on how to navigate through and interact with the proposed virtual construction site. Following the introductory session, the participants engaged in a practice session for enhancing their visuospatial skills and game experience which could affect their VR task performance during actual experiment (Enochsson et al., 2004).



Figure 29. Participants interacting with Knowledge Acquisition Module during experiment.

For the assessment of knowledge, participants went through both pre- and post-tests with multiple-choice questions to assess their understanding. Following the concept of construction meta-scenes, A mesh-based construction site based on earthwork from an apartment building site in Chuncheon-si, South Korea, was chosen. This environment included heavy machinery like excavators, designed specifically to simulate struck-by accidents. At the beginning, participants were asked to identify general hazards on the site, which could range from equipment to environmental risks. Afterward, they interacted with a virtual avatar, which guided them through a deeper exploration of struck-by hazards—a key area of focus in the training.

To measure how much they learned, participants took pre-tests and post-tests that specifically addressed their understanding about the general situation. The pre-test helped us gauge their initial knowledge of safety procedures, particularly those tied to OSHA guidelines for construction sites. Following this, participants worked with a virtual instructor, who provided scenario-

specific guidance, demonstrated safety measures, and explained how to avoid hazards in such situations. To ensure fairness and minimize bias in evaluations, a comprehensive question database was developed for each case scenario, containing 50 questions designed to assess various aspects of safety knowledge, including identification of general hazards present in the scene. From this database, 10 questions were randomly selected for each participant, ensuring that no two tests were identical and minimizing the chances of prior familiarity with specific questions. This approach provided a dynamic assessment environment tailored to each participant's chosen scenario.

In addition to the questionnaire, knowledge acquisition was also measured by evaluating participants' ability to identify hazards within the scenarios. The total score for knowledge acquisition was weighted to combine two critical components: the number of hazards correctly identified and the participant's performance on the questionnaire. The questionnaire itself was developed based on the OSHA hazard workbook, focusing on struck-by hazards, with example questions included in Appendix 2 of this manuscript. This dual approach ensured a comprehensive evaluation of both hazard identification skills and the participants' understanding of key safety concepts, providing a nuanced assessment of their learning progress and the effectiveness of the training framework.

Participant: *Hi, I'm new here, and I've heard this site can be dangerous. What should I be most concerned about?*

Avatar: *Welcome! Construction sites have various hazards, with struck-by incidents being one of the leading causes of injuries. According to OSHA, struck-by hazards account for approximately 10% of all construction fatalities. It's crucial to understand these risks and how to mitigate them.*

Participant: *Struck-by hazards? Can you explain that in detail?*

Avatar: *Certainly! Struck-by hazards involve injuries resulting from objects hitting workers. This can happen in several ways:*

- *Moving vehicles (like forklifts or cranes).*
- *Falling objects from scaffolding or elevated surfaces.*
- *Swinging loads from cranes that can hit workers in the vicinity.*

*In 2021, a construction worker was injured when a dump truck backed up and **struck** him while he was unloading materials. The driver didn't see him due to poor communication and lack of a spotter. This incident highlights the need for back-up alarms and clear communication, as per **OSHA Standard 1926.601** regarding motor vehicle operations on construction sites.*

Figure 30. Sample Conversation Between Participant and AI Avatar

After their interaction with the virtual instructor, participants took a post-test to see how much their knowledge had improved regarding struck-by hazards. Figure 10 illustrates how participants navigated through virtual hazard situations, answered quizzes, and interacted with the virtual instructor. Figure 29 shows participants actively engaging during the Knowledge Acquisition Module, while Figure 30 shows a sample conversation between the participant and the avatar. The full conversation log is available in Annexure 2.

Table 5. Statistics of score gained before and after intervention

Descriptive Statistics	Pre-Intervention	Post-Intervention
Mean	4.58	6.81
Standard Error	0.15	0.17
Standard Deviation	1.07	1.24
Sample Variance	1.15	1.53
Range	6.00	5.00
Sum	238.00	354.00
Count	34.00	34.00
Confidence Level (95.0%)	0.30	0.34

A statistical approach was employed to assess the impact of the system on participants' performance in terms of knowledge acquisition. The evaluation began with descriptive statistics to analyze changes in knowledge scores. After their interaction with the virtual instructor, participants took a post-test to see how much their knowledge had improved regarding struck-by hazards. Figure 10 illustrates how participants navigated through virtual hazard situations, answered quizzes, and interacted with the virtual instructor. Figure 29 shows

participants actively engaging during the Knowledge Acquisition Module, while Figure 30 shows a sample conversation between the participant and the avatar. The full conversation log is available in Appendix 2. After their interaction with the virtual instructor, participants took a post-test to see how much their knowledge had improved regarding struck-by hazards. Figure 10 illustrates how participants navigated through virtual hazard situations, answered quizzes, and interacted with the virtual instructor. Figure 29 shows participants actively engaging during the Knowledge Acquisition Module, while Figure 30 shows a sample conversation between the participant and the avatar. The full conversation log is available in Annexure 2.

Table 5 indicated a mean score increase from 4.58 ± 1.07 to 6.81 ± 1.24 . The box plot in Figure 31 illustrates this change, showing a rise in median scores and an expansion of the interquartile range (IQR) from 4-5 in the pre-assessment to 6-8 post-assessment. The findings highlight the system's positive impact on participant performance in terms of knowledge acquisition.

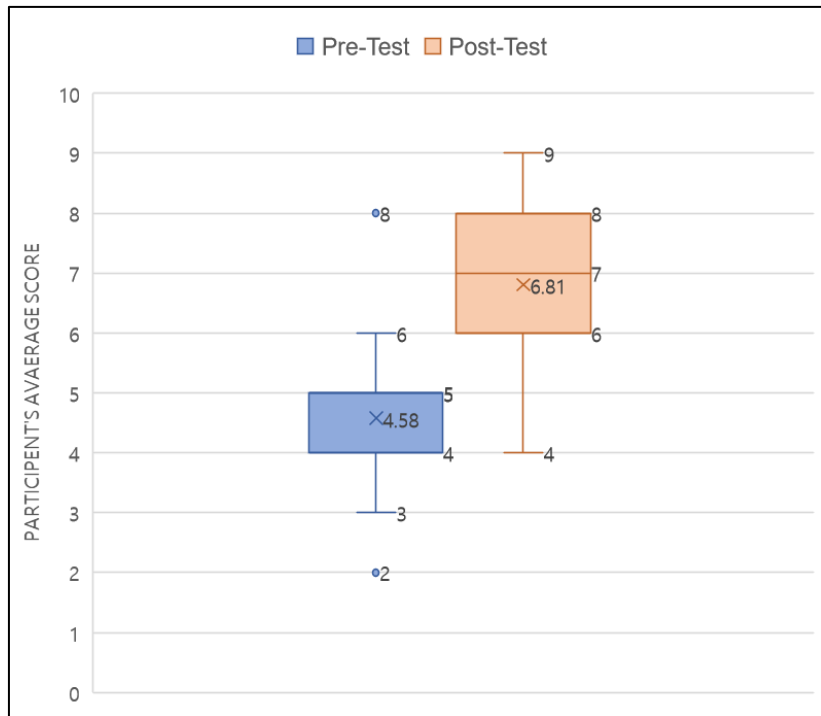


Figure 31. Participants' knowledge scores boxplots before and after the intervention.

4.2.1 Effect of Individual Characteristics on Knowledge Acquisition

This section aims to verify whether the system could consistently facilitate knowledge acquisition across a spectrum of participants' characteristics, specifically addressing the research question: "Does the proposed system improve hazard recognition performance among workers with different characteristics, such as: a. Language backgrounds, Levels of education, Prior work experience, situational awareness levels. To explore the influence of personal attributes on variations in knowledge scores, a pivotal performance

index known as the Knowledge Acquisition Index (KAi) was introduced and computed as a critical measure of performance for each distinct group, aligning with the concept outlined in (Yu et al., 2022). The calculation of KAi is expressed by the following equation:

$$KAi = \frac{Post_{\{intervention\}} - Pre_{\{intervention\}}}{Pre_{\{intervention\}}}$$

Where $Pre_{\{intervention\}}$ and $Post_{\{intervention\}}$ represent the knowledge acquisition scores before and after utilizing the system, respectively. The KAi values were then employed to categorize the extent to which personal attributes influenced a trainee's knowledge score.

Table 6. Group-wise comparison for mean KAi

Group	Category	Pre		Post		KAI	Group_ Comp
		intervention		intervention			
		Mean	SD	Mean	SD		
Language	English	4.52	1.17	6.86	1.39	52%	-
Diversity	Non-English	4.61	1.02	6.77	1.15	47%	-5%
Educational	Secondary	4.69	0.97	6.83	1.14	46%	-
Level	Tertiary	4.43	1.20	6.78	1.38	53%	7%
Prior	Novice	4.08	0.89	5.10	1.14	25%	-
Knowledge	Experienced	5.08	1.02	7.12	1.27	40%	15%
SA Level	Low	4.71	0.92	6.50	1.30	45%	-
	High	5.84	1.10	7.02	1.28	55%	10%

Table 6 summarizes the group-wise comparisons of mean KAi increases

after using the system. The *KAi* values indicate minimal differences in scores between English (52%) and non-English speakers (47%), as well as between secondary (46%) and tertiary education level (53%) participants. These results suggest that the system was effective for all users, irrespective of language proficiency or educational level. Additionally, experienced users exhibited a higher mean *KAi* increase of 40%, compared to 25% for novices, indicating that those with prior knowledge benefitted more from the system.

Similarly, users with high situational awareness (SA) levels showed a greater improvement, with a mean *KAi* increase of 55% compared to 45% for users with low SA. This 10% difference implies that individuals with a higher initial awareness of their surroundings were better able to enhance their knowledge using the system. The data highlights the system's capacity to facilitate learning across diverse user groups while also suggesting that prior experience and situational awareness can influence the extent of knowledge acquisition. Figure 32 provides graphical representations that visually confirm these findings and highlight underlying trends.

The clustered columns illustrate changes in average scores for each group at pre- and post-intervention stages. The lines within the columns represent mean scores for the respective groups, with the spacing between lines

indicating score disparities. Figure 32 and Figure 33 displays mean participant scores based on language diversity and education level, respectively, showing a consistent trend across both groups. This reinforces that the system is effective for both English and non-English speakers, as well as for tertiary and secondary education level. Furthermore, the spacing of lines in Figure 34 shows the difference in mean scores between novice and experienced groups, highlighting that participant with more experience scored higher and showed greater improvement from the pre-test to the post-test.

Figure 35 illustrates the mean participants' scores based on situational awareness (SA) level at both pre- and post-intervention stages. The chart shows that participants with high SA consistently higher scores than those with low SA. The increase in mean scores from pre- to post-intervention for both groups indicates an improvement in knowledge acquisition, with high SA participants demonstrating a greater overall gain (7.02 vs. 6.50). The trend lines for both groups confirm that higher situational awareness is associated with better learning outcomes, suggesting that the system effectively supports knowledge acquisition across different SA levels.

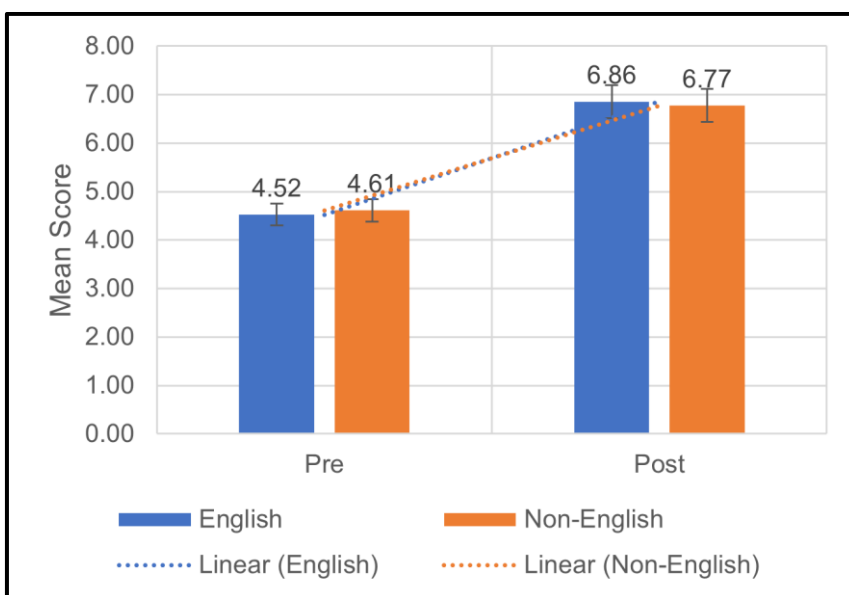


Figure 32. Mean participants' score based on Language Diversity.

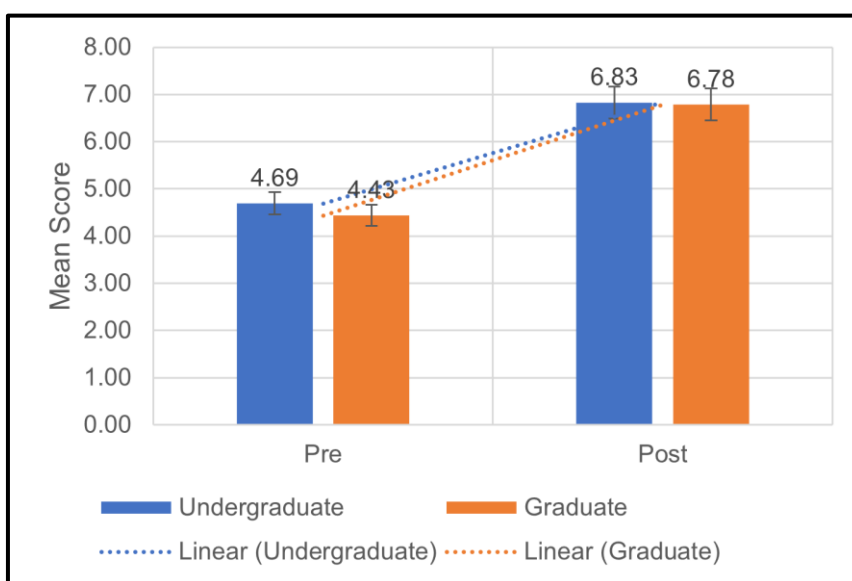


Figure 33. Mean participants' score based on Level of Education.

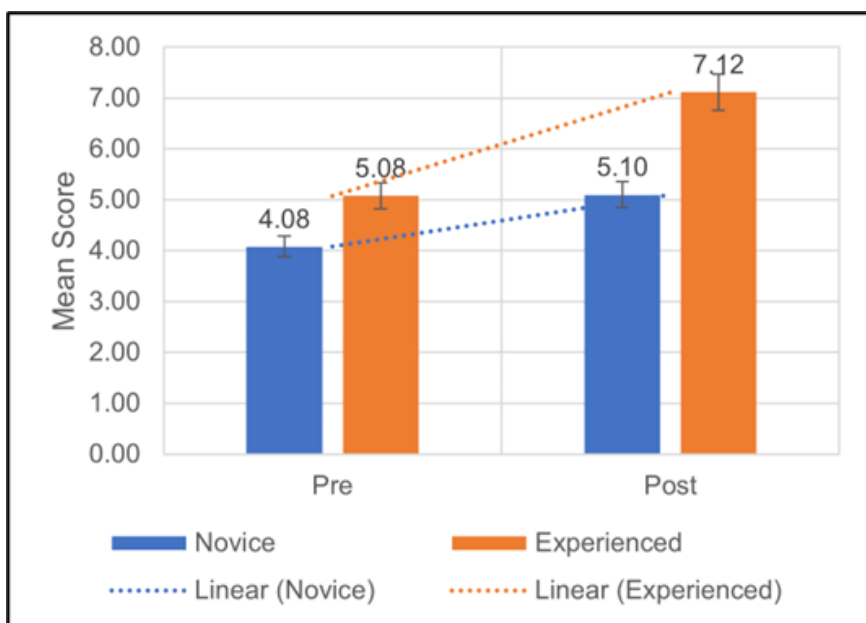


Figure 34. Mean participants' scores based on Level of Experience.

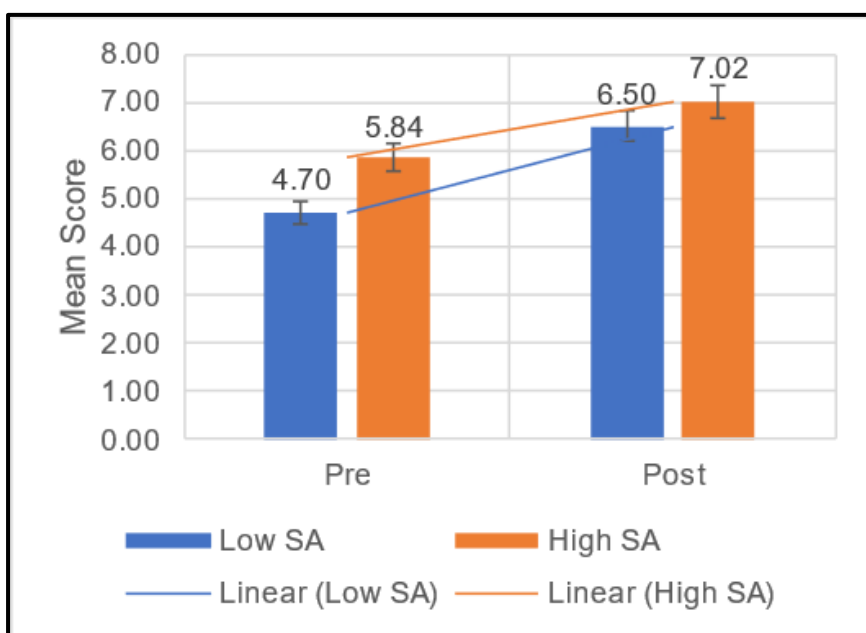


Figure 35. Mean participants' scores based on SA Level.

A moderate positive correlation between participants' initial knowledge levels and their subsequent performance underscores the influence of prior knowledge on the effectiveness of the training system. This suggests that individuals with a stronger foundation in safety principles are more likely to experience greater knowledge improvement after engaging with the system.

4.3 Evaluating Scene Understanding Module

4.3.1 Object Detection: Experimental Validation and Results

An open-source dataset from the Roboflow platform (open-ended tool for CV based dataset management and training) was utilized, containing nine classes for heavy machines of construction such as excavators, cranes, dumpers, mixers and human as workers (Roboflow, n.d.). The YOLO model (version-5), renowned for its high accuracy and efficiency in object detection tasks, was employed as the underlying architecture for our computer vision system. Object detection provides bounding box coordinates for the detected objects, but it is insufficient to assess whether a person is working with object under consideration or to determine how many people are on it at any given time. The dataset contains 4323 images including 3545 for training, 527 for validation and 250 for testing. The Figure 36 shows the annotated data set on roboflow (Roboflow, 2024).

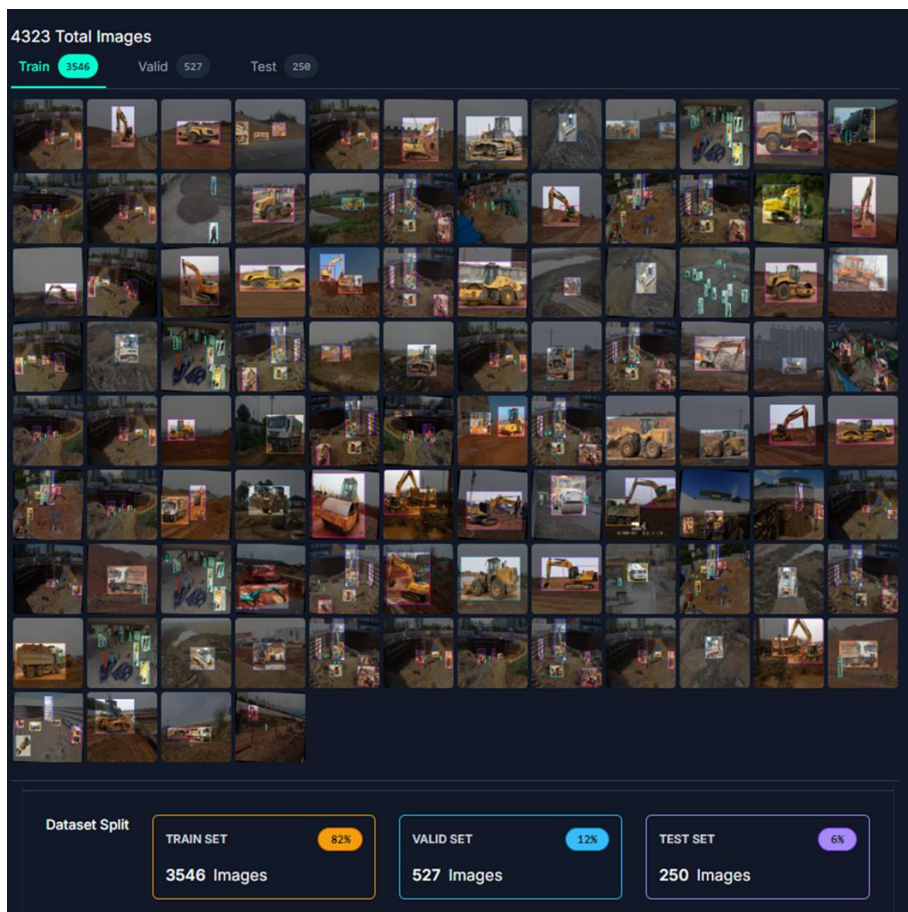


Figure 36. Description of dataset used for experimental purposes.

Computer vision based logical algorithms were adopted and implemented to systematically evaluate adherence to safety rules. These algorithms are designed to identify objects within scenarios and determine potential situations based on their relationships. By leveraging the power of Large Language Models (LLMs), the system can assess compliance with safety rules. For instance, when evaluating the proximity of workers to heavy machinery, the

system first analyzes whether both entities are present in the same frame. By establishing this relationship, the system can predict potential hazards that may arise in the future. Subsequently, the system will suggest possible hazards and precautionary measures to avoid accidents. These suggestions are presented in a clear and understandable text format, enabling workers to comprehend potential risks and their consequences.

The qualitative performance of the system was assessed using well-established performance metrics. To evaluate the performance of the proposed model, we conducted experiments using an available test dataset of construction images. The model's accuracy in detecting and classifying various construction elements was assessed using standard metrics such as precision, recall, and F1-score. The models were evaluated using precision-recall (PR) curves. The model demonstrates effective learning from the training data, generalizes well to unseen data, and achieves high precision (92%), recall (88%), and F1-score (90%) across multiple classes. The Figure 37 shows the PR curve for the proposed model. The PR curve demonstrates the trade-off between precision and recall. The high AUC values for curve indicate the model's strong performance in detecting and classifying construction elements.

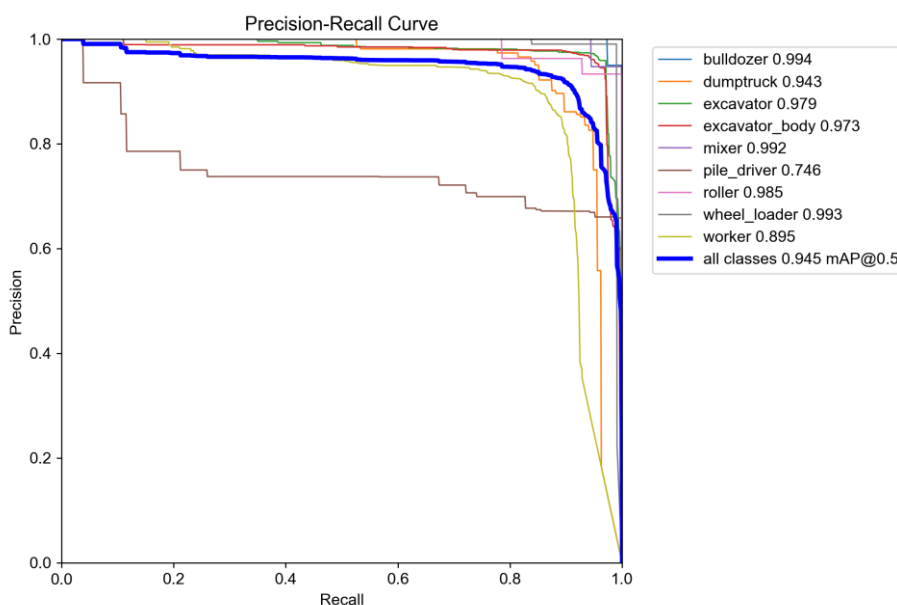


Figure 37. Average Precision Vs Recall for various classes.

To further illustrate the model's performance, we present example images with the detected and classified construction elements (Figure 38). These images visually demonstrate the model's ability to accurately identify various objects and elements within complex construction scenes. The images showcase the model's ability to detect and classify a wide range of construction elements, including machinery (e.g., excavators, bulldozers) and other relevant objects. The model accurately localizes the objects within the images and assigns the correct class labels, demonstrating its effectiveness in real-world construction scenarios. Following the detection of objects, the LLM based chatbot begins in which workers can read the predicted hazards through the

auto generated caption.

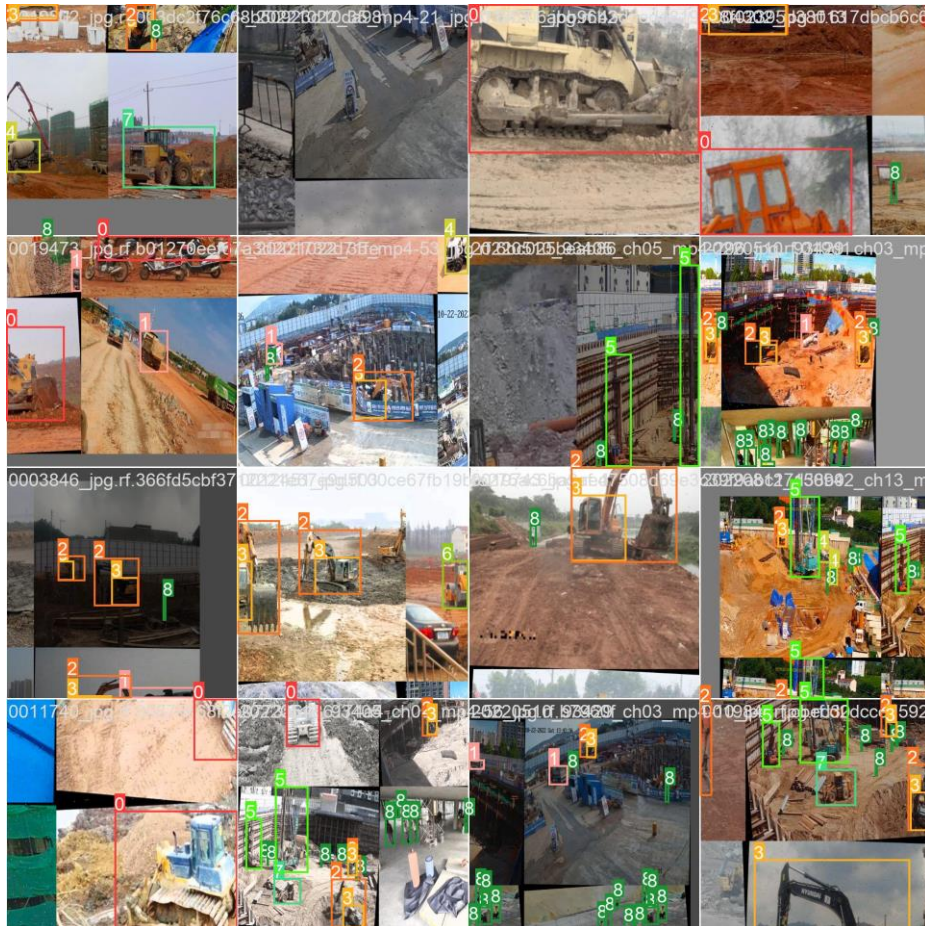


Figure 38. images with the detected and classified construction elements

Beyond caption generation, the interaction module extends its functionality by enabling real-time, interactive dialogues with users. The generated caption, along with other relevant contextual information, is used to initiate a conversation through the system's chatbot interface. This interactive component allows users to ask follow-up questions or seek further clarification

about the safety assessment. For example, the chatbot can respond to queries like "What specific rule is violated in this scene?" by retrieving and presenting the relevant safety regulation and its context. This fosters a deeper understanding of the safety issues present and supports informed decision-making.

Additionally, the chatbot supports various user interactions, including requests for mitigation strategies, explanations of applicable safety regulations, and discussions of resource constraints. Users can inquire about potential solutions to rectify unsafe conditions, such as asking, "How can we make this area safer?" The chatbot responds by suggesting specific actions aligned with best practices, such as installing guardrails or ensuring proper PPE use. The system can also handle questions like "What other safety precautions should be taken in this situation?" by generating recommendations that comply with industry standards. Moreover, in situations where resource limitations are a concern, the chatbot can propose alternative measures that are feasible within the constraints mentioned by the user, such as optimizing the use of existing resources or prioritizing the most critical safety interventions. The LLM-powered chatbot facilitates a continuous learning process for safety inspectors, helping them address specific hazards while also broadening their understanding of general safety practices. The interaction module, therefore,

not only provides immediate insights into site safety but also empowers users to engage actively in discussions about safety management, ultimately contributing to a safer work environment.

The model's ability to identify safety hazards or ensure that the correct equipment is being utilized can lead to proactive risk management and a reduction in workplace accidents. Additionally, the automation of these processes may free up valuable time not only for workers but also for construction managers and inspectors, allowing them to focus on more strategic decision-making rather than manual inspections.

4.3.2 User-Centered Evaluation of System Performance

The effectiveness of the system was evaluated through a user experience assessment utilizing the semantic differential scale, which focused on both technical and visual design aspects. This assessment engaged a diverse group of participants, including nine safety inspectors, eleven construction management graduates, and twelve students/researchers specializing in automation within the construction domain. To facilitate a comprehensive understanding of the system's capabilities, participants were introduced to its mobile application interfaces via a detailed demonstration.



Figure 39. Illustration of task-oriented hazards present with the real construction environment.

Following this introduction, participants interacted with the application within designed construction scenarios, such as understanding the scenes. Various construction real site images were provided to understand the scene by analyzing potential task-related hazards. For instance, Figure 39 shows an image illustrating the hazards of actual construction sites with potential hazards. After interaction, participants were asked to upload the same image in mobile based applications to explore more. System usually uploads pictures

and detect the hazardous objects which includes the defined classes (Figure 40), based on the detected objects, system automatically generates the captions. For instance, the generated caption of the example figure under consideration was *“An excavator and worker are seen in the image, highlighting a potential struck-by hazard as per OSHA Rule 29 CFR 1926.651. It is important to maintain 20-feet clearance from the excavator’s swing area to ensure safety in such work environments.”* Participants can see the results and hence they were also encouraged to ask further questions through the application's interaction module. For instance, in the example case, a user asked, “What are the other measures? The system can respond accordingly by utilizing the power of large language models and input query.



Figure 40. Detected objects using integrated CV-based module

Finally, to quantify the user experience, user feedback was collected using basic questionnaire. This questionnaire was based on a semantic differential scale comprising bipolar adjectives which employed across four critical criteria. Each criterion was rated on a scale from 1 to 7, as presented in Table 7. This dual-dimensional evaluation framework enabled a thorough assessment of system, considering both its technical efficacy and user-centric design considerations.

Table 7. Results of user satisfaction against different criteria

Criterion	Bipolar Adjectives	Mean	Standard Deviation
Reduction in Time and Cost	Time inefficient – Timesaving	6.33	0.47
	Costly – Cost effective	5.87	1.05
Suitability	Not suitable - Highly suitable	5.33	1.15
	Limited - Comprehensive	5.67	0.89
Ease of Use	Difficult to use - Easy to use	6.27	0.8
	Confusing - Intuitive	5.67	0.83
Operability under Dynamic Conditions	Unstable – Stable	4.73	0.79
	Unreliable - Reliable	4.87	0.89

The evaluation of the system, as summarized through statistical analysis in the corresponding table, underscores its significant effectiveness across crucial operational dimensions. Participants rated the system's performance in reducing time and cost with mean scores of 6.33 and 5.87, respectively, which indicates notable efficiency gains and cost savings. From a managerial perspective, the system's ability to streamline inspection processes has substantial implications for resource allocation and project timelines.

Furthermore, the system received favorable ratings for suitability and comprehensiveness, averaging 5.33 and 5.67, respectively. These scores reflect the system's adaptability to diverse inspection scenarios, enhancing operational efficiency while contributing to improved safety protocols and ensuring compliance with industry standards. Participants also expressed satisfaction with the system's usability, achieving a mean score of 6.27, indicating an intuitive design and user-friendly features. This aspect is

particularly crucial from a user experience perspective, as it fosters user adoption and engagement. However, it is important to note that participants raised concerns regarding the system's stability and reliability in dynamic environments, with mean scores of 4.73 and 4.87. These scores suggest areas for improvement to ensure consistent performance and bolster user confidence in the application.

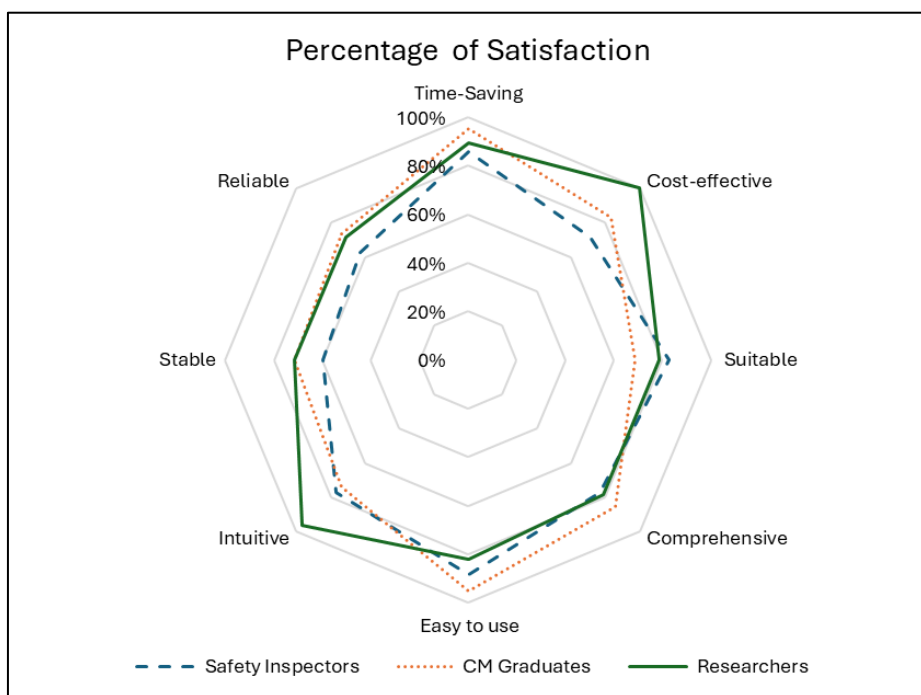


Figure 41. Satisfaction level of Users

Figure 41 illustrates the user satisfaction percentages for various aspects of the Scene Understanding Module system, as reported by three distinct groups: Safety Inspectors, Construction Management (CM) Graduates, and

Researchers. The radar chart shows that the system received high ratings for attributes such as being timesaving, cost-effective, suitable, and comprehensive across all user groups. Researchers particularly rated the system as cost-effectiveness and suitability, while Safety Inspectors and CM Graduates showed a balanced satisfaction across stability, intuitiveness, and ease of use. However, lower scores in stability and reliability for all groups suggest areas where improvements could be made. Overall, these results indicate that Scene Understanding Module is widely appreciated for its usability and efficiency, with some room for enhancement in operational stability. The next section discusses the research findings, detailing the study's main contributions and limitations. It also outlines potential directions for further research to build upon these results.

4.4 Post-Intervention Situational Awareness Level Assessment

The post-intervention phase was conducted to evaluate the improvement in situational awareness (SA) levels among participants following their interaction with the Knowledge Acquisition and Situational Understanding modules. As noted in the initial measurement, by the time of this experiment, none of the trainees had prior experience specific to the construction domain although all had some basic skills in using VR in specific domains. This phase

is designed to evaluate participants' cognitive capabilities and their ability to maintain situational awareness in dynamic environments following the engagement with the Knowledge Acquisition and Situational Understanding modules.

To ensure consistency, participants used the same head-mounted display (HMD) and VR setup during this phase as in the initial experiment (Figure 27). The demographics of the participants remained unchanged, as detailed in Table 4. As in the initial SA measurement, the Situation Awareness Global Assessment Technique (SAGAT) methodology was employed to maintain consistency in evaluation. The key distinction in this phase was that participants had undergone structured training and interaction with the Knowledge Acquisition and Situational Understanding modules—integral components of the proposed framework. These modules were designed to enhance participants' cognitive abilities, situational understanding, and hazard awareness in construction-Meta Scenes.

During this phase, participants were presented with a randomized set of SAGAT questions at freeze moments. There were 18-20 questions for each session, covering all three levels of SA: Level 1 refers to perception of elements in the environment, Level 2 to comprehension of their meaning, and

Level 3 to projection of future actions. The objective was to measure how effectively participants retained and applied the knowledge and strategies learned from the KA and SU modules. The responses were scored, and the total SA score was calculated based on the number of correct answers, consistent with the initial phase. Initially, 34 participants were recruited for the study. However, two participants were dropped from the study during the course of the experiment, resulting in a final sample size of 32 participants in post SA Score evaluation.

Results have shown that there is a significant improvement in the SA scores of participants after interacting with both KA and SU modules. Precisely, the average SA score increased from 35.0 to 47.84-an about 37% increase. Secondly, dispersion in the SA scores reduced significantly, as shown by the reduction in standard deviation from 8.02 to 3.40, which depicts consistency in performance after the intervention.

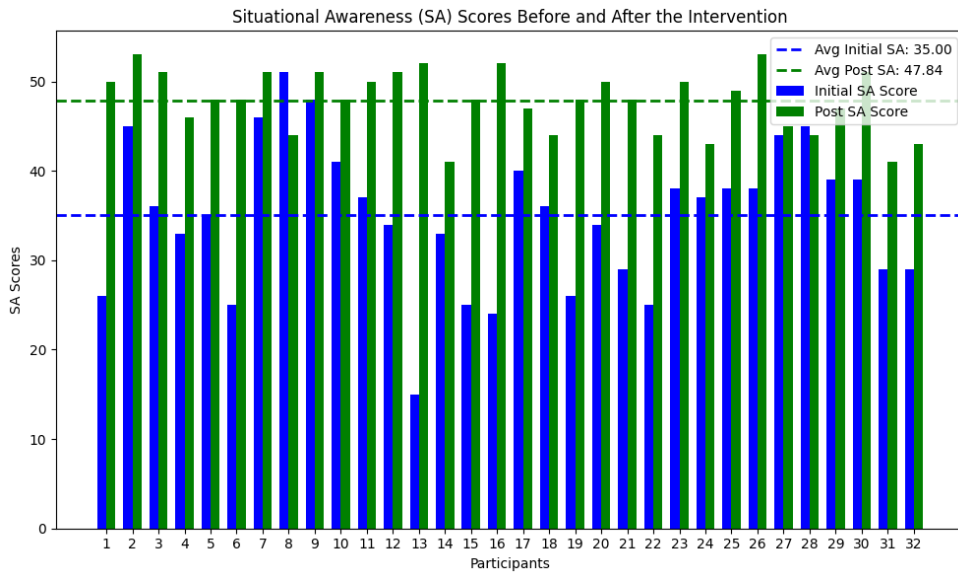


Figure 42. Situational Awareness Score before and after the intervention

Figure 42 illustrates the distribution of participants between high and low SA categories, with a marked shift towards higher SA scores post-intervention. For example, while the maximum initial SA score was 51, the maximum post-intervention SA score increased to 53, with the minimum post-intervention score rising to 41, compared to 15 in the initial phase. This upward shift highlights the effectiveness of the Knowledge Acquisition and Situational Understanding modules in improving participants' hazard recognition, comprehension, and future hazard projection abilities.

By maintaining the same measurement methodology as in the initial phase, the study ensured that the improvements in SA levels could be directly

attributed to the intervention. This again established possibilities for improved situational awareness according to new proposals but implemented using situational awareness principles within any construction environment. This supports several findings on the need for inclusion of structured training aspects, such as Knowledge Acquisition and Situational Understanding, within safety programs focused essentially on hazard identification and associated management at real construction sites.

4.5 Summary of Results

Situational Awareness level, assessed using the SAGAT methodology, revealed a significant improvement following the interaction with the Knowledge Acquisition and Scene Understanding modules. Specifically, the average SA score increased from 35.0 (initial) to 47.84 (post-intervention), reflecting a 37% improvement. The variability of the scores also decreased, with the standard deviation dropping from 8.02 to 3.40. This indicates that the participants became more consistent in their hazard recognition, comprehension, and projection abilities, demonstrating the effectiveness of the proposed framework.

Knowledge Acquisition scores, derived from quiz tests, also showed significant improvement in post-intervention. Initial scores ranged from 1 to

8 out of a possible 10, reflecting a broad range of baseline knowledge. After engaging with the Knowledge Acquisition module, the average score increased from 4.1 to 7.5, marking an 83% improvement. These results indicate that the KA module was highly effective in improving participants' safety knowledge and hazard awareness, which is critical for better situational awareness.

Scene Understanding (SU) module's effectiveness was evaluated through participant feedback. Although this is based on a single data point, the results indicated an increased confidence in applying acquired knowledge to real-world scenarios. Participants felt better prepared to recognize and anticipate hazards in construction environments after completing the SU module. This feedback suggests that the SU module contributed to improving participants' ability to apply theoretical knowledge to practical safety decision-making, aligning with the improvement in their SA scores.

The improvement in knowledge acquisition scores strongly related to participants' SA scores. As participants acquired more knowledge through the knowledge acquisition module, they demonstrated a better ability to recognize hazards, comprehend the risks, and project future scenarios. This correlation between knowledge acquisition and Situational Awareness performance

underlines the importance of knowledge-based training in improving situational awareness.

The feedback from participants who completed the interaction with mobile phone-based scene understanding suggests that applying the learned knowledge in a simulated, dynamic environment helped them improve their real-world hazard recognition skills. Though the feedback is based on a single data point, the positive response reinforces the value of integrating practical, scenario-based learning into the training framework.

The integration of both the KA and SU modules resulted in a comprehensive improvement in situation awareness. The improvement in KA scores was directly linked to better performance in SA assessments, while SU provided an additional layer of practical understanding that enhanced participants' ability to act on their newfound knowledge in real-world environments.

5. Discussion

The research findings reveal that the proposed system effectively enhances the safety knowledge of construction workers, as indicated by the significant improvements in post-test scores compared to pre-test scores. The system showed consistent effectiveness across diverse groups, including English and non-English speakers, as well as participants with different educational backgrounds (tertiary and secondary). These results highlight the proposed training system's potential for overcoming language and educational barriers in construction safety training. Such barriers often prevent workers from understanding essential safety protocols, which can lead to accidents and non-compliance with safety standards. By effectively accommodating various levels of language proficiency and educational backgrounds, the system promotes a more inclusive and comprehensive approach to safety training.

Furthermore, the study's findings underscore the importance of situational awareness (SA) in enhancing safety outcomes. Participants with higher SA consistently achieved better post-training scores than those with lower SA, indicating a correlation between situational awareness levels and learning outcomes. This insight is particularly relevant for construction sites where dynamic and complex environments require workers to adapt quickly to

changing conditions. By incorporating training techniques that focus on situational awareness, the system helps workers develop a sharper understanding of their surroundings, equipping them to identify potential hazards more efficiently and respond to them more effectively.

The ability of the training system to support knowledge acquisition across different SA levels demonstrates its flexibility and adaptability to the varied skill sets of workers. The research shows that individuals with higher initial knowledge and awareness experience greater improvements in their learning outcomes, suggesting that the system can cater to both novices and experienced workers. This versatility is crucial in the construction industry, where workers' familiarity with safety practices can vary widely. For novices, the system provides foundational safety knowledge, while for more experienced workers, it offers advanced techniques and reinforces existing skills.

Several practical construction site problems could be addressed or mitigated by implementing these research findings. For instance, the system can help reduce the incidence of common site accidents such as slips, trips, and falls by enhancing workers' awareness of unsafe conditions, such as cluttered walkways or unstable scaffolding. Additionally, the training can aid in minimizing equipment-related accidents by teaching workers how to properly

operate and maintain machinery, identify faulty equipment, and use personal protective equipment (PPE) correctly.

The presented framework can also address the challenge of reinforcing safety training in hazardous work environments through very simple Gaussian Splatting Technique, where traditional classroom-based training may not sufficiently prepare workers for real-world risks. Through realistic VR simulations, workers can experience high-risk scenarios, such as falling objects, electrical hazards, or working at heights, in a controlled environment, allowing them to practice proper responses without being exposed to actual dangers. This experiential learning approach not only improves safety outcomes but also boosts workers' confidence in their ability to manage emergencies. The study also emphasizes the need for a holistic training lifecycle that encompasses pre-training, training, and post-training phases. The research shows that static training approaches that do not consider dynamic learner information, such as cognitive behavior and attentiveness, may fall short in addressing individual workers' needs. By incorporating advanced technologies like conversational AI and computer vision, the system can continuously monitor workers' progress, detect gaps in knowledge, and provide tailored training content that evolves with the learner's development. This adaptive approach ensures that safety training remains relevant and impactful

throughout the worker's career.

The post-evaluation results further substantiate the effectiveness of the proposed safety training system in significantly enhancing workers' situational awareness (SA) and overall safety performance. Building on the positive improvements observed in the pre- and post-test assessments, the system demonstrates its capacity to dynamically adapt to individual learning needs, yielding substantial improvements in SA across a range of learner profiles. This is particularly noteworthy as it highlights the system's ability to foster both immediate and long-term safety improvements by leveraging continuous, real-time feedback and personalized content. The enhancement in SA is indicative of the system's potential to better equip workers with the skills necessary to identify hazards, anticipate risks, and respond appropriately in complex, dynamic construction environments. The post-training phase, reinforced by mobile-based job-site support, further ensures that workers carry forward the knowledge gained during training, thereby bridging the gap between theoretical learning and practical application. This comprehensive and adaptable approach to safety training represents a significant leap forward, offering more effective, inclusive, and context-aware solutions to construction safety challenges. Current safety training methods often lack mechanisms for ongoing reinforcement of learned skills, especially once workers are back on

the job site. The proposed system addresses this gap by incorporating mobile-based job-site support, which provides real-time feedback and guidance to workers as they navigate their tasks. This feature is particularly valuable in construction projects where conditions are constantly changing, and new risks can emerge at any time. By offering immediate support, the system helps workers apply what they learned during training, ensuring that safety practices are not only understood but consistently followed.

Moreover, the proposed system can help mitigate the challenge of language diversity on construction sites, where multilingual workers may struggle to understand safety instructions delivered in a single language. The system's ability to adapt content based on language proficiency allows for a more personalized learning experience, ensuring that safety messages are clearly communicated and understood by all workers. This capability can help bridge communication gaps that often contribute to misunderstandings and accidents on the job site. The findings also suggest that the system's data-driven approach to training offers valuable insights for safety managers. By analyzing the training data collected from VR sessions and on-site interactions, safety managers can identify common safety challenges, monitor workers' progress, and adjust training programs as needed. This proactive approach allows for continuous improvement of safety protocols and training methodologies,

which is essential for maintaining high safety standards in construction projects.

The relationship between situational awareness (SA), hazard identification, and the practical application of safety knowledge is multifaceted and critical for effective safety training. While high SA is essential for recognizing potential risks, it does not always guarantee that workers can identify and mitigate hazards in real-world scenarios. This gap underscores the need for a training approach that goes beyond awareness to foster a comprehensive understanding of hazards and their contextual implications. To address this, the study integrates three phases of construction meta-scenes—virtual, mesh-based, and real-world environments—which collectively enhance SA, knowledge acquisition, and scene understanding. By allowing participants to visualize simulations and experience dynamic safety scenarios, this framework not only raises awareness but also deepens understanding and prepares individuals to apply their knowledge effectively in on-site practices. This integrated method bridges the critical gap between recognizing hazards and implementing safer behaviors, ultimately improving overall safety performance.

5.1 Research Contributions

The insights gained from this study can be instrumental in enhancing the safety of workers in the construction industry. Developing a comprehensive understanding of the learner's Model development including static and dynamic information of workers can assist safety practitioners in identifying workers who may be at higher risk. This understanding can enable the practitioners to design targeted safety interventions tailored to at-risk workers, thereby helping to mitigate the risk of accidents at job sites and improve overall safety within the construction sector. The experimental modules presented in this study can provide direct support for workers in overcoming challenges related to their situational awareness. For example, engaging in training methods that emphasize the importance of maintaining high levels of SA can empower workers to better recognize and respond to hazards in their environment (Grégoire et al., 2022). Therefore, Occupational health and safety (OHS) organizations all around the globe can use it as a robust tool and technique aimed at addressing critical issues related to workers' safety behaviors.

Furthermore, by integrating advanced AI technologies—such as Whisper for accurate speech recognition, Amazon Polly for generating natural-

sounding voices, and ChatGPT for facilitating interactive and personalized communication—our system significantly enhances the effectiveness of safety training. Whisper effectively transcribes and understands trainees' spoken instructions and queries, addressing common language barriers and promoting inclusivity within the learning environment. This seamless communication is critical for creating an engaging and supportive educational atmosphere. This capability ensures that workers with varying levels of language proficiency, including non-native speakers, can engage with the training content without feeling excluded or disadvantaged. By bridging the communication gap, the system empowers diverse groups of workers to access critical safety information in a manner that is clear and comprehensible. This contribution is particularly significant in construction environments, where language differences often hinder effective training and lead to misunderstandings that can compromise safety. The system not only supports inclusivity but also enhances overall learning outcomes by tailoring responses to meet the unique linguistic needs of each trainee, thereby fostering a more equitable and effective training experience.

Amazon Polly further enhances the training experience by generating clear and natural instructions that resonate with trainees. The use of human-like voices improves comprehension and engagement, facilitating effective

knowledge absorption. ChatGPT serves as a virtual instructor, providing real-time feedback, answering queries, and guiding trainees throughout the training process. Its contextual understanding and ability to tailor responses to individual needs amplify the efficacy of knowledge transfer, resulting in an immersive and impactful learning experience.

While some may question the novelty of utilizing ChatGPT due to its commercial nature, it is essential to emphasize that ChatGPT functions as an integral tool within our unique system, specifically designed to address the challenges of personalized safety training. Moreover, future developments in large language models (LLMs) present an opportunity to create tailored solutions that cater explicitly to the requirements of the construction domain. As technology continues to evolve, the potential exists to employ even more advanced tools than ChatGPT in our training system. Nevertheless, the core innovation of our research lies in its commitment to revolutionizing construction safety education and training. The integration of these advanced technologies fosters enhanced communication, comprehension, and interactivity, ultimately resulting in a transformative learning experience.

In order to assess the efficacy of the designed prompt engineering technique, a test query was generated for evaluation purposes. The results obtained from

the web based ChatGPT and proposed systems were compared for assessing the quality of their responses. Figure 43 visual depicts this comparison. The responses of the two systems differ significantly. The response generated by our system is focused and concise, while the ChatGPT response covers a broader range of struck-by hazards and mitigation strategies. Notably, our system excels in interactivity, by fostering dynamic dialogue that engages trainees and allows for real-time feedback. Moreover, the framework utilizes personalized instructions tailored to the unique needs of individual trainees, by leveraging immersive virtual reality (VR) simulations to create an engaging and realistic training environment.

I need to work around the crane, is there any specific rule to follow?



Yes, working around cranes involves following specific safety guidelines to minimize the risk of accidents. Here are some important rules to follow:

1. Keep a Safe Distance

- Maintain a safe distance from the crane's swing radius to avoid being struck by the counterweight or load.
- Be aware of the "fall zone," where a load could drop if something goes wrong.

2. Stay Clear of Suspended Loads

procedures.

- Attend all safety briefings to stay informed about site hazards and safety practices.

7. Be Aware of Swing Radius and Blind Spots

- Cranes have blind spots where the operator may not see you; avoid these areas, and ensure the operator knows your location if you must work nearby.
- Be cautious of the crane's counterweight swing, as this can extend beyond the base and pose additional risks.

Observing these rules helps prevent accidents and ensures a safer work environment around cranes.

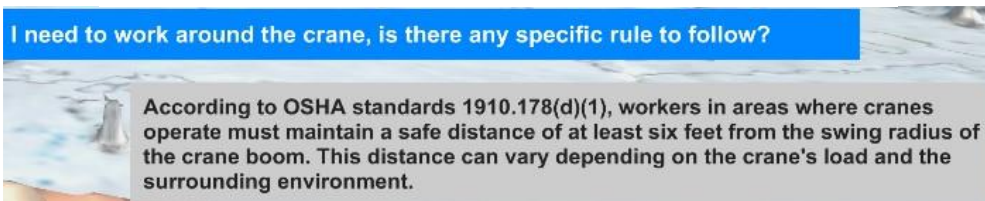


Figure 43. Comparative Illustration of Responses from Web-based ChatGPT and Proposed System to the Same Query.

When comparing our results to previous studies, some disparities emerge. For example, findings from Yu et al., 2022 suggest that novices may outperform experienced individuals in certain training scenarios. Our system represents a novel approach, leveraging AI as a real-time instructor capable of engaging in dynamic conversations and asking probing questions. This interactive feature may enhance the quality of generated outputs, allowing participants with prior knowledge to ask more focused questions, which leads

to contextually relevant responses.

Additionally, while (Adami et al., 2023) indicate that students perform better in VR-based training compared to workers, it is crucial to recognize that training effectiveness can vary based on design and implementation. Adami et al., 2023 acknowledge the limitations of their own study, emphasizing that their findings may not be universally applicable. As such, our research contributes valuable insights into the potential effectiveness of AI-VR-based safety training in the construction industry and serves as an important step in this burgeoning field.

The findings of this study provide valuable insights into the efficacy of our training system in enhancing safety knowledge among participants. The significant mean difference between the pre-test and post-test scores, alongside the higher mean post-test scores, indicates a substantial improvement in participants' understanding of safety practices. This enhancement is consistent across diverse participant groups, regardless of language proficiency or educational background. Table 8 illustrates how interaction with an AI avatar enhances critical outcomes such as hazard identification, situational awareness (SA), and scene understanding. It emphasizes the value of personalized, interactive training systems in improving safety outcomes on construction sites.

The table demonstrates that simply identifying hazards is not sufficient for effective safety training; it is equally important to recognize the various aspects of dealing with and mitigating those hazards. Through the conversational AI system, learners are provided with dynamic insights, allowing them to grasp not only the presence of hazards but also the nuanced actions required to respond to and mitigate them effectively. This approach strengthens the overall effectiveness of safety training by fostering deeper engagement, enabling better decision-making, and improving real-time risk management specifically, focusing on the personalization aspect while generating answers upon query. For instance, when a same query is asked by users having different profiles in terms of SA level and experience, the response would be different as per the profile of the user.

A comparative analysis was conducted to evaluate the variation in AI-generated responses when presented with the same query, tailored to different learner profiles. This was designed to highlight the adaptability of the AI system in providing context-specific guidance based on individual attributes such as situational awareness, experience levels, and educational background. For instance, the query, “Where are the blind spots around an excavator, and how should you position yourself to remain visible?” was posed to two distinct profiles: an experienced worker with high situational awareness, low education,

and English-speaking proficiency, and a novice worker with low situational awareness, low education, and English-speaking proficiency. For the experienced worker, the AI response was concise, emphasizing self-reliance and proactive measures, such as positioning themselves in front of the cab, using hand signals or radios to maintain communication, and dynamically adjusting their position based on the movement of the excavator. The Large language model anticipate by itself that the user could identify risks and independently manage the site safety. Conversely, for the novice worker, the AI provided step-by-step instructions, emphasizing basic safety practices such as staying in front of the cab, wearing high-visibility clothing, and obtaining confirmation from the operator before approaching the excavator.

The example case if user ask the query like “Where are the blind spots around an excavator, and how should I position myself to remain visible? different responses for different profiles could be possible for this query. For Experienced, High Situational Awareness, Low Education, English-speaking User: Blind spots are at the rear, sides, and near the boom. Stay in front of the cab where the operator can see you. Use hand signals or radios to stay connected and adjust your position as the machine moves. And for Novice, Low Situational Awareness, Low Education, English-speaking User: Blind spots are behind the excavator, along the sides, and near the boom. Always

stay in front of the cab and wear high-visibility clothing. Get the operator's attention and confirm they see you before moving closer.

The key differences between the two approaches lie in their emphasis on communication and decision-making styles. The first approach highlights independent decision-making, proactive positioning, and the strategic use of communication tools such as hand signals and radios to ensure clarity and coordination. In contrast, the second answer places greater importance on visual cues, seeking external confirmation by actively getting the operator's attention, and avoiding assumptions to minimize misunderstandings. These distinctions underscore the varying priorities in how information is conveyed, and actions are coordinated in critical scenarios. This response was designed to compensate for the worker's low situational awareness and limited ability to anticipate hazards. Comparative analysis demonstrates the AI's ability to adapt its guidance to the specific needs and capabilities of workers, ensuring that safety information is communicated effectively and tailored to reduce risks for individuals with varying levels of experience and awareness.

Table 8. Conversational AI's role in improving awareness, knowledge and understanding of the scene.

Hazard Identification Aspect	Pre-Interaction User Knowledge	Conversational AI Interaction	Post-Interaction Learning Outcome
Dynamic Swing Radius	The user knows the excavator's arm poses a danger but cannot determine when or how the radius changes during operation.	Question: Why does the swing radius of the excavator vary during operation, and how can it be anticipated? Response: The swing radius changes depending on the boom angle and load weight. Always maintain the maximum clearance regardless of the load size.	The user understands the dynamic nature of swing hazards and learns to apply conservative safety measures during proximity.
Operator Awareness	The user assumes the operator sees all workers within the operation zone but is unaware of blind spots created by the excavator's structure and movement.	Question: Where are the blind spots around an excavator, and how should you position yourself to remain visible? Response: Blind spots are located behind the cab, near the counterweight, and under the boom. Always position yourself where the operator can make direct eye contact or use a camera system.	The user learns how to safely position themselves to minimize the risk of being struck due to blind spots.

Load Swing Impact	The user does not recognize how the excavator's load, particularly irregular or heavy objects, can influence the arm's trajectory unpredictably.	Question: What factors make an excavator arm's movement unpredictable when carrying a load? Response: Factors include sudden weight shifts, operator miscalculations, and wind forces acting on the load. Always stand outside the swing radius.	The user gains awareness of how external and operational factors affect load dynamics, improving their hazard anticipation skills.
Emergency Stop Zones	The user does not know where to stand or move during an emergency involving the excavator, increasing risk during critical incidents.	Question: In case of an emergency, where should you move to ensure maximum safety? Response: Move to the designated safe zones marked on site plans or retreat to a location at least 10 meters away, ensuring no contact with swinging or rolling parts.	The user becomes capable of identifying and moving to emergency safe zones, even in high-pressure situations.

Before Interaction, workers generally possess awareness of potential hazards but lack a critical understanding of dynamic variables such as swing radius, blind spots, and external factors like wind or terrain. This limited knowledge restricts their ability to adapt to changing environmental conditions or respond effectively in emergency scenarios. After Interaction with the AI avatar, workers gain critical hazard awareness, acquiring precise knowledge of dynamic hazards like the swing radius and blind spots. Additionally, they learn proactive risk management strategies, such as safe positioning and emergency response procedures, tailored to site-specific conditions. This leads to advanced situational insight, where workers demonstrate an improved ability to predict and mitigate risks by adjusting their behavior based on evolving environmental factors and site dynamics. An example of implementation of the same content in terms of different aspects of construction safety management is presented in Table 9 with sample queries and expected avatar responses.

Table 9. Examples of implementation cases with Q&A

Scenario	Worker Query	Avatar Response
1. Task Clarification	I'm not sure how to connect this 3-inch PVC pipe to the existing 4-inch steel pipe.	To connect a 3-inch PVC pipe to a 4-inch steel pipe, you'll need to use a reducing couple-ing. This coupling will reduce the diameter from 4 inches to 3 inches, allowing you to connect the two pipes.
2. Safety Rules	I'm unsure about the safety rules for working on scaffolding at heights over 10 feet.	According to OSHA standards 1926.105, workers on scaffolds over 10 feet high must be protected by guardrails, toe boards, and a safety net if the scaffold is more than 25 feet above the ground.
3. Relevant Accidents	I heard about an accident that happened on this site last week involving a forklift. Can you tell me more about it?	Yes, there was an incident where a forklift driver failed to yield to a pedestrian, resulting in a collision. The pedestrian suffered minor injuries. The accident was investigated by OSHA, and the forklift operator was cited for violating safety regulations.
4. Training and Development	I'm interested in learning more about advanced welding techniques. Are there any online courses available?	Yes, there are several online platforms that offer welding courses, such as Udemy, Coursera, and LinkedIn Learning. These courses can provide you with in-depth knowledge and practical skills. Additionally, our on-site safety training coordinator can provide you with information about any upcoming welding workshops or certifications offered by local organizations.
5. Team Collaboration	I'm having trouble communicating with my team members on this project. Do you have any tips for improving teamwork?	If you're facing specific communication challenges, consider using a project management tool or scheduling regular team meetings to discuss progress and address any issues. It's also helpful to establish clear roles and responsibilities within the team.

Furthermore, the Scene Understanding module offers versatile applications and can be expanded in numerous directions beyond its initial use for training transfer. In education, Scene Understanding Module can provide an immersive learning tool for students, enabling them to explore construction sites and identify potential hazards using its computer vision (CV)-based detection system. For training, the module can identify real-world job-site situation

where workers practice responding to identified dangers, ensuring better preparedness. In inspection and hazard identification, Scene Understanding Module CV capabilities can autonomously scan work environments to detect machinery, structural risks, or improper safety equipment usage, with large language models (LLMs) generating comprehensive reports on the findings. For reporting, the system can interpret detected hazards and provide actionable feedback, streamlining the documentation process.

When it comes to problem-solving, Scene Understanding Module can suggest risk mitigation solutions based on recognized threats, guiding workers or site managers through appropriate safety measures. In site assessment for future activities, the module can analyze detected conditions to advise on preemptive safety steps before projects commence. Beyond construction, Scene Understanding Module utility can extend to the manufacturing industry, where it assesses tools and machinery, ensuring they are correctly sized and aligned with specifications to prevent mishandling and potential hazards. By combining CV-based detection with LLM-driven interpretation, Scene Understanding Module not only identifies safety risks but provides users with insightful, context-rich guidance tailored to varied operational scenarios. The next section addresses the possible limitations associated with this study.

5.2 Limitations

Despite the promising results, this study has several limitations that need attention. This section provides a detailed discussion of these limitations, highlighting areas for further refinement, exploration, or adjustments that could be required.

The system focuses only on one dynamic factor, i.e. Situational Awareness (Situational Awareness Module). This singular focus may limit the comprehensiveness of the findings, as other factors influencing safety performance were not examined. To enhance the reliability and applicability of the system, future research should consider incorporating additional dynamic factors that can interact with SA. For instance, incorporating behavioral and mental data captured through advance tools can also bifurcate the population before considering for training content delivery.

While the integration of prompt engineering techniques has improved ChatGPT's relevance to individual needs, there remains potential for enhancement in ensuring that the model generates contextually appropriate discussions during safety training sessions. One possible avenue for improvement is the development of a construction industry-specific fine-tuned/ RAG based LLM that can better address the nuances of this field. The

implementation of our training system on a larger scale must also consider the limitations of technological infrastructure and accessibility. Continuous improvement and refinement of the AI models and simulations used in our system are essential to ensure that it delivers accurate and reliable training outcomes across diverse real-world contexts

The selection of a limited sample consisting primarily of students, which may not fully represent the characteristics and experiences of actual construction workers. While student participants can provide valuable insights, the applicability of these results to the real-world context of construction workers may vary. Furthermore, the subjective categorization of participants into novice and experienced groups based on their domain knowledge may influence the accuracy of assessing learning outcomes. The reliance on self-reported measures could introduce biases and inaccuracies in the data collected.

The number of questions included in the assessments was constrained by the experimental duration and participants' availability. A broader range of questions and longer training sessions could enhance the understanding of our system's impact. Incorporating observational methods alongside quantitative measures will be essential for enhancing the validity and reliability of future studies, thereby addressing concerns regarding methodological robustness.

Integrating advanced technologies such as depth cameras during the scene understanding phase could significantly improve the results of computer vision (CV) applications. Depth cameras can provide accurate distance estimations, enhancing the contextual understanding of spatial relationships among construction elements. This improvement would contribute to a more robust analysis of the environment, ultimately facilitating better situation awareness and safety outcomes for workers. There is a need to test the system on a larger scale, particularly in commercial-level projects before practicing on jobsite. Conducting studies in real-world settings (other than controlled environments like laboratory) will provide a more holistic understanding of how the system performs under various conditions and complexities present in the construction environment. This larger-scale implementation is crucial to ensure that the findings are applicable and beneficial to a broader range of stakeholders within the industry.

Scene Understanding Module system employs the YOLOv5 model for object detection in the scene understanding module, which, despite being a highly efficient real-time object detection system, may still struggle with complex or obscured images where safety-critical elements are partially hidden. The use of more advanced detection models in future versions could enhance the system's precision and robustness, particularly in cluttered or visually

challenging environments. The computational demands associated with real-time CV and LLM integration are also challenging in some cases. Although the system is designed to operate efficiently on mobile platforms, its scalability and performance across devices with varying computational power remain a consideration. Extensive testing across different hardware configurations will be necessary to ensure that the system can be reliably deployed in diverse on-site conditions.

The proposed framework represents a significant departure from traditional safety training methods, leveraging artificial intelligence (AI) and visual technologies to provide a personalized and immersive learning experience. This innovative approach has the potential to revolutionize the construction industry by enhancing worker knowledge acquisition, improving safety performance, and ultimately reducing accidents. To maximize its impact, ongoing research and development are essential. Future exploration could focus on refining AI algorithms to enhance the accuracy of hazard identification and risk assessment, expanding VR scenarios to encompass a broader range of construction environments, and integrating the proposed framework with existing safety systems and tools. Additionally, ongoing optimization of the user interface is essential to improve usability and engagement. Advancements in these areas hold the potential to establish the

system as a vital resource for enhancing worker safety and promoting a safer, more efficient construction industry.

6. Conclusion

This chapter concludes the dissertation by summarizing the study and suggesting research directions for future studies.

Traditional construction training methods often involve a one-size-fits-all approach, where all learners, despite their diverse backgrounds and skill levels, engage with the same content. This uniformity can reduce training effectiveness, as workers may not adequately prepare for unpredictable real-world hazards. Recent efforts aim to improve training by tailoring it to workers' individual needs, a concept known as personalized learning. However, most studies focus on basic personalization using static information, such as demographic traits, and overlook dynamic factors like situational awareness. Additionally, current training systems fall short in addressing the entire training lifecycle, including pre-training, training, and post-training phases. The need remains for an intelligent system that dynamically adapts to workers' evolving needs, offering real-time support and comprehensive training throughout all stages.

The research aims to develop and evaluate an intelligent framework for personalized safety training that enhances safety performance by addressing individual worker characteristics. The study has three main goals: (1) to assess

situational awareness using advanced VR techniques to customize training based on each worker's safety needs; (2) to deliver personalized safety content that adapts to both static and dynamic learner characteristics; and (3) to provide continuous real-time job-site support through mobile technologies, offering immediate feedback and guidance to improve the transfer of training to actual job performance. Therefore, this study adopts a systematic approach to design, implement, and evaluates a personalized safety training system that addresses the shortcomings of traditional training methods, particularly the lack of personalization. The study begins with a literature review that identifies critical gaps in existing safety training practices and emphasizes the need for a comprehensive learner model encompassing all phases of the training lifecycle—pre-training, training, and post-training.

The proposed framework utilizes advanced techniques to collect personalized data and tailor training content to individual characteristics and job-specific needs. The training phase evolves intelligent avatars, powered by large language models, engage learners in real-time, providing adaptive feedback and scenario-based training. The post-training phase reinforces learning by applying acquired skills in real jobsite contexts. Advanced technologies, including virtual reality for situational awareness assessments, conversational AI for interactive content delivery, and computer vision for

real-time hazard analysis, enhance the system's effectiveness. Evaluation methods include both objective assessments of safety performance and subjective evaluations of the system's relevance, usability, and overall effectiveness, ensuring a holistic approach to training and knowledge retention.

The study demonstrated significant improvements in situational awareness (SA), with average SA scores increasing by 37%. Knowledge acquisition also improved substantially, highlighting the effectiveness of the Knowledge Acquisition module in enhancing safety knowledge and hazard awareness. Additionally, feedback on the Scene Understanding module indicated increased confidence in applying learned knowledge to real-world scenarios, further supporting the framework's ability to prepare participants for practical safety decision-making.

In conclusion, this research contributes to the current body of knowledge by providing a foundation for personalized construction safety training using advanced technologies. The framework developed in this research will aid construction practitioners, safety trainers and content developers in establishing safety culture in a diverse construction workforce and also help to align construction safety practices within the industry.

6.1 Future Research

To further enhance the capabilities of the proposed system, several promising avenues for future research could be explored. First, developing a dedicated language model using advanced techniques like Retrieval-Augmented Generation (RAG) or fine-tuning specifically for construction safety could significantly improve the system's ability to generate accurate and contextually relevant responses. This industry-specific model should be trained in a comprehensive corpus of construction-related texts to ensure it provides tailored, domain-specific guidance to users.

Additionally, expanding proposed framework into a multi-user platform that embraces the metaverse concept would facilitate collaboration and knowledge-sharing among construction professionals worldwide (Mystakidis, 2022). This transformative environment should integrate virtual reality (VR), augmented reality (AR), and immersive technologies, allowing users to collaborate, engage in realistic simulations, and exchange knowledge (Le et al., 2015). For instance, creating a virtual collaborative space would enable individuals from diverse regions and backgrounds to interact and share their experiences related to construction safety. AR could overlay virtual safety instructions and information onto real-world construction environments,

providing real-time guidance, hazard identification, and risk mitigation strategies to workers. For example, if a worker encounters a potentially hazardous situation, a virtual instructor could visually highlight the area of concern and offer step-by-step guidance on how to address it.

Furthermore, integrating blockchain technology into system could facilitate the implementation of an incentivization and reward system (H. Liu et al., 2023). By utilizing smart contracts, trainees' progress and achievements in construction safety training could be tracked transparently, allowing for automatic distribution of incentives and fostering a culture of continuous improvement among users. Given construction specific interest in financial incentives, this approach could effectively motivate and engage them in the safety training process, leading to improved outcomes (Tian et al., 2021). Another advanced concept for future exploration is the integration of wearable technologies with the proposed framework. Wearables such as smart helmets, safety vests, or wristbands equipped with sensors could provide real-time data on workers' movements, body posture, environmental conditions, and potential hazards. This data could be analyzed using AI predictive models integrated into the system to deliver personalized feedback, identify risky behaviors, and provide real-time safety alerts. Exploring the incorporation of machine learning algorithms to analyze training data could enhance the acceptance of

proposed framework. By leveraging data analytics, the system could continuously improve its training modules based on user performance and feedback, ensuring that the training remains relevant and effective.

The integration of Vision and language models with construction meta-scenes to improve the personalization aspect of content delivery process by considering all phases of the life cycle, is a novel approach for the current body of knowledge. In contrast to previous investigations about the safety training framework for construction workers, this dissertation is anticipated to provide significant additions to the existing body of knowledge within the Architecture, Engineering, and Construction (AEC) sector in the context of all phases of safety training i.e., Design, Delivery and Application. The adaptability of proposed framework to transform the safety domain by enabling effective communication and real time job site support could bring significant value to organizations operating across various regions. This will not only enhance the accessibility and effectiveness of safety training but also promote the dissemination of the best practices across industries on a global scale.

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Appendix I

Quiz sample questions used for SA level measurement.

Hazard 1: Excavator Scenario		
Level 1: Perception		
1.1	Where are you currently located in relation to the excavator?	a. In front of the excavator b. Behind excavator c. To the right of excavator d. To the left of excavator
1.2	Can you see the excavator from your current location?	a. Digging a trench. b. Laying pipes. c. lifting a load. d. Other
1.3	What task is the excavator currently doing?	a. Yes b. No
1.4	Is the excavator's movement impacting your task?	a. Yes b. No c. Not sure
1.5	What is the excavator's current activity?	a. Digging b. Moving dirt. c. Lifting a load. d. Other
1.6	Are there any other workers in the vicinity of the excavator?	a. Yes b. No c. Not sure
Level 2: Comprehension		
1.1	Are you aware of the excavator's proximity to your task area?	a. Yes b. No c. Not sure
1.2	Are you aware of the excavator's movement and direction?	a. Yes b. No c. Not sure
1.3	Are you aware of any potential hazards related to the excavator's proximity?	a. Yes b. No c. Not sure
1.4	Have you assessed the impact of the excavator's movement on your task?	a. Yes b. No c. Not sure
1.5	Are you aware of the excavator operator's signals or communication?	a. Yes b. No c. Not sure
1.6	Are you aware of any posted safety guidelines related to the excavator?	a. Yes b. No c. Not sure

1.7	Are you aware of any safety equipment or devices that are being used in relation to the excavator?	a. Yes b. No c. Not sure
Level 3: Projection		
1.1	Are you able to predict the excavator's future location and movement?	a. Yes b. No c. Not sure
1.2	Are you able to anticipate any potential hazards related to the excavator's proximity?	a. Yes b. No c. Not sure
1.3	Have you planned for contingencies related to the excavator's movement?	a. Yes b. No c. Not sure
1.4	Are you able to predict the excavator operator's next move?	a. Yes b. No c. Not sure
1.5	Have you identified any potential hazards related to the excavator's activity?	a. Yes b. No c. Not sure
1.6	Can you tackle the potential hazards with the excavator?	a. Yes b. No c. Not sure
Hazard 2: Crane Scenario		
Level 1: Perception		
1.1	Where are you currently located in relation to the crane?	a. In front of crane b. Behind crane c. To the right of crane d. To the left of crane
1.2	What task are you currently completing?	a. Lifting a load b. Building a wall c. Welding d. Other (please specify)
1.3	Can you see the crane from your current location?	a. Yes b. No
1.4	Is the crane's movement impacting on your task?	a. Yes b. No c. Not sure
1.5	What is the crane's current activity?	a. Lifting a load b. Moving horizontally c. Lowering a load d. Other (please specify)
1.6	Are there any other workers in the vicinity of the crane?	a. Yes b. No c. Not sure
Level 2: Comprehension		

1.1	Have you identified the crane's proximity to your task area?	a. Yes b. No c. Not sure d. N/A
1.2	How close is the crane to your task area?	a. Within 5 feet b. Within 10 feet c. More than 10 feet d. Not sure
1.3	Have you determined the crane's movement and direction?	a. Yes b. No c. Not sure d. N/A
1.4	Have you recognized any potential hazards related to the crane's proximity?	a. Yes b. No c. Not sure d. N/A
1.5	Have you evaluated the impact of the crane's movement on your task?	a. Yes b. No c. Not sure d. N/A
1.6	Have you been informed of the crane operator's signals or communication?	a. Yes b. No c. Not sure d. N/A
1.7	Have you been familiarized with any posted safety guidelines related to the crane?	a. Yes b. No c. Not sure d. N/A
1.8	Have you been briefed about any safety equipment or devices that are being used in relation to the crane?	a. Yes b. No c. Not sure d. N/A
Level 3: Projection		
1.1	How confident are you in predicting the crane's future location and movement?	a. Very confident b. Somewhat confident c. Not confident d. Not sure
1.2	How familiar are you with any potential hazards related to the crane's proximity?	a. Very familiar b. Somewhat familiar c. Not familiar d. Not sure
1.3	Have you taken any steps to plan for contingencies related to the crane's movement?	a. Yes, I have taken steps. b. No, I haven't taken steps. c. Not sure
1.4	How are you able to adjust your task to accommodate the crane's movement?	a. Very able b. Somewhat able

		c. Not able d. Not sure
1.5	How confident are you in predicting the crane operator's next move?	a. Very confident b. Somewhat confident c. Not confident d. Not sure
1.6	How familiar are you with any potential hazards related to the crane's activity?	a. Very familiar b. Somewhat familiar c. Not familiar d. Not sure
1.7	Can you tackle the potential hazards with the crane?	a. Yes b. No c. Not sure d. N/A

Hazard 3: Truck Scenario

Level 1: Perception

1.1	What is the current location of the truck in relation to your Location?	a. In front of truck b. Behind truck c. To the right of truck d. To the left of truck
1.2	What is the truck's current activity?	a. Loading b. Moving c. Unloading d. Other (please specify)
1.3	Are there any other workers in the vicinity of the truck?	a. Yes b. No c. Not sure
1.4	Are there any blind spots on the truck that could impact your task?	a. Yes b. No c. Not sure
1.5	Are there any warning lights or signals on the truck that you should be aware of?	a. Yes b. No c. Not sure
1.6	Are there other potential hazards in surroundings related to truck movement?	a. Yes b. No c. Not sure
1.7	Is the truck's load properly secured?	a. Yes b. No c. Not sure

Level 2: Comprehension

1.1	How close is the truck to your task area?	a. Within 5 feet b. Within 10 feet c. More than 10 feet d. Not sure
1.2	How familiar are you with the truck's movement and direction?	a. Very familiar b. Somewhat familiar

		c. Not familiar d. Not sure
1.3	Have you recognized any potential hazards related to the truck's proximity?	a. Yes b. No c. Not sure
1.4	Have you been informed of the truck driver's signals?	a. Yes b. No c. Not sure
1.5	How familiar are you with any posted safety guidelines related to the truck	a. Very familiar b. Somewhat familiar c. Not familiar d. Not sure
1.6	How familiar are you with any safety equipment or devices that are being used in relation to the truck?	a. Very familiar b. Somewhat familiar c. Not familiar d. Not sure
1.7	Are you aware of the truck's maximum speed and weight capacity?	a. Yes b. No c. Not sure
1.8	Are you aware of the truck's braking distance and turn radius?	a. Yes b. No c. Not sure
1.9	Are you aware of any specific safety procedures related to the truck, such as loading and unloading procedures?	a. Yes b. No c. Not sure
1.10	Are you aware of the emergency situations while crossing the track?	a. Yes b. No c. Not sure
Level 3: Projection		
1.1	Are you able to predict the truck's future location and movement?	a. Yes b. No c. Not sure
1.2	Are you able to anticipate any potential hazards related to the truck's proximity?	a. Yes b. No c. Not sure
1.3	Have you planned for contingencies related to the truck's movement?	a. Yes b. No c. Not sure
1.4	Are you able to adjust your task to accommodate the truck's movement?	a. Yes b. No c. Not sure
1.5	Are you able to predict the truck driver's next move?	a. Yes b. No c. Not sure
1.6	Are you able to anticipate any potential hazards related to the truck's load?	a. Yes b. No c. Not sure

1.7	Have you planned for contingencies related to the truck's load shifting or falling?	a. Yes b. No c. Not sure
1.8	Are you able to identify the truck's blind spots and adjust your task accordingly?	a. Yes b. No c. Not sure
1.9	Are you able to anticipate the truck driver's next move based on their signals and communication?	a. Yes b. No

Appendix II

Construction Focus Four: Struck-By Hazards (Example Questions)

NAME: _____

DATE: __/__/__

1. Struck-by injuries are produced by forcible contact or impact between the injured person and a/n _____.

- a. High voltage power lines or other energy sources.
- b. Object or piece of equipment
- c. Co-worker or employer

2. The following are examples of struck-by hazards. Which one is an example of a struck-by flying hazard?

- a. Hit by a nail from a nail gun
- b. Hit by a load dropped from a crane
- c. Run over by a vehicle in a roadway work zone

3. As a load is mechanically lifted, the materials _____

- a. May strike workers if the load swings, twists or turns
- b. Will not be affected by windy conditions or bad weather
- c. Can weigh any amount without causing a problem with the equipment

4. Among the list of ways workers can protect themselves when working on or near any construction zone, is to _____

- a. Direct traffic in and out of the work zone
- b. Work behind moving vehicles
- c. Wear high-visibility reflective clothing

5. A struck-by hazard can be described as anytime a worker _____.

- a. Falls from a height of greater than ten feet
- b. Is hit by a falling, swinging, flying or rolling object
- c. Can get any part of his/her body caught in or in between objects

6. Employers must protect workers from being struck by hazards by _____.

- a. Providing PPE such as hard hats and safety glasses
- b. Establishing guidelines that allow only contractors access in the crane work zone
- c. Ensuring guards on tools and equipment are removed when it is necessary to get the job done

Appendix III

Sample Conversation log between Participant and AI Avatar

Participant: *Hi, I've heard this site can be dangerous. What should I be most concerned about?*

Avatar: *Welcome! Construction sites have various hazards, with struck-by incidents being one of the leading causes of injuries. According to OSHA, struck-by hazards account for approximately **10%** of all construction fatalities. It's crucial to understand these risks and how to mitigate them.*

Participant: *Struck-by hazards? Can you explain that in detail?*

Avatar: *Certainly! Struck-by hazards involve injuries resulting from objects hitting workers. This can happen in several ways:*

- *Moving vehicles (like forklifts or cranes).*
- *Falling objects from scaffolding or elevated surfaces.*
- *Swinging loads from cranes that can hit workers in the vicinity.*

*In 2021, a construction worker was injured when a dump truck backed up and struck him while he was unloading materials. The driver didn't see him due to poor communication and lack of a spotter. This incident highlights the need for back-up alarms and clear communication, as per **OSHA Standard 1926.601** regarding motor vehicle operations on construction sites.*

Participant: *How can I protect myself from these hazards?*

Avatar: *There are several key strategies to enhance your safety:*

Training and Awareness: Familiarize yourself with the site's layout and the equipment in use. Attend all safety training, including those specific to struck-by hazards.

- *Personal Protective Equipment (PPE): Always wear appropriate PPE, including Hard hats (OSHA regulation 1926.100), High-visibility vests,*

Safety glasses

- *Safe Work Practices: Maintain a safe distance from operating machinery. For example, when cranes are in operation, be aware of the swing radius and stay clear of it.*
- *Use barricades and warning signs to delineate hazardous areas.*
- *Proper Communication: Use clear hand signals and verbal communication with equipment operators. For instance, during a lifting operation, ensure you have established a spotter to monitor the area.*

Participant: *Okay, but if something does fall, what should I do?*

Avatar: *If an object falls, your immediate action should be to take cover. Find shelter under a sturdy structure or move away from the potential path of the falling object. Always report any near misses or incidents to your supervisor. For example, in a database review, a worker was injured when a tool fell from scaffolding, which could have been avoided if proper toe boards were in place as per OSHA regulations.*

Participant: *Is there any formal training I can take to learn more about these hazards?*

Avatar: *Yes, we provide regular safety training sessions that align with OSHA standards. These sessions cover recognized hazards, including struck-by incidents, proper equipment usage, and emergency procedures. I recommend signing up for these sessions. Additionally, I can provide you with access to OSHA training materials and resources for further learning.*

Participant: *Thanks! This sounds important. Is there anything else I should keep in mind?*

Avatar: *Absolutely. Always maintain a proactive mindset towards safety. If you see any unsafe conditions, report them immediately. According to OSHA,*

effective communication and reporting can significantly reduce accident rates. Remember, your safety and the safety of others depend on vigilance and compliance with safety protocols.

Participant: *Got it! I'll be sure to stay alert and ask questions if I'm unsure. Thanks for all the helpful information!*

Avatar: *You're very welcome! Stay safe and remember: a well-informed worker is a safe worker. Let's keep our site accident-free together!*

Glossary

AI Detector	A computer vision algorithm that is used for detecting specific hazard scenario
Scene Understanding Module	Intelligent Scene understanding System for real time job site safety support.
Situational Awareness Module	Intelligent System to measure dynamic factors particularly individuals Situation Awareness level
Knowledge Acquisition Module	Intelligent Safety Content Delivery System by utilizing Conversational AI techniques.

국문 초록

메타장면 기반 LLM을 활용한 개인화된 안전 교육 프레임워크

라훗 후센

건축공학과, 건축시공 및 건설관리 전공

중앙대학교 대학원

건설 산업은 지속적인 안전 개선 노력에도 불구하고, 여전히 사고와 부상의 발생률이 높은 위험한 산업 분야로 남아 있습니다. 안전 교육은 이러한 위험을 최소화하는 가장 효과적인 해결책 중 하나로 인식되고 있습니다. 그중에서도 개별 작업자의 특성과 필요에 맞춘 학습 콘텐츠를 제공하는 맞춤형 교육은 참여도와 지식 유지율을 향상시키는 데 큰 잠재력을 보이고 있습니다. 그러나 기존의 전통적 안전 교육 연구는 주로 연령, 언어 능력, 직무 역할과 같은 정적 학습자 정보에 초점을 맞췄으며, 교육 생애 주기의 제한된 단계를 다루는 데 그쳤습니다. 이러한 접근법은 실제 위험에 노출되기 전에 작업자가 위험을 인식할 수 있는 능력을 평가하는 사전 훈련 단계의 핵심 요소인 상황 인식(SA)과 같은 동적 정보를 간과했습니다. 또한, 안전 실천을 강화하기 위한 사후 직장 지원 역시 충분히 탐

구되지 않았습니다.

이 연구는 이러한 한계를 해결하기 위해 정적 정보(예: 개인 특성)와 동적 정보(예: 상황 인식)를 통합하여 건설 메타 씬에 맞춘 안전 콘텐츠를 제공하는 포괄적인 안전 교육 프레임워크를 제안했습니다. 제안된 프레임워크는 세 가지 주요 모듈로 구성됩니다. 1) 상황 인식 모듈: 가상 현실 환경에서 작업자의 상황 인식을 평가하여 개인화된 학습자 모델을 개발합니다. 2) 지식 습득 모듈: 대화형 AI를 활용하여 학습자 개인 정보와 성과에 기반한 맞춤형 교육 콘텐츠를 제공합니다. 3) 씬 이해 모듈: 직장에서 실시간으로 비전 기반의 위험 요소를 인식할 수 있는 모바일 솔루션을 제공합니다. 프레임워크의 효과는 건설 현장에서 흔히 발생하는 사고 유형인 충돌 사고 시나리오를 통해 평가되었습니다. 연구 결과, 언어 능력, 교육 수준, 이전 경험 또는 초기 상황 인식 수준에 관계없이 모든 참가자가 유의미한 지식 향상을 달성한 것으로 나타났습니다. 특히, 높은 상황 인식(SA) 수준을 가진 사용자들은 낮은 SA 수준의 사용자들보다 지식 습득에서 약 10% 더 높은 향상을 보여 초기 인식 수준이 시스템의 효과를 향상시키는 중요한 요소임을 나타냈습니다. 또한, 씬 이해 모듈에 대한 사용자 피드백에서는 사용 용이성과 예측 가능성 측면에서 높은 수용도를 보였습니다. 사용자 만족도는 시간 효율성과 사용 용이성에서 높은 점

수를 받았지만, 동적 조건에서의 조작 가능성에서는 비교적 낮은 점수를 보였습니다.

지식 습득 및 씬 이해 모듈과의 상호작용 이후 참가자들의 사후 SA 수준은 초기 평균 35.0에서 47.84로 37%의 유의미한 향상을 보였습니다. 본 연구는 안전 실천을 촉진하고 교육 내용을 실제 건설 현장에 효과적으로 적용하도록 함으로써 글로벌 건설 커뮤니티에 중요한 기여를 합니다. 또한, 몰입형 기술과 AI 기반 교육 방법론의 새로운 연구 방향을 제시하여 다양한 건설 환경에서 사고를 줄이고 안전 결과를 개선하려는 노력에 기여합니다.

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